Chapter 12
Reliability and Reliability Testing

The reliability of power electronic devices and components has been mentioned several times in the previous chapters. It is so important, because it is a prerequisite for the performance in applications: Reliability is the ability of a system or component to perform its required functions under stated conditions for a specified period of time [SAE08]. The requested lifetime of power electronics systems is seldom below 10 years and can reach up to 30 years.

The reliability tests that will be discussed in detail in the following sections have been developed and established during more than 30 years of experience on power packages with implemented Si device technology. With the growing maturity of wide bandgap devices like SiC and GaN, it seems obvious that the same test procedures should be applied to modules equipped with these novel device technologies, since applications will require a comparable reliability level. However, wide bandgap devices exhibit specific differences with respect to the established Si devices, which have to be taken into account in reliability testing. These special requirements will be discussed in the respective sections.

12.1 The Demand for Increasing Reliability

Applications of power electronic devices face an increasing requirement for high reliability for several reasons:

- Power electronics faces a continuous demand for an increase in power density – often expressed in terms of controlled power per unit volume. This demand results in an increasing current density in power chips and in an increasing package density in power modules, with the consequence of higher temperatures and temperature gradients in the package.
- New fields of applications define more severe ambient conditions for the power packages, i.e. automotive hybrid traction systems, in which the combustion

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engine cooling system, featuring cooling liquid temperatures up to 120 °C, extracts the losses of power electronic components. This requirement has led to an extension of the operation temperature range from a maximum T_j = 150 °C to a specification limit of 175 °C.

- The number of interdependent frequency inverters is continuously growing in some areas of industrial automation. In automobile assembly for example, several hundred process steps are linked together in a single production line, each has to remain functional to keep the line running. For the same operational availability of the assembly line as for a single inverter, this results in mean time between failure (MTBF) values for each inverter divided by the number of interlinked inverters, which reduces the acceptable failure rate easily by orders of magnitude.

These general trends have been the boundary conditions for the progress of power electronics packages in the past and will continue to do so in the future.

It is obviously impossible to test the reliability of power modules under field conformal stress conditions, because these tests would last as long as the expected service lifetime in the field – 10 to 30 years. Thus, manufacturers of power modules have developed a canon of accelerated test procedures during the last 30 years, which are derived from experience and which are considered as a base line for the product qualification to verify the expected functionality over the total field lifetime.

This historic dimension might make it more comprehensible, that the general test categories seem identical for all power module manufacturers at the first glance, but exhibit considerable differences at a closer look. International standards are defining the general test setup, but the procedural details remain ambiguous. Therefore, every manufacturer of power modules has established its internal test philosophy, which is capable of defining, maintaining and improving the internal quality level, but which makes it difficult to compare qualification test results between different manufacturers.

Table 12.1 collects a common set of (accelerated) qualification tests for IGBT and MOSFET power modules with test conditions from [LV324]. The standards mentioned in the table are background for the specific conditions.

This compilation also indicates the progress of modern IGBT- and MOSFET-modules compared to conventional diode and thyristor modules: By way of example is the high temperature reverse bias test performed at 100% of the nominal blocking voltage and the power cycling requirement is increased from 10,000 to 20,000 cycles.

Before we will discuss each test in more detail, we have to add an important definition for every test: the failure criteria. It must be emphasized, that the exact definition of failure criteria is essential for the evaluation of any test. Table 12.2 states the common failure criteria for qualification and endurance tests, as they are specified by international standards.

The failure criteria in Table 12.2 allow a certain increase relative to specification limits or initial measured values. Since the accelerated test conditions aim to

Table 12.1 Reliability tests for qualification of IGBT/MOSFET-modules for industrial applications with reference to conventional modules

	Name	Conditions	Standards
HTRB	High temperature Reverse bias test	MOS/IGBT: 1000 h, T_{vjmax}, $0.8*V_{Cmax}$ partially V_{Cmax} ($\leq$2.0 kV), Conv.: 1000 h, T_{vjmax} −20 K, $V_R/V_D = 0.8*V_{RRM}/V_{DRM}$ resp. $0.66*V_{RRM}/V_{DRM}$[a]	IEC60747-9:2007 IEC 60747-2/6
HTGS (HTGB)	High temperature Gate stress test	1000 h, V_{Gmax}, T_{vjmax}	IEC60747-9:2007
H3TRB (THB)	High humidity High temperature Reverse Bias test	1000 h, 85 °C, 85% RH, $V_C = 0.8*V_{Cmax}$, however max. 80 V, V_G = 0 V	IEC60749 -5:2003
LTS	Low temperature Storage test	$T = T_{stgmin}$, 1000 h	JESD-22 A119:2009
HTS	High temperature Storage test	$T = T_{stgmax}$, 1000 h	IEC60749-6:2002
TST	Thermal shock	T_{stgmin} − T_{stgmax}, typ. −40 °C to +125 °C, $t_{storage} \geq$ 15 min, $t_{change} \leq$ 30 s 1000 cycles Conv.: 25 cycles[a]	IEC60749-25:2003
PC_{sec}	Power cycling	Internal heating and external cooling t_{on} < 5 s; $I_L > 0.85 \cdot I^b_{nom}$	IEC60749-34:2011
PC_{min}	Power cycling	Internal heating and external cooling t_{on} > 15 s; $I_L > 0.85 \cdot I^b_{nom}$	IEC60749-34:2011
V	Vibration	Sinusoidal sweep, 5 g, 10–1000 Hz, 2 h per axis, (x, y, z)	IEC60068-2-6 Test Fc
MS	Mechanical shock	Half sine pulse, 30 g, 18 ms, 3 times each direction ($\pm x$, $\pm y$, $\pm z$)	IEC60068-2-27 Test Ea

[a]Conventional devices—thyristors, diodes
[b]For min. one data point in [LV324]

simulate the stress applied in the total service lifetime, the specification limits are permitted to be exceeded within certain limits. For parameters less critical for the performance of the module, a greater increase can be allowed as shown for the leakage current. Other more critical parameters for the performance, as the forward voltage drop or the thermal resistance have to remain within closer limits, because they have a direct impact on the chip temperature.

The qualification tests in Table 12.1 can be classified into three groups. The first three tests are chip related qualification tests, which are also part of every chip

Table 12.2 Failure criteria for acceptance after endurance tests

Failure criteria IEC60747-9(2001):	
Gate leakage current I_G	+100% USL
Collector/Drain leakage current I_D/I_C	+100% USL
On-state voltage $V_D/V_{C(sat)}/V_F$ [a]	+20% IMV or 0% USL
Threshold voltage V_T	+20% USL
	−20% LSL
Thermal resistance R_{thjc}/R_{thjh} [b]	+20% IMV or 0% USL
Isolation voltage V_{ISOL}	Not below specification limit

USL Upper specification limit, *LSL* lower specification limit, *IMV* initial measured value
LV324 for power cycling tests [a]+5% IMV [b]ΔTj <+20% IMV

qualification. But since the chips are exposed to different substances during the module assembly process (i.e. solder flux, cleaning solvents and silicone soft mold), a confirmation of the chip reliability in the assembled module is inevitable. This set of chip related tests is followed by a group of seven tests related to stability of the package in the specified operation and storage temperature range and under external and internal temperature swings. Especially the power cycling test is important for the lifetime of power modules in application. The last two tests are confirming the mechanical integrity of the package.

12.2 High Temperature Reverse Bias Test

The high temperature reverse bias test (HTRB) – sometimes also referred to as hot reverse test – verifies the long term stability of the chip leakage currents. During the HTRB-test, the semiconductor chips are stressed with a reverse voltage at or slightly below the blocking capability of the device at an ambient temperature close to the operational limit. No degradation can be expected in the bulk silicon of the devices at these temperatures, but the test is able to reveal weaknesses or degradation effects in the field depletion structures at the device edges and in the passivation.

The electrical field has to be expanded at the edges of a power device to reduce the tangential field at the chip surface by a field depletion structure. This can be achieved by a field ring structure, by a variation of lateral doping or by a suitable geometric contour. Nevertheless, in silicon the electrical fields at the surface are in a range of 100–150 kV/cm. Movable ions can accumulate in these high field areas and can generate a surface charge. The source of these ions can be contaminations during the assembly process or residues of process agents, for example solder flux. The high temperature accelerates the process. The surface charge can alter the electrical field in the device and generate additional leakage currents. It can even produce inversion channels in device regions with low doping profiles and produce short circuits paths across the pn-junction.

The failure criteria limit the allowed leakage current increase after the test – when the device is disconnected from the voltage supply and cooled down – to identify such degradation effects. Additionally, most manufacturers of semiconductor devices also continuously monitor the leakage current during the 1000 h test and require a stable leakage current throughout the test.

Figure 12.1 shows an example of the recorded leakage current in a high temperature reverse bias test. Eight devices were monitored for the test duration. The devices are initially stable but after approx. 200 h the leakage current starts to increase. The test was aborted after 920 h due to the massive increase of leakage current of some devices. The test failed for these devices, because the applied junction passivation was not capable of fulfilling the requirement.

The test conditions apply a considerably higher stress than the typical application. The nominal DC-link voltage will be in the range of 50–67% of the specified device blocking voltage in real systems; it will be exceeded only by temporary voltage peaks. Furthermore, the device will reach the maximum operational temperature only occasionally in normal applications. Thus, the test is a highly accelerated procedure to generate stress within a test duration of 6 weeks for applications designed for a lifetime of 20 years and more.

But even if the junction passivation is compatible with the requirements and the assembly process, the test can reveal flaws in the package design. Figure 12.2 shows an example of a temporarily increased leakage current for a single device after nearly 100 h of test time. After reaching its maximum, the leakage current decreased and showed only a small increase at the end of the test. The following investigation showed, that a single wire bond of this device had a wrong geometry: it had no loop and was directly laying on the guard ring passivation of the IGBT.

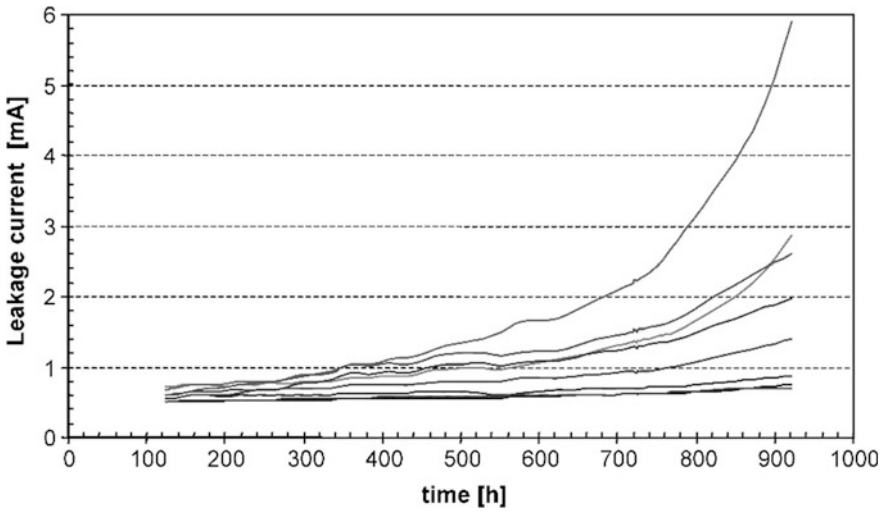


Fig. 12.1 Recorded leakage current during a high temperature reverse bias test – an example for a failed test

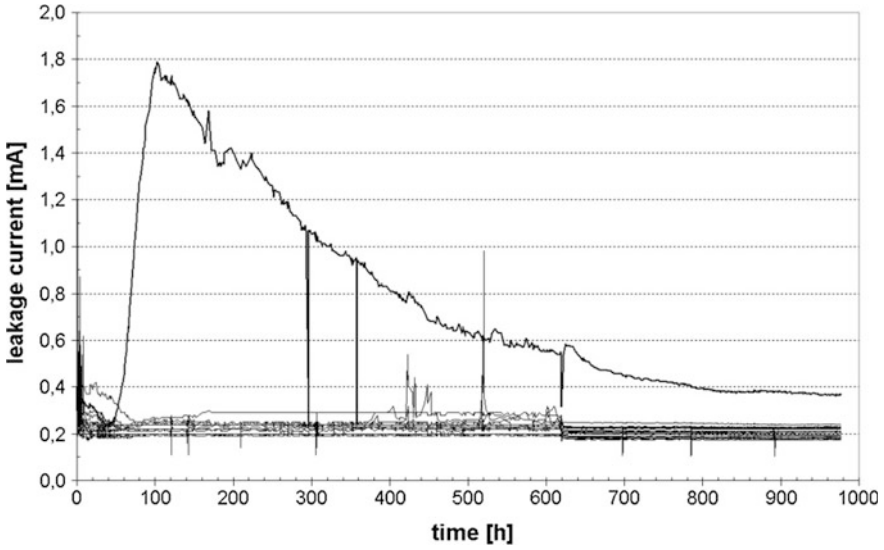


Fig. 12.2 Recorded leakage current during a high temperature reverse bias test – the test was passed but revealed a design flaw of the wire bond geometry

The test was repeated after the correction of the bond wire layout and the formally observed leakage current increase was eliminated. This result confirms the relevance of the HTRB-test for the package development.

In SiC, in most cases this junction termination is a "reduced surface field" (RESURF) structure which is fabricated as junction termination extension by a low doped p-layer, similar to Fig. 4.25. Due to the material properties of SiC, the critical electric field strength is in the range of 2.5 MV/cm, even up to 3 MV/cm at the edge of cells [Rup14]. The high critical field strength is not fully used in SiC nowadays and, compared to Si, higher safety margins are used in state-of-the-art devices. Nevertheless, electrical fields at the SiC surface in the range of >1 MV/cm are expected.

Furthermore, the acceptors implanted in the junction termination region are typically partially compensated (or enhanced) by surface charges generated by the polarity of the hexagonal axis of SiC (Si-face or C-face behave differently) and by processing techniques (e.g. oxidation or dry etch processes, see [Yan00]). Those surface charges typically are in the range of 10^{12} cm^{-2}, i.e., they have an influence on the optimum junction termination design in order to allow a maximum breakdown voltage. As discussed above, mobile ions can accumulate and can modify the initial surface charge. The surface charge alters the electrical field in the device and changes the breakdown voltage of the junction termination. Depending on the initial state both an increase and a decrease is possible. This is illustrated in Fig. 12.3 which shows the dependency of the breakdown voltage on the junction termination extension (JTE) dose as used for design of SiC devices.

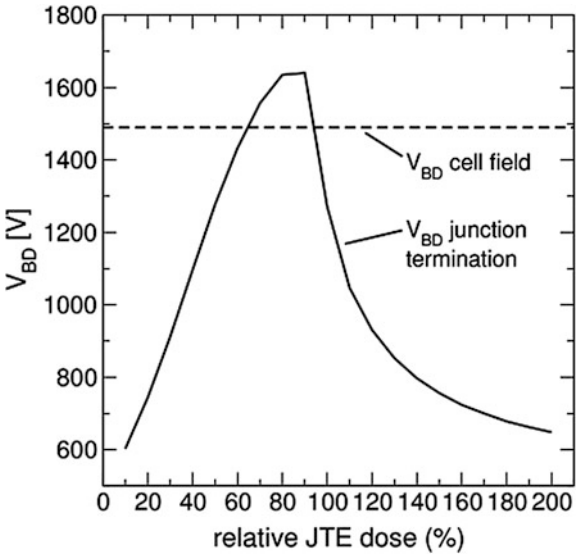


Fig. 12.3 Simulated breakdown voltage versus JTE edge dose in comparison to cell field breakdown **for a 1200 V device** indicating the process window for a single zone JTE implantation. Figure from R. Elpelt, Infineon

Even if a device is designed to have the avalanche breakdown in the cell field [Rup14], a change of the JTE charge via mobile ions may even lead to a shift of the avalanche onset from the cell field to the periphery, accompanied by a decrease of the avalanche current withstand capability. This becomes even worse if this effect does not occur homogenously on the chip circumference, but is concentrated on specific areas leading to punctiform avalanche locations. Such surface charges in the edge termination region can even produce inversion channels in device regions with low doping density and produce short circuit paths across the pn-junction. Since the base doping in SiC is about 100 times higher than in Si for the same blocking voltage, 100 times more surface charges are also required to create an inversion channel. In general, hard failures in HTRB testing are therefore not very likely in case of mature SiC device designs, but it is also important to carefully monitor any shift of breakdown voltage and leakage current triggered by the stress test. Such a drift is a good indicator for a JTE design which is not rugged enough. This holds not only for HTRB but – even more pronounced – for high voltage H3TRB stress tests (see Sect. 12.4).

12.3 High Temperature Gate Stress Test

The high temperature gate stress test (HTGS) or high temperature gate bias test (HTGB) confirms the stability of the gate leakage current. Even though the maximum allowed gate voltage is limited to ± 20 V, this voltage is applied to a not more than 100 nm thick gate oxide layer in state-of-the-art IGBTs and MOSFETs.

This results in an electrical field of 2 MV/cm across the gate oxide. For a stable leakage current, the gate oxide must be free of defects and only a low density of surface charges is tolerable. The boundary condition of maximum operational temperature for the devices again accelerates the test.

Since the leakage currents are very small (< 10 nA), this test is also extremely sensitive to surface contaminations on the chip. In test modules, where thermocouples were glued to the emitter contact of an IGBT for measurement purposes, the gate leakage was found to be considerably increased. This increase was caused by the residues of the solvent of the glue, which remained on the chip surface between the gate and the emitter and caused a measurable increase of the leakage current. Therefore, the gate leakage test also ensures the cleanness during the assembly process of a module.

While Si MOSFETs and IGBTs are usually very stable in the gate stress test, the gate oxide reliability has always been a great challenge for MOS (metal-oxide-semiconductor) structures fabricated on SiC substrate materials [Lip99]. As SiC technology has matured, SiC MOS devices have exhibited gradual improvements in time-dependent dielectric breakdown (TDDB) characteristics. However, even until now the gate oxide reliability of large devices (5–50 mm^2) did not succeed in achieving the same low rates of early failures as Si devices with comparable gate oxide area [Lut17].

Especially the trade-off between R_{on} and allowed electric field in the gate oxide is more challenging with SiC. The channel resistance is a major part of R_{on}. R_{ch} of the MOSFET is given by Eq. (9.3), with Eq. (9.1) it can be written as

$$R_{ch} = \frac{L}{W \cdot \mu_n \cdot C_{ox} \cdot (V_G - V_T)} = \frac{L \cdot d_{ox}}{W \cdot \mu_n \cdot \varepsilon_0 \cdot \varepsilon_r \cdot (V_G - V_T)} \tag{12.1}$$

with the entire width W and the length L of the channel; μ_n is the mobility of free electrons, V_G is the applied gate voltage, V_T the threshold voltage and d_{ox} the gate oxide thickness. The channel mobility in SiC is, despite some progress, much lower than in Si. Low R_{ch} might be achieved by reducing d_{ox} in SiC-MOSFETs in comparison to Si-MOSFETs and Si-IGBTs due to the impact of oxide thickness on the channel resistance. Increasing the recommended V_G is another possibility for low R_{ch}, see Eq. (12.1). Both measures increase the electric field in the gate oxide and are a risk for its durability.

The qualification conditions in Table 12.1 require a test of 1000 h at V_{Gmax} and 125 °C, the recent specification [LV324] requires T_{vjmax} which can be up to 175 or 200 °C. However, for a high-quality oxide layer it will not be possible to reach end-of-life within reasonable time. Intrinsic failures are caused by broken SiO_2-bonds or due to Fowler-Nordheim tunneling and are expected at 10 MV/cm for SiO_2. More detailed models [McP85, Kim97] allow the calculation of oxide breakdown depending on electric field, temperature and time. Assuming oxide thickness d_{ox} = 80 nm, V_{Gmax} = 20 V, $E = V_{Gmax}/d_{ox} = 2.5 \times 10^6$ V/cm with the parameters of [Kim97] leads, even for T_{vjmax} = 200 °C, to an intrinsic lifetime of about 32 years, which is not applicable as end-of-life test method.

A new test method is suggested in [Bei16] and [Bei17]. It is based on the distinction between extrinsic and intrinsic failures. Extrinsic failures are attributed to defects or weak points in the oxide, whereas intrinsic failures represent the true capability of the oxide itself [Coo97]. Simplified, any extrinsic failure is considered as gate oxide thinning [Lee88]. Figure 12.4 shows gate oxide with d_{ox} as nominal thickness. d''_{ox} and d'_{ox} are the reduced gate oxide thicknesses for the extrinsic failures. For the lowest thickness d''_{ox} the breakdown of gate oxide thickness will occur first due to the highest electrical field at a given gate voltage.

In the test at a temperature of 150 °C the applied gate voltage V_G of the devices is increased during the test every 168 h (one week). After each interval of 168 h, the test was interrupted for a measurement of the threshold voltage at room temperature. The gate voltage in the first step is the rated gate voltage V_{GUSE} according to the data sheet. In the second step, the voltage was set to the maximum use gate voltage V_{GMAX}. After this step, the gate voltage was increased in defined steps.

A graphical demonstration of the test strategy is shown in Fig. 12.5. In this image, the voltage steps increasing each interval of 168 h is shown. To be able to compare different devices, the difference between applied gate voltage V_G and the use-gate-voltage V_{GUSE} recommended by the manufacturer, $V_G - V_{GUSE}$, is chosen for characterization. By applying the $V_G - V_{GUSE}$-method, the test can be

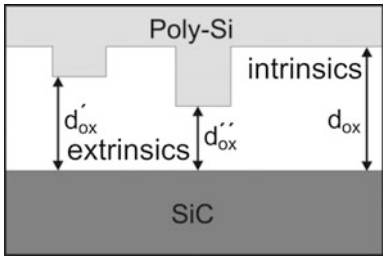


Fig. 12.4 Model for extrinsic and intrinsic failures. d''_{ox} and d'_{ox} symbolize gate oxide thinning for extrinsic failures

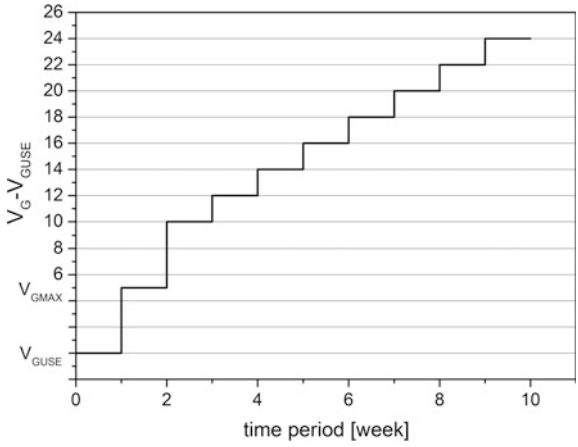


Fig. 12.5 Example for the test process with stepwise increased $V_G - V_{GUSE}$

reproduced in every laboratory. $V_G - V_{GUSE}$ is relevant for evaluating the impact of gate overvoltages in applications.

The test results depicted in Fig. 12.6 show a considerable difference in the gate stability for both manufacturers. In both test, a pronounced increase of the failure rate is observed. This increase could be attributed to the change of failure mode from extrinsic to intrinsic failures. The devices from manufacturer 2 reached the intrinsic limit at the step $V_G - V_{GUSE}$ = 20 V (Fig. 12.6a), while no failures were

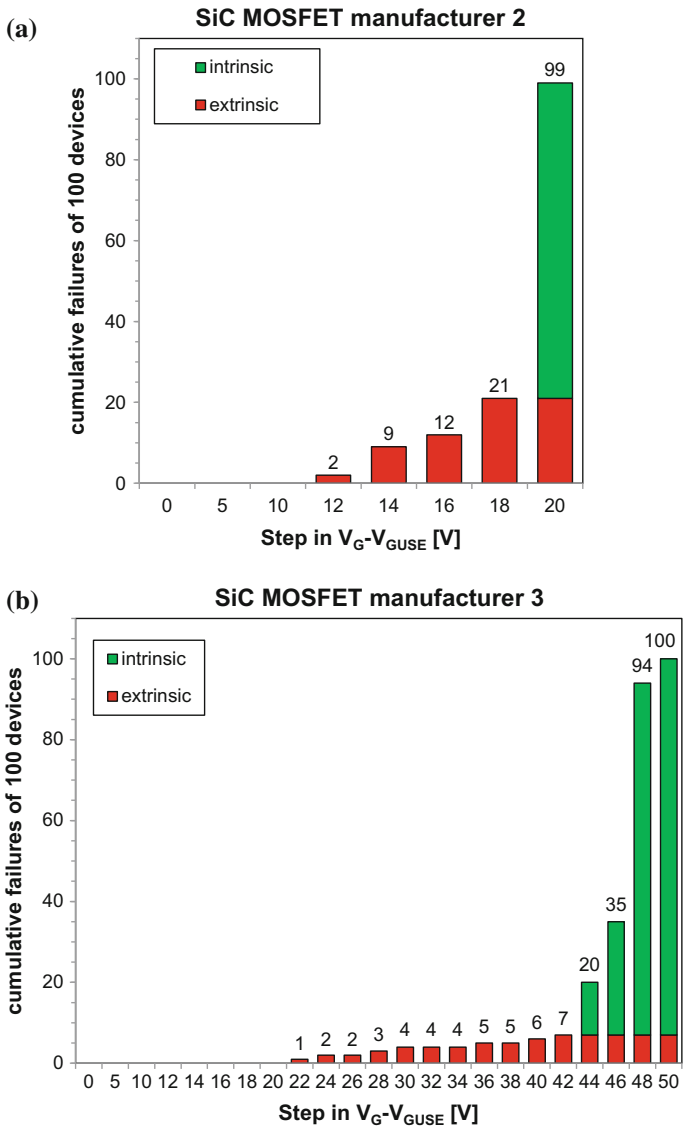


Fig. 12.6 Test results with stepwise increased $V_G - V_{GUSE}$ for two SiC MOSFETs from different manufacturers

observed for manufacturer 3 up to this stress level. The devices from manufacturer 3 reached the intrinsic limit at $V_G - V_{GUSE} = 44$ V (Fig. 12.6b).

Si-IGBTs of two professional manufacturers have been exposed to a similar test method in [Bei17]. They show a smaller rate of extrinsic failures. The intrinsic limit for the first IGBT manufacturer is found between 45 and 55 V, for the second IGBT manufacturer it is detected at 75 V. The differences in intrinsic failure limit can be explained by a different thickness of the gate oxide, see discussion of Eq. (12.1). From an application viewpoint, no evidence for significant gate oxide failures of IGBTs from established manufacturers in the field is available, even though there are many millions of IGBTs working in various applications. It must be considered that the applied gate voltage in the tests is much higher than used in practical applications.

Compared with IGBTs, only the SiC MOSFET of manufacturer 3 shows a similar behavior. The first intrinsic failures are observed in the same range of the gate voltage with a marginally higher level of extrinsic failures compared to Si IGBTs. This could be taken as evidence, that SiC MOSFETs are capable to reach a gate oxide reliability comparable to Si IGBTs. However, in state-of-the-art SiC MOSFETs huge differences between the investigated manufacturers were confirmed by the tests.

It was pointed out in [Lut17] that the difference between SiC and Si MOSFETs is a three to four orders of magnitude higher defect density of SiC MOS structures at the end of the process. This much higher defect density is most likely linked to substrate defects, metallic contaminations and particles. If further investigation would confirm that the extrinsic failures in the presented accumulated gate stress test are early life failures, then efficient screening tests could be developed on this basis. The data on failures as a function of test time reported in [Sie17] exhibit a decreasing failure rate over time and therefore identify the failure category as early life failures. The test procedure could also be applied to define the necessary gate oxide thickness to achieve a comparable gate oxide reliability for SiC MOSFETs as is established today in commercial Si switches.

12.4 Temperature Humidity Bias Test

The temperature humidity bias test (THB), also known as high humidity high temperature reverse bias test (H3TRB), is focusing on the impact of humidity on the long term performance of a power component.

Capsules are – when defect-free assembled – hermetically sealed against the environment. This is not the case for the majority of power module packages. Although bond wires and chips are completely embedded in silicone soft mold, this material is highly permeable for humidity. Therefore, humidity can intrude the package and can reach the chip surface and junction passivation. This test aims to detect weaknesses in the chip passivation and to initiate humidity related degradation processes in the packaging materials.

Sometimes, proposals are made to suppress the humidity access by an additional protective layer. But two strong arguments have to be taken into account: First, the silicone soft mold exhibits a high linear coefficient of thermal expansion ($\sim$ 300 ppm/K). This yields a considerable increase in volume during temperature swings and makes it difficult to apply a hermetically tight layer on top of the soft mold. Moreover, if a hermetically sealing is not achievable, then an additional layer can only reduce the diffusion rate of the humidity. However, a reduced diffusion rate works in both ways: It increases the time for the humidity to soak in, but it will also increase the time for the intruded humidity to be driven out. Therefore, a non-hermitical protection layer would increase the duration for the humidity to influence the device in operative condition. This is not desirable. If the humidity cannot be kept from intruding the package, then it should be driven out fast when the module is going in operational mode. For this reason, a highly penetrable embedding compound should be preferred.

The applied electrical field during the test acts as a driving force to accumulate ions or polar molecules at the semiconductor surface. On the other hand, the power losses generated by the leakage current must not heat up the chip and its environment and thus reduce the relative humidity. Therefore, standards require a limitation of the self-heating of the chip to not more than 2 °C. Consequently, the reverse voltage had been limited to 80% of the blocking voltage for low blocking voltage MOSFETs and was restricted to a maximum of 80 V for higher blocking capabilities in the past.

A number of field experiences in the past years have shown that this test condition is not sufficient for all application conditions. Field failures, which could clearly be attributed to the influence of humidity, have raised a discussion about this 80 V maximum applied voltage. Since the leakage currents of modern semiconductor chips are low enough to maintain the temperature increase within 2 °C even at 80% of the nominal blocking voltage for blocking voltages of 1200 V and more, the restriction to 80 V seems to be outdated.

Accelerated humidity bias tests [Zor14] have confirmed that the failure rate is considerably increased at elevated voltage levels. Furthermore, degradation of the blocking voltage was observed at bias levels of 65 and 90% of the nominal blocking voltage (Fig. 12.7), which was not observed at 80 V bias voltage.

For power semiconductors, the most prominent corrosion mechanisms induced by an exposition to humidity are the electrochemical migration and the aluminum corrosion [Zor15]. On the other hand, humidity can also enhance the mobility of mobile ions resulting from processing, polyimide passivation or mold compound [Beu89]. This can cause an alteration of the junction termination blocking capability faster than in the HTRB test.

A model to simulate the moisture inside the converters and power modules is presented in [Bay16]. This model describes the diffusivity, permeability and storage of humidity in polymers by equivalent R-C-networks. It was found that the moisture inside the converter cabinets can exceed the ambient conditions significantly and that condensation of water can occur under unfavorable conditions of operation.

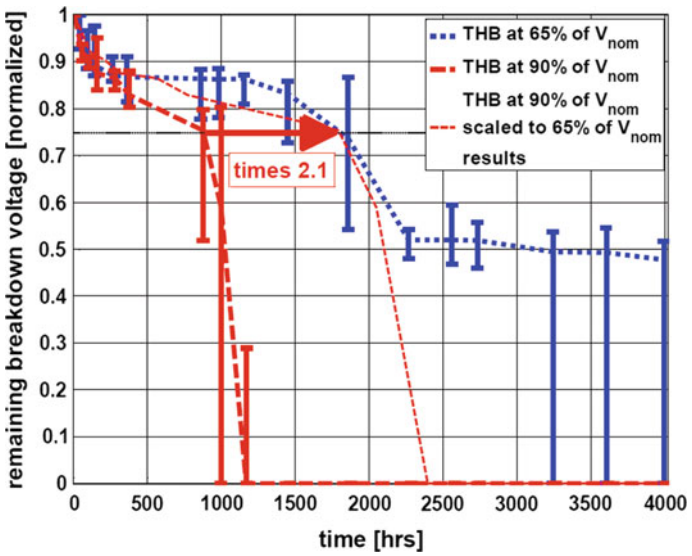


Fig. 12.7 Degradation of blocking voltage in humidity test biased at 65 and 90% of the nominal blocking voltage [Zor14]. Reprint with permission of VDE Verlag

SiC devices were found to be in the same way sensitive to humidity. Figure 12.8 shows a test of a phase leg of a 1200 V module with SiC MOSFETs executed at 1080 V, which is 90% of the rated voltage. Up to a time of 122 h the leakage current at 85 °C was below the measurement resolution. Then an increase becomes visible, after 134 h follows a fast increase and the module fails.

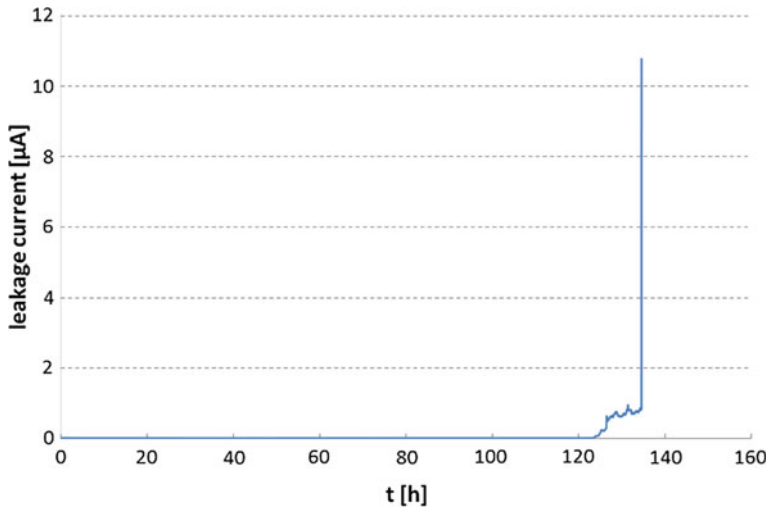


Fig. 12.8 Leakage current of one phase leg of a 1200 V SiC MOSFET module. Relative humidity 85%, temperature 85 °C, DC voltage 1080 V. Test executed by Chemnitz University of Technology

Many applications of power devices are outdoor applications, like photovoltaic inverters. The humidity might be more critical when the temperature inside the converter enclosure and modules housing is varying with daily temperature variations and weather constraints [Sad16]. A test where half of the power modules are placed indoor and half outdoor and both are operated in inverter mode is executed in [Sad16]. The disadvantage is that there is no acceleration of failure mechanisms.

Currently, activities have been started for improving the humidity sensitivity of power devices at high bias levels. In parallel, high voltage humidity testing (HV-H3TRB) is implemented in power module qualification programs. This progress will result in a higher reliability of non-hermetical power modules in high humidity application environments.

12.5 High Temperature and Low Temperature Storage Tests

The storage test at the maximum and minimum storage temperatures have been implemented to verify the integrity of the plastic materials, rubber materials, organic chip passivation materials, glues and silicone soft molds utilized in most state-of-the-art power module packages. These materials must maintain their characteristics in the complete specified storage temperature range.

At this point, a remark is necessary to prevent a misunderstanding of the term ‘storage temperature’. The denomination ‘storage temperature’ has been established in the early days of semiconductor power modules. It refers to the non-operational temperature limits for power modules assembled in a power electronic system. It does not refer to storage conditions of the unassembled power module, as the denomination might suggest. The description ‘non-operational temperature limits’ would be more appropriate, but the heritage of the early days of power modules impedes this transition.

Long term storage at high temperatures is critical for the mechanical strength of all thermoplastic housing materials. It is also minatory for the flame retardant additives to the thermoplastic materials required for the resilience against fire hazards. Conventional silicone soft mold starts to degenerate at temperatures above 180 °C, so that specially developed high temperature silicone gels must be implemented for junction temperatures above 175 °C.

Long term storage at low temperatures is critical for the softening agents in plastic materials and rubber materials, it can destroy the elastic capabilities of these materials and thus impair their functionality. Silicone soft mold compound is also limited in the minimum storage temperature; most standard soft molds are limited to −50 °C. Below this temperature, cracks might appear in the soft mold, which will not be cured by increasing temperatures and thus will compromise the insulating environment for high blocking voltage devices.

The standard storage temperature limits are −40 °C/+125 °C and a variety of materials are available for reliable performance in this temperature range. Extensions of the temperature range, which are demanded by many applications today, will make these qualification tests more challenging in the future.

12.6 Temperature Cycling and Temperature Shock Test

Temperature swings are an essential stress condition for every power electronic component in application. The temperature cycling test and the temperature shock test are two test methods to simulate ambient temperature swings during the field lifetime.

The test conditions are discriminated by the change rate of the externally imprinted temperature. If the rate of temperature change is slow in the range of 10–40 °C/min, the test is called temperature cycling test. In a temperature shock test, the ambient temperature is changed typically in less than 1 min. For power modules, this is typically achieved by a two chamber equipment, in with the air is permanently heated or cooled to the maximum or minimum test temperature, while an elevator carrying the devices under test moves between the two chamber in a time interval below one minute. Since the heat exchange rate is rather slow for a gas environment, the duration for reaching an equilibrium temperature distribution inside the module can vary from 30 min to 2 h, depending on the total thermal capacity of the devices under test.

A more extreme version of the temperature shock test is the liquid-to-liquid thermal shock test. In this test, the ambient is formed by appropriate liquids, heated or cooled to the desired temperature limits, for example oil at 150 °C or more and liquid nitrogen at −196 °C. Such test conditions are not common for modules, but are often performed for package elements as DBC substrates. In a liquid ambient, the heat transfer is much faster than in a gaseous ambient, so that an equilibrium temperature distribution can be achieved in minutes rather than hours.

Due to the wide range of heat transfer rates, the denomination of temperature swing test is somewhat ambiguous. While some manufacturers refer to the two chamber air ambient test as temperature shock test to distinguish it from the rather slow single chamber test with ambient temperature change rates in the range of 20 °C/min, other supplier label the two chamber air environment test as Temperature Cycling Test to discriminate it from the much faster liquid-to-liquid test. This ambiguity of denomination must be kept in mind when comparing qualification requirements and test results from different manufacturers.

A common boundary condition for all types of temperature cycling tests is the requirement, that the cycle time must be long enough, so that all parts of the assembly reach the maximum or minimum temperature – which are typically the storage temperature limits – so that the assembly is in a thermal equilibrium condition. Since the test simulates the impact of temperature changes by external sources, for example the change of ambient temperatures or an increase of heat sink

temperature by other heat sources, the power modules are not actively stressed by current or voltage. The changes in parameters are checked by an initial and final measurement and have to comply with the failure criteria.

The combination of different materials with different coefficients of thermal expansion results in high mechanical stress in the system. More so, the bimetal effect causes a cyclic deformation of the module. Simulations of the thermo-mechanical behavior of a power module have shown [Mik01], that if this bimetal bending is reduced – for example by mounting the module on a heat sink – the stress is reduced and the lifetime is extended. Therefore, modules should be mounted on assembly plates during the test to simulate as close as possible the application conditions.

The cyclic mechanical deformation generated by temperature cycles due to the difference in coefficients of expansion of the material layers causes stress in the functional layers themselves and in the interconnection layers. This will lead over time to the initiation of cracks and cause growing delaminations in these layers. Scanning Acoustic Microscopy (SAM) is the appropriate detection method of identifying delaminations in power semiconductor modules.

An example of the damage caused during temperature cycling with classical 34 mm base plate modules show the SAM images in Fig. 12.9. Two different solder materials for the substrate-to-base plate interface were compared in this test: a RoHS compatible SnAg(3.5) solder and the classical SnPb(37) solder. Both solder interfaces – the interface between the substrate and the base plate and the interface between chip and substrate – were investigated by choosing the appropriate time-of-flight windows in the SAM signal for each solder type. The comparison between the initial measurement and the SAM image after 200 temperature cycles (−40/+125 °C) in a two-chamber test equipment reveals growing delaminations in the substrate-to-base plate solder layer for both solder versions, indicated by the white areas (= regions of high reflection) which move inward from the corners and short edges of the substrates. The classical eutectic SnPb solder shows more damage along the outside short sides of the substrates, where the terminals are connected at the topside of the substrate.

The SAM images of the chip-to-substrate interface show no indications of any fatigue in the chip solder interfaces, but it presents black areas in the regions, were solder delaminations are found in the substrate-to-base plate solder layer. This artefact is produced by a lack of acoustic energy in these regions, because most of the signal was already reflected in the delaminations found in the substrate-to-base plate solder layer. Since the SAM signal is injected form the base plate surface, reflections nearer to the base plate reduce the signal propagating into deeper layers. However, this common artefact allows conveniently evaluating how close the delamination has come to the chip position. As Fig. 12.9 clearly shows, the delaminations have propagated much deeper under the chip areas for the SnPb solder system as for the SnAg solder system with a stronger unfavorable impact on the thermal resistance of the chips.

The lifetime under temperature cycles is determined by the combinations of different materials (with different coefficients of thermal expansion) and the stability

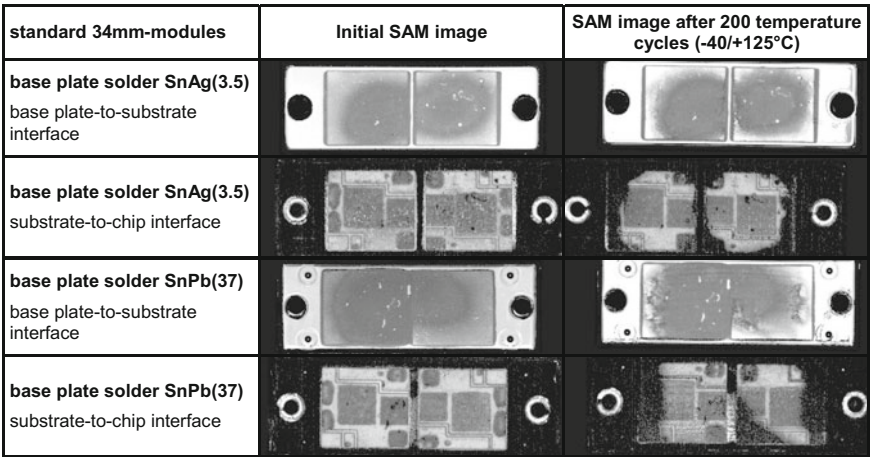


Fig. 12.9 Scanning Acoustic Microscope (SAM) images of standard 34 mm modules before and after 200 temperature cycles (−40/+ 125 °C) – different delamination patterns are found for different base plate solder materials

of the interconnect layers. Due to the mechanical deformation, smaller packages as the TO-family are more stable than larger modules that are more complex. Since the source for the passive temperature cycles is located outside of the module, only the used materials and interconnection layer decide about the reliability of a package.

12.7 Power Cycling Test

In contrast to the temperature cycling test, the power chips are actively heated by the losses generated in the power devices themselves in a power cycling test. This accounts for a fundamental difference between the two tests: The durability of a power module in temperature cycling tests is determined by the CTE of the implemented materials and the geometry of the layers. In a power cycling test, the amount of losses can be affected by the chip technology and by the silicon area implemented in the module. Therefore, any power cycling lifetime requirement can be met just by implementing sufficient silicon area to reduce the temperature swing generated by the chip losses. However, commercial aspects limit this option in practical applications.

12.7.1 Power Cycling Test Execution

During power cycling test, the device under test is mounted on a heat sink as in a real application. A constant DC load current is conducted by the power chips and

the power losses are heating up the chip. When the maximum target temperature in the chip is reached, the load current is switched off and the system cools down to a minimum temperature. The reaching of the minimum temperature completes the cycle and the next cycle begins by starting the load current again. During each cycle, considerable temperature gradients are generated inside the module.

An exemplary test setup as established for Si IGBTs is shown in Fig. 12.10. The device is heated up by a DC constant current load pulse. At the end of the load pulse the upper temperature T_{vjhigh} is reached and the device cools down. The lower limit of the junction temperature T_{vjlow} is reached at the instant when the load current is turned on again, and the cycle is repeated. The characteristic parameter for power cycling tests, the temperature swing ΔT, is given by the temperature difference between maximal junction temperature T_{vjhigh} at the end of the heating phase and the minimal junction temperature T_{vjlow} at the end of the cooling interval:

$$\Delta T = T_{vjhigh} - T_{vjlow} \tag{12.2}$$

In Fig. 12.10, ΔT_j can be determined as 78 K.

Another important parameter for the power cycling test is the medium temperature T_m of the swing:

$$T_m = T_{vjlow} + \frac{T_{vjhigh} - T_{vjlow}}{2} \tag{12.3}$$

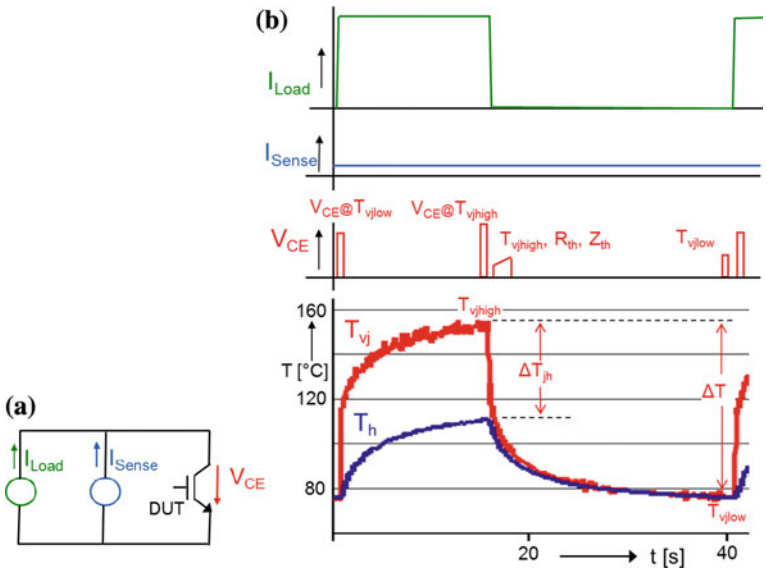


Fig. 12.10 **a** Basic test setup, **b** course of time for load current, sense current, executed voltage measurements, virtual junction temperature and heat sink temperature for one cycle of a power cycling test

Because of these relations, any pair of the parameters ΔT_j, T_m, T_{vjhigh}, and T_{vjlow} can be used as characteristic parameters for the temperature swing. Further parameters, e.g. the duration of the load pulse, are of importance as shown below. A long power pulse duration (15 s in Fig. 12.10) usually represents a higher stress for the devices.

The measurement of the junction temperature is executed with the constant sense current I_{Sense} which must be small enough to create only negligible losses, $I_{Sense} \approx 0.001\ I_{Load}$. A pn-junction is used as temperature sensor, its junction voltage is the temperature-sensitive electrical parameter (TSEP). This method is known as the determination of the virtual junction temperature T_{vj}. It is based on the physics of a pn-junction whose characteristic is described in Chap. 3 with Eqs. (3.50) and (3.51). It leads to a voltage V_j which is strongly depending on temperature given in Eq. (3.52). An example is given in Fig. 3.12.

The voltage V_j is decreasing with T due to the domination of n_i in Eq. (3.51). The linear dependency is lost if the voltage drop in the base is to be taken into account.

First, a calibration function $V_j(T)$ is determined, see Fig. 11.19. When a current I_{Sense} is applied, the temperature can be read out. It is done in Fig. 12.10b just after turn-off of I_{load} to determine T_{vjmax} and just before the next turn-on to determine T_{vjmin}. The method to determine T_{vj} is established since the beginning of power device development. The pn-junction of a diode or the base-emitter junction of a bipolar transistor is used in [Oet73]. Thermal resistance in data sheets of European manufacturers is determined with the $V_j(T)$ method.

The so-determined virtual junction temperature T_{vj} deviates significantly from the maximum and minimum temperatures on the chip surface. The lateral temperature gradient across the die (see Fig. 11.20) is especially pronounced for chips > 1 cm^2 and under the condition of forced water cooling. The temperature at the center position of the die can be 20 K higher and at the edges 40 K lower than the temperature measured with the $V_j(T)$ method. A detailed investigation of the geometrical interpretation of the virtual junction temperature value for an IGBT is reported in [Scn09]. It shows that T_{vj} correlates to the area related average of the chip surface temperature. This is important when comparing temperature measurements to simulation results. The comparison with an IR camera measurement in this investigation revealed, that the shading of the temperature radiation of the IGBT surface by wire bonds affects the temperature measurement and leads to a lower value of the area related average of the junction temperature.

Details of measurements with an IGBT at turn-off of I_{load} are shown in Fig. 12.11. Just before turn-off, I_{load} (= I_{CE}) and V_{CE} are measured, leading to $P_v = I_{CE} \cdot V_{CE}$. After turn-off and a time interval t_d, V_{CE} at I_{sense} (= V_j) is measured. With the help of the calibration function, it gives $T_{vjhigh} = f(V_{CE})$. The case temperature T_{case} is measured by a thermocouple, hence obtaining $\Delta T = T_{vjhigh} - T_{case}$, see Fig. 12.10. The thermal resistance follows with

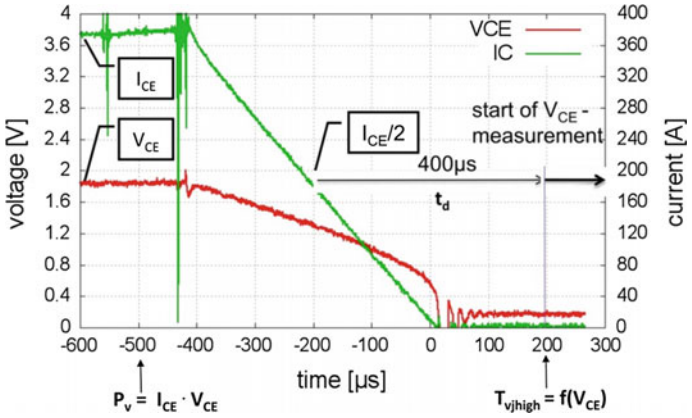


Fig. 12.11 Details of measurements at turn-off of the load current in Fig. 12.10

$$R_{thjc} = \frac{\Delta T}{P_V} \tag{12.4}$$

Collecting more measurement points after t_d, the thermal impedance can be calculated from the cool-down curve.

The time interval t_d must be set because of possible oscillations at turn-off, and also because of internal recombination processes if bipolar devices are used. The cooling-down in this interval for Si devices lies in the range of 2 K for usual power density, and up to 4 K for modules with high power density and advanced cooling systems. Usually, this cooling is neglected. A correction can be made with a simulation model, however, it should be mentioned in test evaluations, and the t_d should be given if a detailed evaluation is done.

The German automotive standard [LV324] is fixing much more details on the measurement process than the international JEDEC standard. It requires:

- the virtual junction temperature T_{vj} of the device under test must be determined with the $V_{CE}(T)$ method according to [Scn09].
- the supervision of the failure criteria has to be done with the parameters voltage drop (IGBT: V_{CE}, MOSFET V_{DS}, Diode V_F) and the temperature swing of T_{vj}. Both parameters must be monitored during the whole test for each cycle and have to be documented accordingly.
- the EOL criteria are to be checked by continuous supervision. Care has to be taken that the measurement data are with sufficient granularity according to the expected lifetime to ensure a valuable and exact determination of EOL.

The failure criteria are

- an increase of $V_{CE}/V_{DS}/V_F$ by 5%.
- an increase of R_{th} by 20% resp.ΔT_j by 20%.

- failure of one of the functions of the device, e.g. failure of the blocking capability or of the gate to emitter (gate to source) insulation capability for IGBTs and MOSFETs or by gate bond lift-off.

After the desired test conditions are adjusted in an initial setup phase, the heating time t_{on} and the cooling time t_{off} are used as constant control parameters for the total test duration. This is the preferred control strategy, however, different control strategies are still applied today with a considerable impact on the test results, as will be discussed in more detail at the end of this section.

The different coefficients of thermal expansion of materials together with the vertical and lateral temperature gradients generated in this layer system create mechanical stress at the interfaces during the temperature swing. This thermal stress leads in the long run to fatigue of materials and interconnections. Figure 12.12 shows the result of a power cycling test with a standard module. During the test, the forward voltage V_C of an IGBT is monitored. Additionally, it is possible to feed a defined sense current of some mA through the device after the turn-off of the load current, which allows to determine the upper temperature T_{vjhigh} with use of a calibration function (compare Fig. 11.19). The power losses P_V are also measured online. From junction temperature T_{vjhigh}, upper heat sink temperature T_h and P_V the thermal resistance is calculated using Eq. (11.4). Since the measurement of the fast changing heat sink temperature is difficult due to the response time of the applied sensor, the thus measured thermal resistance can deviate from the true stationery value, especially for short cycles below 10 s. However, even then this relative value can be used to monitor relative changes in R_{thjh}.

In Fig. 12.12 one can recognize that the on-state voltage drop at the IGBT remains almost constant over a large number of cycles, and the thermal resistance begins to increase slowly after approx. 6000 cycles. This is an indication for an increase of thermal resistance in the thermal path, mostly attributed to solder

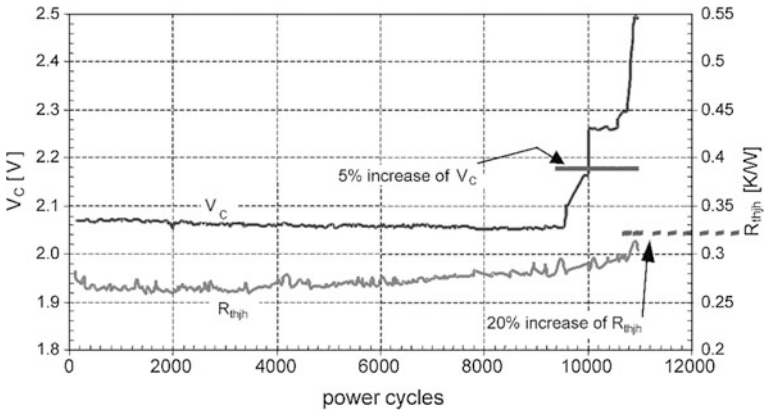


Fig. 12.12 Behavior of on-state voltage drop V_C and thermal resistance R_{thjh} at a power cycling test with ΔT_j = 123 K

fatigue. After more than 9000 cycles a first step in the V_C characteristic is observed, which is an indication for wire bond degradation. Shortly after that, the further steps are observed until finally all bond wires failed, the power circuit is open and the test can no longer be continued.

Bond wire degradation and solder fatigue are the main failure mechanisms in standard power modules. But from the shape Fig. 12.12 it is difficult to determine the primary failure mechanism. The failure limit of R_{thjh} would be reached at after approx. 11,000 cycles. But increase of R_{thjh} results in increasing temperature T_{high} and this will escalate the thermal stress for the bond wires. Therefore, solder fatigue is a significant failure mechanism in this test; it could be even the main failure mechanism. On the other hand, bond wire lift-off leads to increased V_C, which together with the constant current causes increasing losses and raises the upper junction temperature T_{high}, resulting in more thermal stress in solder layers. Due to the interdependency of the failure modes, power cycling tests require a careful failure analysis.

As mentioned before, different control strategies can be applied in power cycling tests. To evaluate the impact of different control strategies on the test result, a special test setup was designed which allowed the adjustment of test conditions during the power cycling test [Scr10]. In this special test setup, a single IGBT chip was subjected to power cycling with identical start conditions, but with 4 different control strategies (see Fig. 12.13):

1. Constant timing (t_{on} and t_{off} both constant): This strategy means that the operating conditions are kept constant throughout the test without reaction to an increase in thermal resistance or dissipated losses. This strategy provides no compensation for degradation effects and yields the lowest number of cycles to failure.

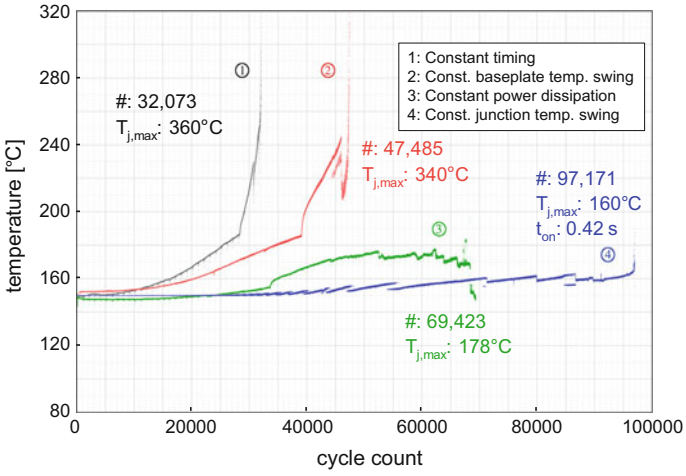


Fig. 12.13 Evolution of the maximum chip temperature during power cycling tests with 4 different control strategies from [Scr10]

2. Constant base plate temperature swing: In this strategy the signal of a thermocouple pressed to the baseplate underneath the center of the chip was used to define switching levels for turn-on and turn-off of the load current. This strategy was applied in the early days of power cycling testing, when several PC test setups shared the same cooling loop. If for example the cooling liquid temperature would increase due to losses generated in other test setups, the load pulse duration would be decreased and the cooling phase would be increased to maintain the same temperature swing. On the other hand, degradation in the thermal path between the baseplate and the heat sink is also compensated by this method resulting in an increased numbers of cycles to failure as shown in Fig. 12.13. Since today each PC test setup is equipped with an individual cooling system, this strategy has become obsolete.
3. Constant power dissipation: The aim of this strategy is to keep the power losses constant throughout the test. This can be achieved by controlling the gate voltage. The test is started with a reduced gate voltage and when losses increase due to degradation during the test, this is compensated by increasing the gate voltage. This compensation of degradation increased the number of cycles to failure for the investigated test vehicle by a factor of 2.
4. Constant junction temperature swing: This strategy aims for a constant temperature swing throughout the test. In [Scr10] this was achieved by reducing the load pulse duration when the temperature increased during the power cycling test. At the end of the test, the initial load pulse duration of 2 s was reduced to 0.42 s and the lifetime was a factor 3 higher than without compensation of degradation.

It should be mentioned that the temperature displayed in Fig. 12.13 is the maximum chip temperature measured in the chip center with a pyrometer in contrast to all other power cycling results showing T_{vjhigh}. This change was necessary to allow an active control of the load pulse duration based on the junction temperature value.

This example shows the significant impact of the control strategy on the power cycling lifetime. This must be kept in mind when comparing results obtained with different test strategies.

12.7.2 Power Cycling Induced Failure Mechanisms

12.7.2.1 Bond Wire Degradation

A typical fault image after power cycling of a standard module with base plate is shown in Fig. 12.14. All bond wires are lifted off from the IGBT chip. The bond wire in the background was the last one to fail and the current was flowing in a short time via an arc flash-over which caused a crater below the bond wire stitch. The bond wire failure in the foreground shows a characteristic feature of the lift-off

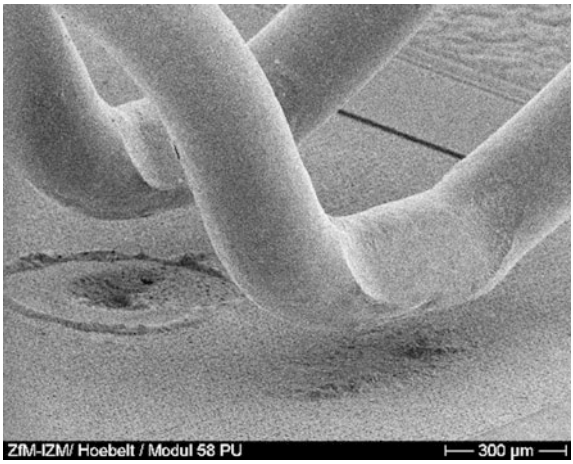


Fig. 12.14 Lifted bond wires after power cycling test with ΔT_j = 100 K of a standard IGBT module. The total failure occurred between 10,791 and 13,000 cycles

failure mode. The dissection did not occur at the interface between bond wire and chip metallization, but it emerges partially in the volume of the bond wire. Residues of bond wire material can still be detected on the surface of the chip metallization.

Figure 12.15 shows a magnified image of the lift-off area on the chip metallization. This example shows no adherence in the center of bond area. Improvements of the wire bond process can enhance the quality of the adherence significantly. In [Amr06] it was shown, that an improved wire bond process in combination with a replacement of the chip solder by a silver sinter technology can achieve a very high power cycling capability for power cycles up to T_{high} = 200 °C. In the viewpoint of high temperature applications, bond wires seem not to be the main limiting factor.

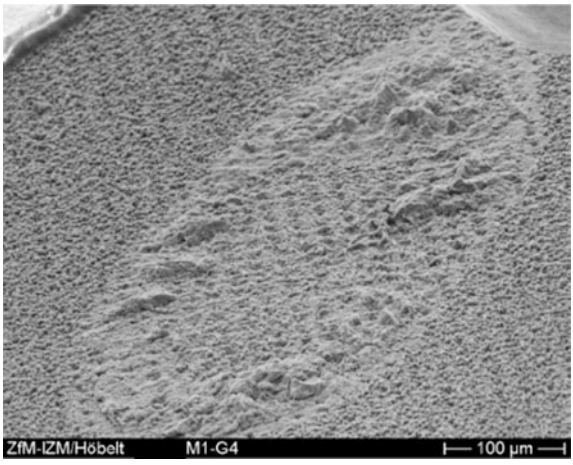


Fig. 12.15 Lift-off pattern as a result of power cycling

Special attention has to be paid to gate wire bonds on chips with a center gate contact. For power devices with a field effect gate structure, the leakage current of the gate is so small, that is takes days to discharge the gate via the leakage current. Therefore, a wire bond lift-off of the gate wire bond will not be noticed if the gate is continuously switched on and the load current is controlled by an external switch. In this case, a special functional test must be performed during the cooling phase of each cycle to verify the gate functionality. This can be done by switching of the gate voltage during the cooling phase. The constant current source, that supplies the sense current during this phase, must then go into voltage limitation mode. This technique ensures that a gate bond wire lift-off will not remain undetected.

It was found that bond wires have a higher lifetime if they are coated with a polyimide cover layer [Cia01]. Another proposal to increase the lifetime of Al wire bonds was the improvement of the bond loop geometry. Investigations on Al wire bonds under cyclic mechanical stress had shown that the lifetime increases with increasing loop height [Ram00]. Thus, increasing the aspect ratio, i.e., the ratio of loop height divided by the distance between the bond stitches, should increase the mechanical stability and therefore the lifetime in power cycling tests. However, first power cycling tests with increased aspect ratio of the Al wire bonds performed soon after publication of the mechanical tests did not confirm this expectation (Fig. 12.16).

It took another ten years until this discrepancy could be explained. Active power cycling tests were performed on baseplate power modules with Al wire bonds with different aspect ratios [Scn11]. Two groups of modules were produced for each aspect ratio, one group with soldered chips and one group with Ag diffusion sintered chips. A comparison of the test results for ΔT_{vj} = 70 K (T_{vjhigh} = 150 °C, T_{vjlow} = 80 °C, t_{on} = 1.2 s) displayed in Fig. 12.16 shows, that the chip solder fatigue limits the increase of lifetime in power cycling for higher aspect ratios. If

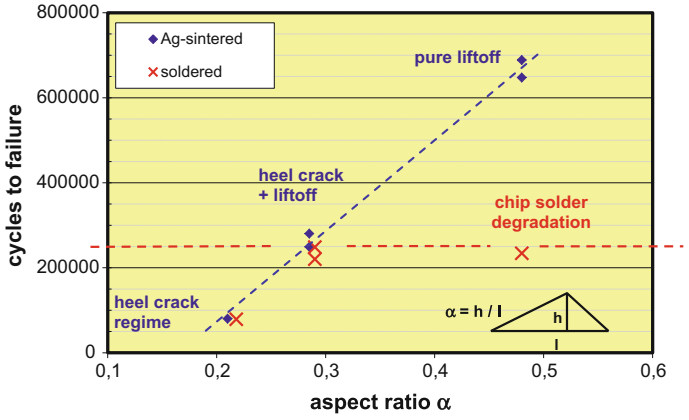


Fig. 12.16 Power cycling lifetime as a function of the Al wire bond aspect ratio at ΔT_j = 70 K for soldered chips and Ag diffusion sintered chips [Scn11]

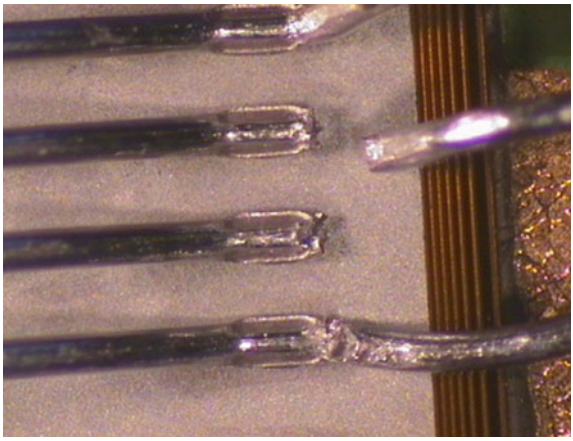


Fig. 12.17 Bond wire heel cracks after active power cycling [Scn11]

solder fatigue is eliminated by implementing Ag diffusion sintering for the chip-to-substrate interconnection, considerable gain in lifetime can be achieved with higher bond loops.

The test results revealed also that the failure mode of the Al wire bond is affected by the aspect ratio; while wire bond lift-off is the dominating failure mode for high aspect ratios, heel cracks become the predominant failure mode for low aspect ratios. As illustrated by Fig. 12.17, these heel cracks occur at the first stitch on the chip of the wire bond connection between chip and substrate.

The replacement of Al by Cu as wire bond material as described in Sect. 11.6.3 further enhances the power cycling lifetime of the topside chip interconnection. It has the capability to eliminate the failure mode of wire bond degradation in power modules completely and reveals a fatigue of the copper metallization of the DBC substrates as a new failure mode [Heu14].

12.7.2.2 Reconstruction of metallization

A phenomenon observed during active power cycles with a high temperature swing is the reconstruction of the topside chip metallization. This contact metallization is conventionally made of a vacuum-metallized aluminum layer, which is formed in a grain structure. Due to the difference in thermal expansion between silicon and aluminum, this layer suffers from a considerable stress during repeated temperature swings. While the silicon chip is only marginally expanding with increased temperature (2–4 ppm/K), the grains of the Al metallization expand considerably (23.5 ppm/K). Thus, the metallization layer is subjected to a compressive stress during the heating phase of temperature cycles.

The surface reconstruction of aluminum films on silicon was first reported in the late 1960s [Pad68] followed by detailed investigations of this degradation effect. Comparison between temperature cycled samples with annealed (uncycled) samples

for equivalent time-at-temperature revealed, that the surface reconstruction is increased by thermal cycling by a factor 2–5 depending on temperature and grain size of the aluminum film [San69]. Analysis of reconstruction phenomena at high temperatures (above 175 °C) and low temperatures (below 175 °C) suggest different fatigue mechanisms for these temperature ranges. Diffusional creep and plastic deformation involving conservative motion of dislocations are assumed as dominant contributions for high temperatures, while for low temperatures the only possible mechanism of mass transport is plastic deformation caused by compressional fatigue [Phi71].

While the former investigations were triggered by surface reconstruction observed in integrated circuits, the same effect was found in power electronic devices [Cia96]. The periodical compressive stress during the heating phase in temperature cycles results in plastic deformation of grains when the elastic limits are exceeded. According to [Cia01] this is the case for junction temperature above 110 °C. The plastic deformation can lead to the extrusion of single grains. This process leads to an increasing surface roughness of the metallization with the macroscopic observable effect of a dull non-reflective surface appearance. In the cooling phase of the temperature cycle, tensile stress can lead to cavitation effects at the grain boundaries if the elastic regime is exceeded. This can explain the observed increase in electrical resistance of the surface metallization [Lut08].

Reconstruction of the aluminum contact layer was also found in failed devices after repetitive short circuit operation of IGBTs [Ara08]. Recently, a similar effect of increasing surface roughness was observed also in thick aluminum layers of 'Direct Bonded Aluminum' (DBA) substrates after temperature cycling between −55 and 250 °C. After 300 cycles the surface roughness of 300 µm thick aluminum layers on AlN substrates increased by more than a factor of 10, while a multitude of voids where observed in a cross section. The authors attribute this effect to grain boundary sliding [Lei09].

Figure 12.18 shows the optical image of such a metallization reconstruction after the power cycling test. The reconstruction appears as a milky white discoloration of the diode metallization. It is concentrated at the center of the chip. Especially interesting is the area around the solder void, which can be seen in the X-ray image. The reconstruction is also very pronounced in this region, corroborating that the reconstruction is generated by the temperature swing. The maximum temperature has its peak in the center of the die, while the thermal resistance is locally increased in the area of the solder void. The dissymmetry of the reconstruction clearly follows this temperature profile [Scn99].

The impact of the maximum temperature during power cycling is illustrated in Fig. 12.19. Figure 12.19a shows an IGBT metallization after 3.2 millions of power cycles between 85 and 125 °C, the figure is taken from [Cia02]. Figure 12.19b shows an IGBT metallization after 7250 power cycles with $\Delta T = 131$ K and $T_{high} = 171$ °C. Figure 12.19c finally shows the metallization of a diode after 16,800 cycles at $\Delta T = 160$ K with $T_{high} = 200$ °C, where significant grains of approx. 5 µm diameter appear on the surface leaving voids in the metallization layer.

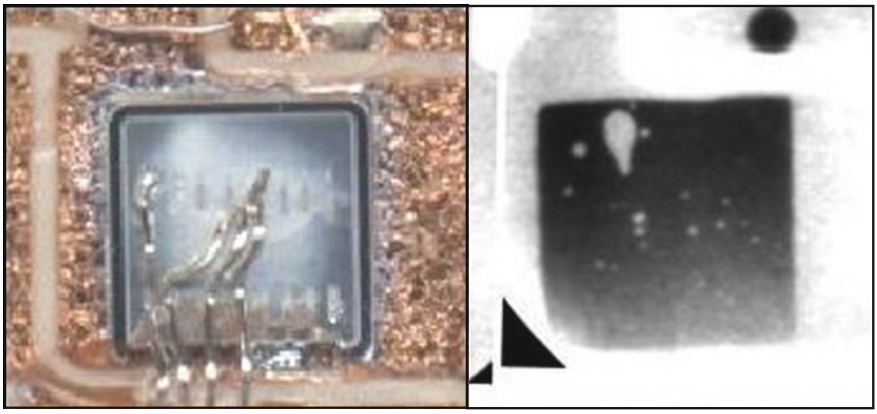


Fig. 12.18 Optical image (left) and X-ray image (right) of a diode after active power cycling – the reconstruction of the metallization appears as a milky non-reflecting discoloration, which is most pronounced in the chip center and in the area of the solder void [Scn99]

The reconstruction effect is suppressed beneath wire bond stitches. Figure 12.20 shows a detail of the metallization at the edge of a bond stitch after wire bond lift-off. This diode survived 44,500 cycles with ΔT_j = 130 K, T_{high} = 170 °C. The high number of cycles was achieved because of single side silver sinter technology [Amr06]. Other investigations have shown that a polyimide cover layer suppresses the reconstruction effect as well [Ham01]. This is an expected phenomenon, because any cover layer will restrict the movement of the grains out of the contact layer. Nonetheless remains the high stress in the layer and it can be expected, that the initiation and the growth of fractures in the interface of bond stitches is driven by the same CTE mismatch that generates the metallization reconstruction.

The reconstruction of the chip metallization reduces the density of the contact layer and therefore increases the specific resistivity of the contact. Since the layer thickness is typically in range of 3–4 μm, the movement of grains sized in the range of the layer thickness as in Fig. 12.19c can be expected to change the conductivity of the layer considerably.

The layer resistivity can be measured according to the method of van der Pauw [Pfu76]. For the example of the power cycled diode in Fig. 12.19c, a specific resistance of 0.0456 mΩ m was measured. The resistivity of an unstressed diode of the same type was found to be 0.0321 mΩ m. This is close to the literature value for pure bulk Al of 0.0266 mΩ m, a slightly higher value can be expected because the chip metallization has a grain structure and contains some small admixture of Si. The observed resistivity increase of 42% during power cycling [Lut08] is a severe change of the device characteristics. Even though only a very small increase of V_F was detected during the power cycling tests of the devices in Figs. 12.19c and 12.20, an impact on the current distribution in the device must be expected which will reduce the lifetime of the power device under high stress conditions.

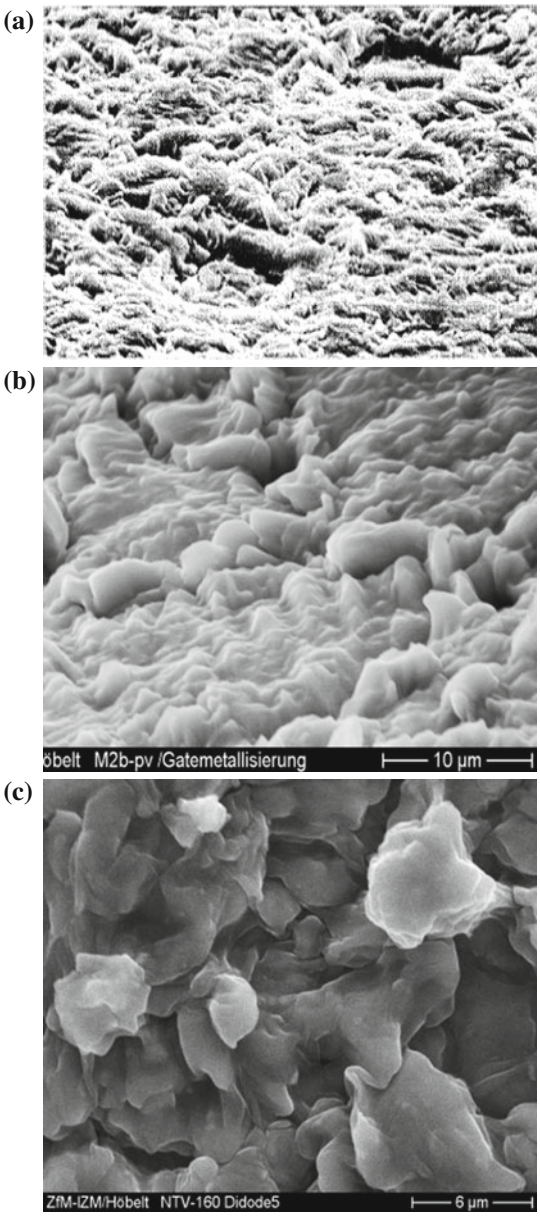


Fig. 12.19 REM images show the augmentation of contact reconstruction with increasing maximum cycle temperature T_{high} during power cycling

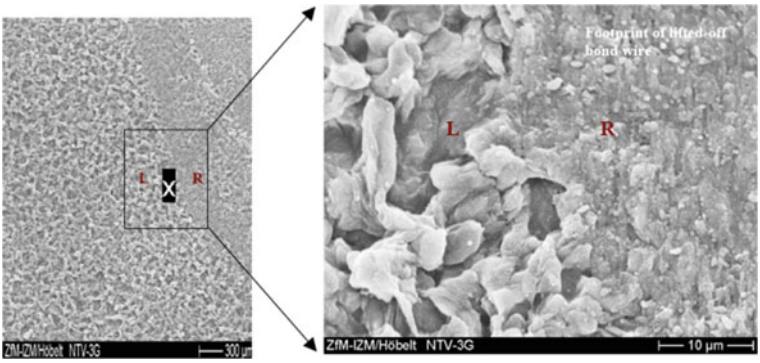


Fig. 12.20 REM image of the edge of bond stitch area after a wire bond lifted-off – the reconstruction is more pronounced outside of the stitch area. The diode failed after approx. 44,500 cycles with ΔT_j = 130 K and T_{high} = 170 °C

Latest investigations on device failure under repetitive short circuit exposure below the critical thermal destruction level have shown that the increase in contact resistance caused by reconstruction of the contact layer is the root cause of the failure [Ara08]. It must be expected, that this reconstruction also has an impact on the surge current capability of freewheeling diodes.

12.7.2.3 Solder Fatigue

The degradation of the solder interface is a fundamental failure mode during active power cycling. The so-called solder fatigue is caused by the formation of fractures in the solder interface, which leads to an increase in thermal resistance and is thus accelerating the total failure of the device. In devices with a positive temperature coefficient of the forward voltage drop (i.e. IGBT and MOSFET), the increasing temperature leads to increasing power losses and therefore expedites the degradation in a positive feedback loop. However, the final end-of-life failure of the conventional module design will typically be the breakdown of the wire bond contacts. If no investigation of the solder interface is performed by Scanning Acoustic Microscopy (SAM) after a power cycling test, the impact of solder fatigue cannot be evaluated and the wire bond breakdown is often mistakenly assumed to be the root cause of failure.

Figure 12.21 shows the evolution of the forward voltage drop and the maximum temperature reached at the end of each heating phase of the IGBTs in the center phase leg of a sixpack module without base plate. The maximum temperature starts to increase after 65,000 cycles and generates increasing power losses in the device due to the positive temperature coefficient. After 85,000 cycles, the temperature swing has already increased to ΔT_j = 125 K and it would continue to grow fast until the liquidus temperature of the solder is reached, if the wire bond contacts would not have failed first. This example illustrates, that only marginal lifetime can

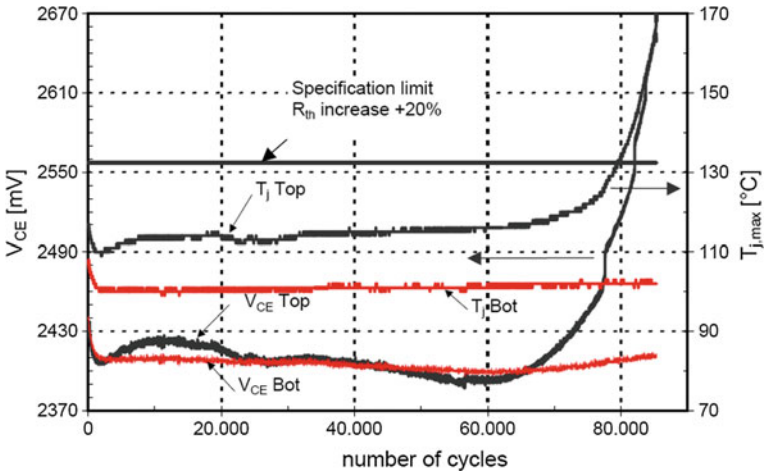


Fig. 12.21 Power cycling results of an IGBT phase leg in a module without base plate with ΔT_j = 80 K [Scn02b]. The temperature evolution is a strong indication of solder fatigue

be gained by improving the top side chip contact (i.e. the wire bonds) without improvement of the bottom side contact (i.e. the solder layer), because the increasing temperature amplitude would destroy any top side chip contact.

The geometry of the solder fatigue depicted in the SAM image in Fig. 12.22 follows the expected deterioration pattern. The discontinuity at the chip edges is responsible for a stress peak at the edges and especially at the chip corners. Therefore, the fractures start at the outside corners and edges and propagate towards the chip center. In this case, the arrangement of the four chips in parallel generates a temperature distribution with the maximum temperature being located at the chip corners close to the center of this quadruple chip assembly. Thus the degradation of the solder layers starts at the corners pointing toward the center of the group and moves outward.

The continuous improvement of the package thermal resistance and the enhanced heat extraction capability of advanced cooling systems have considerably increased the power density in modern IGBT chips. Assisted by the trend to minimized chip thicknesses (down to 70 µm for latest generation 600 V trench chips), which reduces the lateral thermal conductivity and thus diminishes possible heat spreading effects, the lateral temperature gradient in the chip has become more and more pronounced. Figure 12.23 illustrates this effect, where the diagonal temperature gradient exceeds 40 °C for a 12.5 × 12.5 mm^2 IGBT on a water cooled copper heat sink with 9 °C cooling liquid temperature [Scn09].

For such pronounced lateral temperature gradients, the stress in the chip center generated by the high temperature exceeds the stress induced by the edge discontinuity and the degradation begins in the center instead of the edges and corners. This is confirmed by the SAM image of a state-of-the-art power module after the power cycling test in Fig. 12.24. Here the damaged region is located in the chip

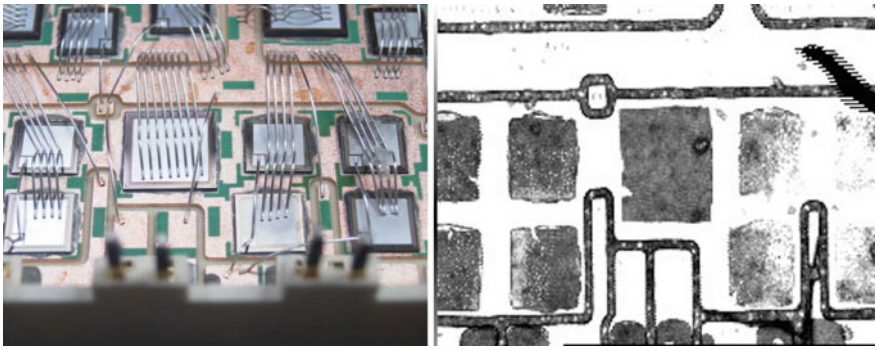


Fig. 12.22 Optical image and SAM image of the IGBT module in power cycling test shown in Fig. 12.21. The four IGBTs in parallel of the failed TOP switch (right side of the images) show the effect of solder fatigue

center, indicated by the light areas of high reflection, while the chip edges seem unaffected.

This observation was also reported from other authors. While some publications attribute this phenomenon to special solder types used [Moz01], other authors have observed this effect with lead free and lead-rich solder systems [Her07]. The characteristics of the solder interface layer might have a small impact on the solder fatigue progress; however the driving force for the deterioration of the solder interface results from the high stress induced by the differences in thermal expansion.

This change in the failure mechanism could have a considerable impact on the evolution of the degradation process. Fractures in solder interface increase the local thermal resistance of the affected chip region and thus raise locally the chip temperature. If the fractures start at the edges, the temperature of relatively cool chip regions is increasing, while the maximum temperature remains unchanged in the chip center. The situation is different, when the fractures start at the center, which has the highest temperature to begin with. Therefore, with fractures in the center of the chip, the maximum chip temperature is immediately increased and it can be expected, that this positive feedback loop will accelerate the fatigue progress and thus reduce the power module lifetime.

The considerable temperature gradients on the chip level also generate temperature gradients in the solder layer between the substrate and the base plate. Test results from a power cycling test at ΔT_j = 67 K and T_{high} = 150 °C on a module with AlSiC base plate show degradation effects in the substrate to base plate interface, that start below the chip position as shown in Fig. 12.25.

These results illustrate the complexity of solder fatigue effects due to the interaction of thermal and mechanical characteristics of the interconnection interface. The problem of solder fatigue must be solved to exceed the limits in reliability of the classical module design.

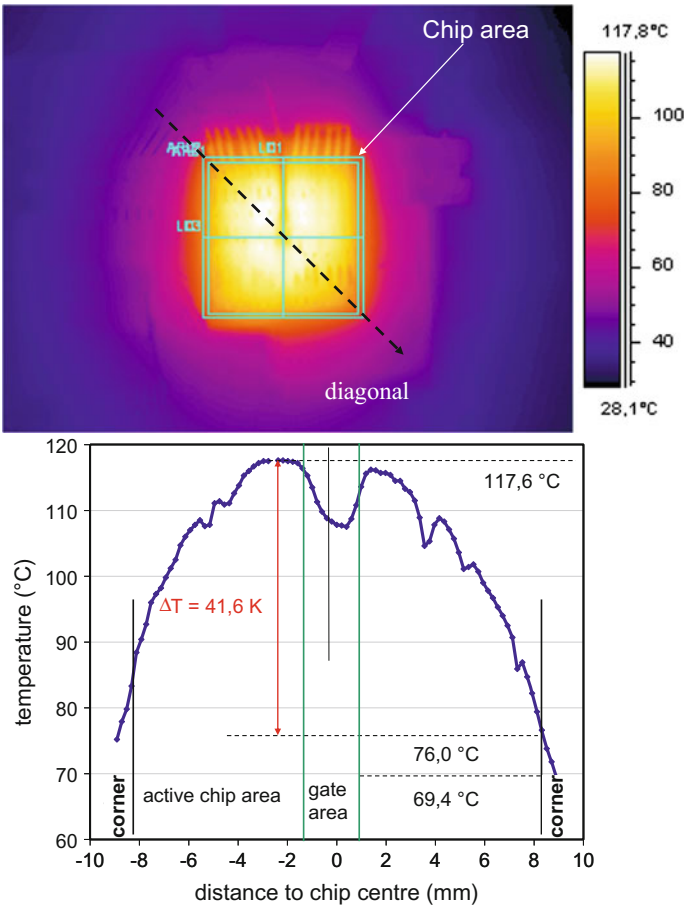


Fig. 12.23 Infrared image of a 12.5 × 12.5 mm^2 1200 V IGBT chip heated by a continuous DC current of 150 A. The module was mounted on a Cu water cooler with 9 °C cooling liquid temperature. The equilibrium temperature profile along a diagonal line through the chip center (bottom diagram) shows a temperature gradient between center and corner of more than 40 °C

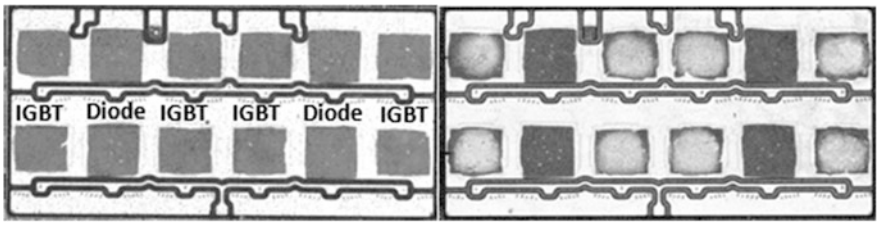


Fig. 12.24 SAM images of the initial solder layer (left) in comparison to the end-of-life image (right) after > 1 10^6(more than 1 million) power cycles of the IGBT with T_{high} = 150 °C, ΔT_j = 70 K, heating phase 0.2 s – the unstressed diode solder remains unchanged

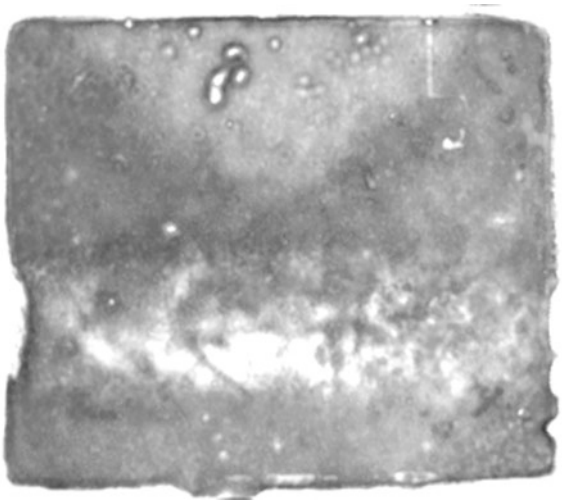


Fig. 12.25 SAM image of solder degradation below active power cycled IGBTs in a lead-free substrate solder joint of an AlSiC base plate module

12.7.3 Models for Lifetime Prediction

Since a standard technology is established for the construction of power modules with base plate and technologies and materials are very similar even for different suppliers, a research program for determination of lifetime for standard power modules was implemented in the early 90s. In this project named LESIT, modules from different suppliers from Europe and Japan have been tested; a common feature was the standard package according to Fig. 11.13 with use of an Al_2O_3 ceramics according to Table 11.1, left row. Tests were executed at different ΔT_j and different medium temperatures T_m, the results have been summarized in [Hel97] and are shown in Fig. 12.26 in form of a characteristics of cycles to failure N_f depending on ΔT_j and for different medium temperatures T_m.

The lines in Fig. 12.26 represent a fit from [Scn02b] to the experimental data: The expected number of cycles to failure N_f at a given temperature swing ΔT_j and an average temperature T_m, the absolute medium temperature in K, can be approximated with the equation

$$N_f = A \cdot \Delta T_j^{\alpha} \cdot \exp\left(\frac{E_a}{k_B \cdot T_m}\right) \tag{12.5}$$

with the Boltzmann-constant $k_B = 1.380 \times 10^{-23}$ J K^{-1}, activation energy $E_a = 9.89 \times 10^{-20}$ J and the parameters $A = 302{,}500$ and $\alpha = -5.039$.

Equation (12.5) consists of a Coffin-Manson law, i.e. the number of cycles to failure (N_f) is assumed to be proportional to $\Delta T_j^{-\alpha}$ [Hel97]. It appears as a straight line when plotting $\log(N_f)$ over $\log(\Delta T_j)$. In addition, an Arrhenius factor containing

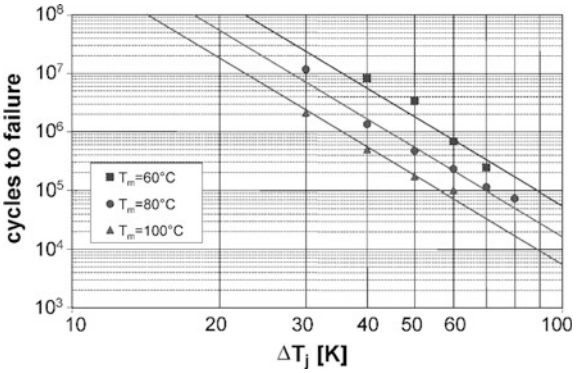


Fig. 12.26 LESIT-results

an exponential dependency on an activation energy is added to the Coffin-Manson law [Hel97].

This LESIT model was the first lifetime model which contained more parameters than the temperature swing ΔT_j. It can be considered as an empirical lifetime model, since it is based on experimental power cycling results on the one hand and it contains only parameters that are directly related to the test conditions on the other hand. The values for ΔT_j and T_m for the experimental data points in Fig. 12.26 are the initial test conditions of the associated power cycling tests, i.e., they represent the initial test condition before any impact of degradation can be observed.

Using Eq. (12.5) it is possible to calculate the number of cycles to failure for given ΔT_j and T_m according to the LESIT model. If the typical cycles in the application are known, it is possible to calculate the expectation for the lifetime of a module under these conditions.

Technologies for standard modules have been improved since 1997. Power cycling results for more recent power modules of two different suppliers are shown in Fig. 12.27. They are compared with the Eq. (12.5) for the condition T_{low} = 40 °C, extrapolated to higher temperature swings outside of the data range in Fig. 12.26. It is visible that the number of cycles to failure is increased by a factor of 3 to 5 in the range of ΔT_j > 100 K compared to the prediction of Eq. (12.5).

However, the temperature swing ΔT_j and the medium temperature T_m alone are not sufficient to uniquely characterize the test conditions of a power cycling test. The target ΔT_j is a function of the dissipated energy – which for a given chip technology is determined by the forward current and the t_{on} duration in the test – and of the thermal resistance of the test setup. Different combinations of load current, pulse duration and cooling condition can be found that generate the same temperature swing and medium temperature in a given power module. If the specific selection of test conditions have an impact on the test results they have to be included in a power cycling lifetime model.

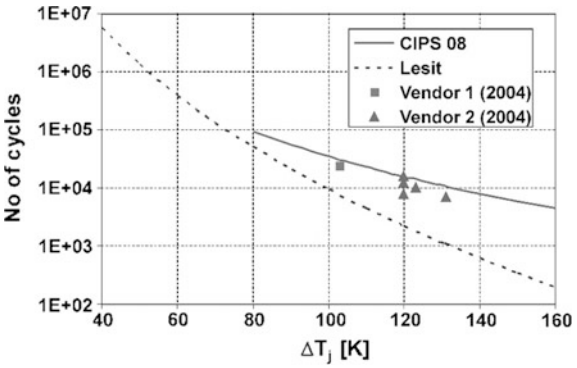


Fig. 12.27 Comparison of experimental power cycling results of state-of-the-art 2004 modules with predictions by the extrapolated LESIT model Eq. (12.5) and CIPS 08 model Eq. (12.6). T_{low} = 40 °C

This was the motivation to present an extended model for the lifetime of standard power modules [Bay08]. Based on a large number of power cycling results of modules, the following equation was derived:

$$N_f = K \cdot \Delta T_j^{\beta 1} \cdot \exp\left(\frac{\beta 2}{T_{low}}\right) \cdot t_{on}^{\beta 3} \cdot I^{\beta 4} \cdot V^{\beta 5} \cdot D^{\beta 6} \tag{12.6}$$

As parameter K we use the value 9.30×10^{14}, the other parameters ß2 … ß6 are given in Table 12.3 [Bay08]. Equation (12.6), which we denote as CIPS 08 model, contains additionally the dependence on the heat-up time t_{on} in seconds, the current per bond stitch on the chip I in A, the voltage range of the device V in V/100 (reflecting the impact of the semiconductor die thickness), and the bond wire diameter D in µm. The prediction by the new CIPS 08 model is also shown in Fig. 12.27 for t_{on} = 15 s. The CIPS 08 model holds for standard modules with Al_2O_3 substrates, it is not valid for high-power traction modules which are built with the materials AlN and AlSiC, see Table 11.1.

Equation (12.6) was a result of purely statistical analysis and is not a result of physics-based models [Bay08]. The dependency of the cycling time t_{on} in Eq. (12.6) – higher number of cycles to failure for short cycling times – may be explained by the fact that degradation mechanisms like viscoplastic deformation or fracture growth are time dependent processes, i.e., a short duration of the stress will generate less damage than a long duration. Furthermore, the temperature gradients in the module will depend on the load pulse duration. For short load pulses thermo-mechanical stress occurs mainly at the interface between semiconductor and

Table 12.3 Parameters for calculation of power cycling capability according to Eq. (12.6)

ß1	−4.416
ß2	1285
ß3	−0.463
ß4	−0.716
ß5	−0.761
ß6	−0.5

bond wire, while in layers closer to the heat sink the temperature increase is marginal and less stress is generated there. The dependency on current per bond stitch on the chip I can be attributed to improved current distribution on the chip with more bond stitches and presumably to a positive impact of the thermal capacity of the bond stitches. The dependency on the Al bond wire diameter D is related to the greater mechanical stress applied to the bond stitch by thicker bond wires. The dependency on the voltage class V (blocking voltage divided by 100) is in fact a dependency on device thickness, which increases from 600 to 1700 V. With thinner devices, the mechanical stress induced in the solder interface by the Si-material will be reduced. Note, that the used devices are produced in a thin-wafer technology for 1200 V (V = 12) and 600 V (V = 6). For devices fabricated from epitaxial wafers, e.g. PT-IGBTs or Epi-diodes, Eq. (12.6) is not applicable.

As a consequence of including additional test parameters (load pulse duration t_{on} and load current per bond stitch I) in the CIPS 08 model, the model parameters are no longer statistically independent, which was pointed out and discussed by the authors themselves [Bay08]. This can be illustrated by the fact, that it is in general not possible to adjust different values of the temperature swing ΔT_j in a series of power cycling tests while keeping the current magnitude and the load pulse duration constant. Although the variation of gate voltage allows to modify the dissipated losses for a given current in a limited range for IGBTs (see Fig. 10.3) or MOSFETs, this option is not available for power cycling tests with diodes. The influence of different heating times t_{on} is shown in Fig. 12.28. Figure 12.27 gives the impression that for a $\Delta T_j < 60$ K the new model predicts less cycles to failure N_f than the former LESIT model. But the dependency on the heating time shows that the lifetime N_f is higher for state-of-the-art 2008 modules, if a short heating time t_{on} for low ΔT_j is assumed.

Despite the fact that data for Eq. (12.6) were only generated with modules of one manufacturer, the equation seems also useful for lifetime calculation of modules of other manufacturers. If lifetime calculations are of vital importance in an application with high reliability requirements, the manufacturing company should always be consulted.

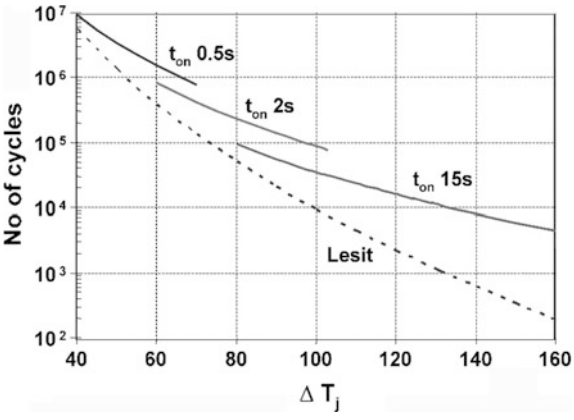


Fig. 12.28 CIPS 08 model according Eq. (12.6) for different heating times t_{on}, compared to the LESIT model. T_{low} = 40 °C

12.7.4 Separation of Failure Modes

The empirical lifetime models presented in Sect. 12.7.3 do not differentiate between the different failure modes discussed in Sect. 12.7.2. Since the materials involved in the degradation process of each failure mode are different, have distinct physical properties and feature divergent thermal behavior, each failure mechanism exhibits individual degradation driving factors resulting in different lifetime limitations. Thus, the extrapolation of the lifetime data gained by accelerated testing to the range relevant to applications may lead to wrong lifetime estimations, if the lifetime-limiting failure mechanism in the accelerated test differs from the failure mechanism dominant in the application. As a consequence, solder fatigue and wire bond degradation should be investigated independently to isolate the individual impact on empirical lifetime models. However, this was not feasible until advanced more reliable alternatives for these interconnections became available.

Today, high reliable interconnection technologies allow to isolate the degradation of different interconnections: by combining these advanced technologies with classical technologies, lifetime of classical interconnections can be studied without interaction. The SKiM63 lifetime model is based on several power cycling test results performed with the SKiM63 module from Semikron [Scn13]. This module comes without baseplate and is assembled in pressure contact technology [Scn08]. The inside connection technology is characterized by sintered chips and the topside die attachment is realized by aluminum wire bonds with an improved loop geometry. Since the load current terminals are implemented by pressure contacts and the auxiliary contacts by spring technology, the standard SKiM63 is a 100% solder-free module. Thus, the degradation of the wire bond connection to the chip determines lifetime. The SKiM63 lifetime model can therefore be regarded as an Al wire bond lifetime model. Following function describes the dependencies of the lifetime on the load conditions:

$$n_f\left(\Delta T_j, T_{jm}, ar, t_{on}\right) = A \cdot \Delta T_j^{\alpha} \cdot ar^{\beta_1 \cdot \Delta T_j + \beta_0} \cdot \left(\frac{C + t_{on}^{\gamma}}{C + 1}\right) \cdot exp\left(\frac{E_a}{k_B \cdot T_{jm}}\right) \cdot f_{diode} \tag{12.7}$$

A is a general scaling factor. The impact of the temperature swing ΔT_j in K and the medium junction temperature T_{jm} in K is represented by a Coffin-Manson relationship and an Arrhenius term with the activation energy E_a and the Boltzmann constant k_B, respectively. The wire bond aspect ratio ar, i.e. the ratio of loop height to distance between the stitches of the bond wire from DBC to the chip, was found to have an influence on lifetime as discussed in Sect. 12.7.2. The SKiM63 lifetime model also considers the impact of load pulse durations t_{on} in s. The factor f_{diode} reflects the influence of the chip thickness of the tested 1200 V diodes since the diodes were thicker and exhibited a lower lifetime than the tested 1200 V IGBTs. The derived equation parameters are presented in Table 12.4.

Table 12.4 Parameters of SKiM63 lifetime model in Eq. (12.7)

α	−4.923
β_1	-9.012×10^{-3}
β_0	1.942
C	1.434
γ	−1.208
E_a [eV]	0.06606
f_{diode}	0.6204

To investigate the lifetime of chip solder interconnections the method of separation of failure modes has been proposed [Sct12a]. Additional investigations [Jun15, Jun17, Sct13] focused on the impact of test conditions on the lifetime of the chip solder interfaces in comparison to the lifetime of the Al wire bond interconnections. For this purpose, two groups of modules were subjected to power cycling tests simultaneously; one group eliminates solder degradation by using the commercial SKiM63 module with sintered chips, the other group eliminates wire bond lift-off in a modified version of the SKiM63 module with SnAg3.5 soldered chips and high reliable Al-cladded copper wires [Sct12b]. Alternatively, other high reliable interconnection technologies like diffusion soldering for the chip interconnection to the DBC or copper wire bonds for the topside chip connection could be implemented, though the latter would require a copper metallization on the chip topside surface.

In [Sct13] load time t_{on} and temperature swing ΔT_j were adjusted identically in each test in order to investigate the impact of the medium junction temperature T_{jm} on the power cycling lifetime of the interconnections near to the chip. In [Jun15] t_{on} and either T_{jmax} or T_{jmin} were equal in all tests (T_{jmax} = 150 °C or T_{jmin} = 40 °C, t_{on} = 2 s) and solely ΔT_j and consequently T_{jm} were varied. The results (Fig. 12.29) revealed that sintered modules with standard aluminum bonds exhibit a lower lifetime at high ΔT_j than the soldered modules with Al-cladded copper bonds. Solder fatigue in contrast becomes the lifetime determining mechanism at low ΔT_j. Furthermore it was demonstrated that the mechanism of solder fatigue is much more affected by the medium junction temperature than the wire bond degradation. This is also reflected in the higher activation energy of E_A = 0.097 eV for the solder fatigue compared to E_A = 0.042 eV for the wire bond degradation. The divergent temperature dependency of both failure mechanisms can be explained by the thermo-mechanical behavior of the involved materials. During testing, the solder layers are subjected to temperatures near the melting point of the solder. Crack formation and propagation is accelerated with increasing temperature level. The melting point of aluminum in contrast is much higher. However, due to the large difference in the CTEs of the silicon of the chip and the aluminum of the wire bond, the interconnection between chip and wire bond is more sensitive to temperature swings as illustrated by the results.

In a further step the impact of load pulse duration was analyzed [Jun17]. The load pulse duration was varied in a wide range and the other test conditions were identical for all tests (T_{jmax} = 150 °C and ΔT_j = 70 or 110 K, t_{on} = 0.07 s…0.60 s). Figure 12.30a illustrates the lifetime results for load pulses $t_{on} \leq$ 2 s and lifetime

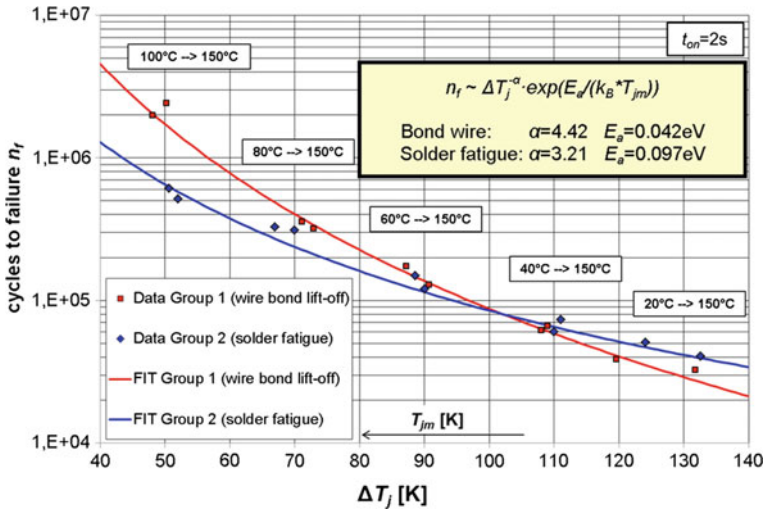


Fig. 12.29 Isolated lifetime of Al wire bonds and SnAg3.5 chip solder over ΔT_j (t_{on} = 2 s, T_{jmax} = 150 K) [Jun15]

results for $t_{on} \geq 2$ s are presented in Fig. 12.30b, along with the lifetime estimation by the SKiM63 lifetime model. The solder lifetime increases with decreasing load pulse durations, though not as pronounced as the lifetime of the wire bond interconnection. Both, the lifetime of solder and wire bond interconnection, decreases with extended load pulses but with a declining rate.

The collected power cycling data separated for Al wire bond degradation and SnAg3.5 chip solder fatigue can be considered as a data base for future improved lifetime models for classical power modules.

The experimental power cycling test results presented here were derived based on a specific architecture: the 1200 V SKiM63 module. It must be considered when applying this model to other module designs with baseplate, that potential baseplate related failure modes (e.g. fatigue of the substrate-to-baseplate solder interconnection) are not contained in these investigations.

However, the extended data base collected by the separation of failure mode investigations can be useful for scaling and testing physics-of-failure lifetime models derived from simulations based on constitutive material models.

Currently, many research groups are working on so-called physics-of-failure lifetime models. These lifetime models are implemented in a simulation procedure and are based on constitutive material models. While empirical lifetime models are mainly built on experimental results and statistics, the goal of simulation models is to reproduce the deformation of the material as an answer to the stress conditions and to describe the failure mechanism on material level in order to deduce impact on lifetime. The thermally induced mechanical stress is hereby considered as an indicator to activate and foster a degradation mechanism. As will be discussed in

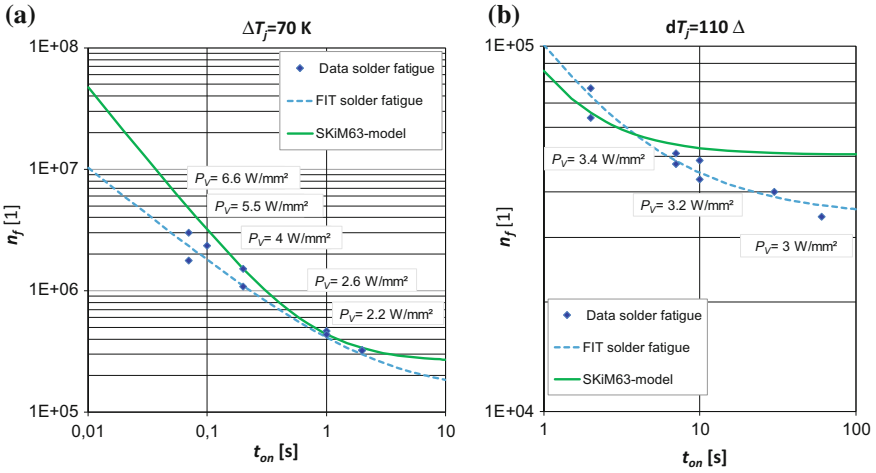


Fig. 12.30 Isolated lifetime of Al wire bonds and SnAg3.5 chip solder over t_{on} **a** ΔT_j = 70 K, T_{jmax} = 150 K, $t_{on} \leq$ 2 s and **b** ΔT_j = 110 K, T_{jmax} = 150 K, $t_{on} \geq$ 2 s)

the following examples, the simulation models need parametrization by experimental test results.

In [Kov15] a simulation model was proposed to describe lifetime of planar solder joints within power semiconductor modules. The accumulated deformation energy Δw_{hys} in the solder interconnection per cycle is calculated by a numerical approach based on Clech's algorithm [Cle97], which calculates the stress-strain hysteresis as material response to a given thermal load profile. For that, suitable solder constitutive equations describing the time-independent elastic and plastic deformation and the time-dependent viscoplastic deformation of the solder are required. With Morrow's type of fatigue law [Cle02], which defines a failure as exceeding a certain critical accumulated deformation energy W_{crit}, the lifetime for a given temperature profile in the solder layer can be estimated. Power cycling test results are required to parametrize the simulation model. The change of properties of the solder interconnection due to fatigue and the impact on the stress-strain hysteresis is not considered in the modeling process.

By contrast, crack propagation in solder joints was directly determined in [Dep06]. Using the crack growth model by Paris [Par63] crack initiation and propagation in solder joints were simulated in dependence on the accumulated creep strain along defined crack propagation lines. Latter was calculated by constitutive solder equations proposed by Déplanque in [Dep07]. The crack model was finally scaled on an empirical database which was generated using SAM-measurements of the crack length generated by thermal cycling.

Crack propagation models are also used to simulate fatigue in Al wire bond interconnections. In [Yag13] damage of the bonding interface is emulated by a damage-based crack propagation model. The analysis is performed in the time domain instead of a cycle-dependent description of the damage in order to capture

time sensitive effects. Furthermore, both damage accumulation and damage removal resulting from thermally activated processes are taken into account.

The presented constitutive lifetime model approaches require the knowledge of the temperature evolution inside the investigated interconnection. This can be obtained by finite element simulation for the case of repetitive cyclic testing. However, it is not possible to determine this internal temperature evolution for a complex mission profile of a real application, which might contain millions of non-repetitive load changes. Therefore, empirical models will be indispensable to evaluate complex mission profiles. On the other hand, it is impossible to validate empirical lifetime models derived from accelerated tests for very low stresses in applications operating for 20 years or more. In this regard, constitutive lifetime models could be very helpful to consolidate extrapolations to application specific stress levels from a material point of view.

12.7.5 Mission Profiles and Superposition of Power Cycles

Additionally to the already discussed restrictions, lifetime models are derived from the repetition of identical power cycles, but in real applications various different temperature swings are superimposed.

To calculate the lifetime under realistic application conformal conditions with superimposed power cycles, a linear accumulation of damage is assumed as discussed for example in [Cia08]

$$Q_i(\Delta T_j, T_m, \ldots) = \frac{N_i(\Delta T_j, T_m, \ldots)}{N_{fi}(\Delta T_j, T_m, \ldots)} \tag{12.8}$$

The parameters are given by the applied lifetime model. If the number of cycles N_i for a specific parameter set reaches N_{fi} – the number of cycles to failure for this parameter combination calculated by the lifetime model – Q_i is equal to 1. Based on this assumption – which is often called 'Miner's Rule' [Min45] – contributions from different cycles in a complex mission profile can be summed up to obtain the accumulated damage. The inverse of the accumulated damage is the estimated repetition rate N_r for the considered mission profile (Eq. 12.9). If this repetition rate N_r is equal to 1 the lifetime is consumed by a single run of this mission profile. If it is larger than 1 the mission profile can be repeated N_r times; if it is smaller than 1 the power module will not survive a single run.

$$N_r = \frac{1}{\sum_{i=1}^{n} Q_i} = \frac{1}{\frac{N_1}{N_{f1}} + \frac{N_2}{N_{f2}} + \cdots + \frac{N_n}{N_{fn}}} \tag{12.9}$$

An investigation with two superimposed power cycles is reported in [Fer08]. A cycle with high ΔT_j of 140 °C is superimposed with short cycles with low ΔT_j.

However, the cycle with high ΔT_j was dominant for the failure under the chosen superposition. It is generally difficult to define two superimposed power pulses in such a way, that each of them contributes with exactly 50% to the accumulated damage.

A different approach was reported in [Scn02b], where first conventional power cycling tests with uniform cycles were performed at two different temperature swings and then a test with both cycling condition interleaved was conducted. The test results showed that in contradiction to the assumption of linear fatigue accumulation, the number of cycles to failure was equal to the sum of cycles for each single test condition. A possible explanation for this result is that the test conditions are initiating different failure mechanisms, which do not interact with each other.

The method applied for cycle counting in application conformal temperature evolutions known as mission profiles is of fundamental importance for the lifetime estimation. The trivial approach of simply collecting the maximum-minimum temperature swings is underestimating the thermo-mechanical stress and it is very sensitive to the resolution of the analysis. A more robust approach, which is also consistent with the strain-stress characteristic in physics-of-failure oriented models, is the widely accepted Rainflow counting method [Dow82]. It is less sensitive to resolution changes and evaluates the fundamental frequencies with a higher weight.

The importance of an adequate cycle counting method can be illustrated by a simple example. Let us suppose a reliability engineer has a problem with the lifetime of a power module in his application. For a cycle with T_{low} = 40 °C and T_{high} = 120 °C a lifetime of 5.1 × 10^4 cycles can be expected from the LESIT model in Eq. (12.5). The application requires 8 times the number of cycles, so the engineer has an idea: He adds a small temperature swing in the slopes of his large temperature swing so that he divides it in smaller swings as shown in Fig. 12.31.

Applying the trivial counting method and using Eqs. (12.5), (12.8) and (12.9), the engineer calculates for his modified cycle in Fig. 12.31b the repetition rate as displayed in Table 12.5. Since he reduced the maximum temperature swing from ΔT_j = 80 K to ΔT_j = 42 K by his small modification, he now calculates a repetition

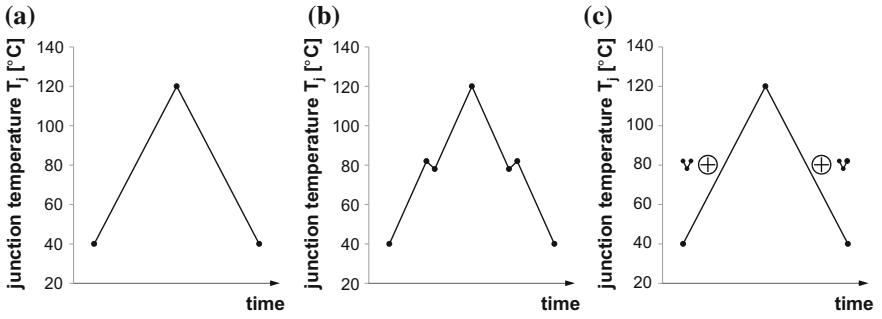


Fig. 12.31 Simple temperature cycle (**a**), modified cycle (**b**) and decomposition of modified cycle with the Rainflow algorithm (**c**) (lines are only guide to the eye)

Table 12.5 Lifetime calculation for the example of superimposed cycles in Fig. 12.31 with the LESIT lifetime model in Eq. (12.5)

Profile	T_{low} (°C)	T_{high} (°C)	N_f	Q_i	$\sum Q_i$	N_r
Simple cycle	40	120	5.1×10^4	2.0×10^{-5}	2.0×10^{-5}	5.1×10^4
Modified cycle trivial counting	40	82	4.1×10^6	2.4×10^{-7}	2.4×10^{-6}	4.2×10^5
	78	120	4.6×10^5	2.2×10^{-6}		
	78	82	1.8×10^{11}	5.5×10^{-12}		
Modified cycle rainflow counting	78	82	1.8×10^{11}	5.5×10^{-12}	2.0×10^{-5}	5.1×10^4
	40	120	5.1×10^4	2.0×10^{-5}		
	78	82	1.8×10^{11}	5.5×10^{-12}		

rate $N_r = 4.2 \times 10^5$ which would meet his requirements. However, by comparing the temperature profiles in Fig. 12.31a, b it seems not plausible that such a small modification would increase the lifetime by more than a factor 8.

This problem is resolved by applying the Rainflow algorithm to the modified profile in Fig. 12.31b. The algorithm decomposes the modified profile as shown in Fig. 12.31c resulting in a fundamental cycle with $\Delta T_j = 80$ K and two superimposed small cycles with $\Delta T_j = 4$ K. Since the damage contributed by these small cycles can be neglected, the repetition rate of the modified cycle is the same as the original simple cycle.

Based on the assumption of linear damage accumulation resulting from thermal cycles, together with the extraction of temperature cycles by a Rainflow algorithm and the lifetime model, application specific mission profiles can be evaluated as shown in Fig. 12.32. The process begins with the specific requirements of the application, e.g. the evolution of rotational speed and the torque for a motor drive or the wind distribution as a function of time for wind turbines. Then the dynamical characteristics of the motor or generator has to be taken into account, resulting in a sequence of values for the time step, the output voltage, the output current, the output frequency, the phase shift between voltage and current, the switching frequency and the DC link voltage. This sequence is the electrical mission profile for a specific application. The generated losses can be calculated with the power module data sheet for a known temperature. This requires an iteration loop, because the dynamical characteristics of the cooling system has to be taken into account for the calculation of temperatures. From the resulting temperature profile the cycles of junction temperature can be extracted by the Rainflow algorithm. The accumulated damage of the different cycles is then evaluated by the lifetime model and a repetition rate for the characteristic mission profile can be determined as an indicator for the estimated lifetime.

When the thermal characteristics of the devices in a power module and heat sink including interface are estimated by one dimensional Foster networks, mission profiles of a million time steps can be calculated within minutes on a state-of-the-art laptop.

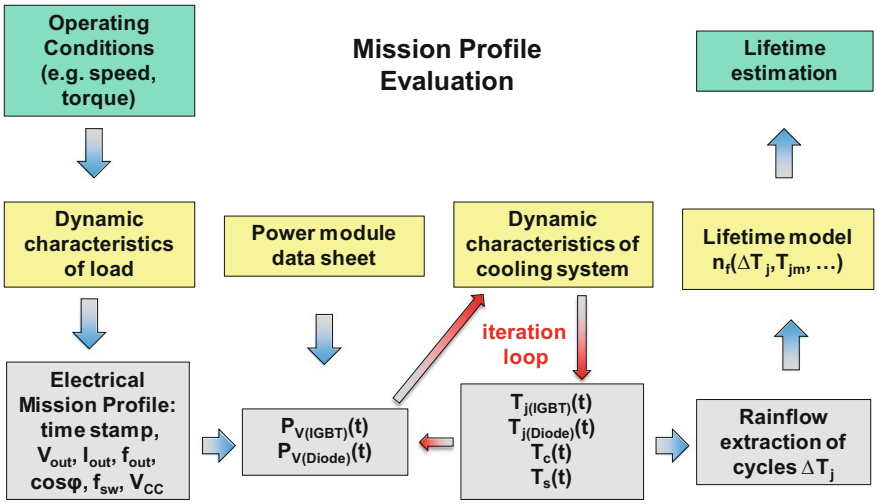


Fig. 12.32 Mission profile evaluation process [Scn15a]

However, a consistent definition of the junction temperature is required for this estimation process. This requirement can be fulfilled by applying the *Vj*(*T*) method (see Sect. 12.7.1) for the thermal impedance measurement as well as for the junction temperature measurement in power cycling tests. Furthermore, the control strategy for power cycling tests should provide no compensation for degradation. By defining the initial temperature swing ΔT_{vj} (after stable thermal conditions are reached) and initial losses as the characteristic test parameters, all degradation effects are inherently included in the lifetime model. If this is not the case, power losses and thermal impedance of the system will become a function of consumed lifetime, which makes the mission profile estimation process considerably more complex.

Additionally, the lifetime model must provide the number of cycles to failure as a function of operating parameters like virtual junction temperature, current, voltage and load pulse duration. It therefore has to be an empirical lifetime model. Constitutive models, which require the knowledge of the temperature in interconnections cannot be implemented in this mission profile estimation method. When FEM simulation is required to obtain input parameters at each time step, just the iteration loop for the temperature calculation would consume minutes to hours of calculation time so that the evaluation of extended mission profiles is not feasible. However, a serious problem of empirical lifetime models remains: they cannot be validated for very low temperature swings. Even if a test would be conducted with a predicted duration of 20 years, the technological progress would render any result obsolete. If constitutive lifetime models could be developed, which correctly predict the experimental results of accelerated cyclic tests for a wide range of test conditions, they could supply supporting evidence for the extrapolation of empirical lifetime models.

When the boundary conditions are fulfilled, lifetime estimation based on mission profiles is a powerful method to validate that a system design meets the lifetime requirements. It should be noted, that this estimated lifetime only considers the impact of application specific thermo-mechanical stress. Other lifetime relevant factors like high humidity, corrosive environments and cosmic ray failures can lead to a much lower lifetime. Therefore, this estimated lifetime defines the upper limit of the lifetime of power modules in a specific application.

12.7.6 Power Cycling Capability of Molded TO Packages

A high power cycling capability was found for DBC-based transfer molded TO-housings, which were described in the context of Fig. 11.6 [Amr04]. The power cycling results in Fig. 12.33 were gained in power cycling test with this package type for ΔT = 105 °C and T_m = 92.5 °C. The number of cycles to failure (Weibull 50% accumulated probability) is about a factor of 10 higher than predicted by the LESIT model (Eq. 12.5) and still significantly above the CIPS 08 model (Eq. 12.6). The statistical methods will be discussed in detail in Sect. 12.9.

Figure 12.34 shows a picture of bond wires in a device which survived 75,000 power cycles under said conditions. On the dark area in the foreground, a bond foot was initially attached. In the first of the still attached bond wires, a heel crack is visible.

The stiff mold-material has a similar effect as a bond wire coating; it additionally hinders mechanically the detachment of bond wires from the chip surface. Even though the bond wire shows signs of heavy deterioration, the electrical contact is still maintained. In contrast to standard TO packages with copper lead frames, the implementation of Al_2O_3 substrates accounts for a lower thermal mismatch between the semiconductor material Si and the assembly layer.

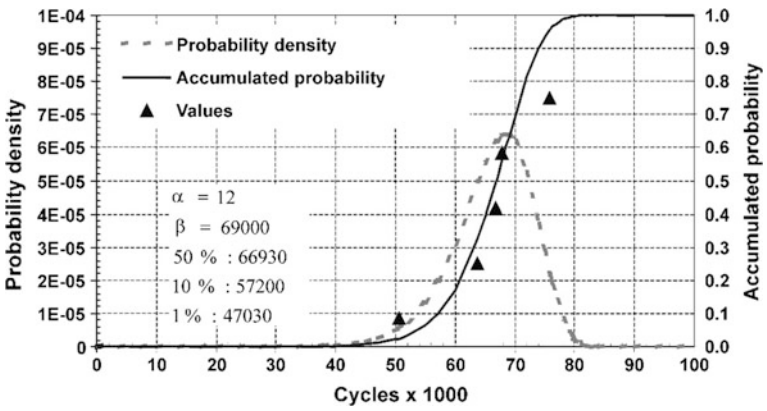


Fig. 12.33 Weibull analysis of a set of power cycling tests on a series TO package

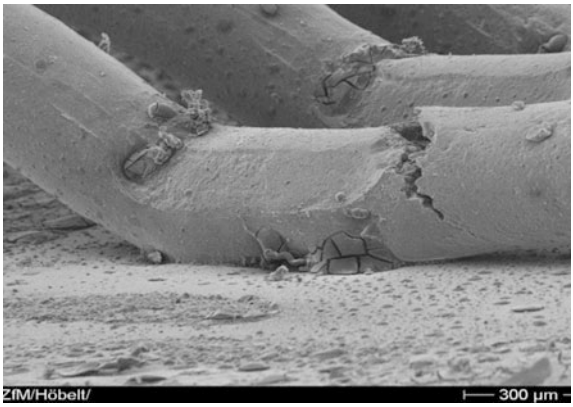


Fig. 12.34 Failure analysis of a DCB-based soft-molded TO package after 75,000 power cycles with ΔT = 105 K, showing a footprint of a detached bond wire in the foreground (left). and a heel crack in a still attached bond wire

The classical TO package with Cu lead frames (Fig. 11.5) exhibits a great mismatch in thermal expansion between copper and silicon. For large chip sizes, the power cycling capability was found to be clearly inferior to packages with ceramic substrates. In a power cycling test of standard TOs with chips of 63 mm^2 area, two of six chips lost their blocking capability after only 3800 cycles with ΔT_j = 110 °C and T_m = 95 °C. The failure analysis of these TOs revealed cracks in the silicon device as the root cause. Figure 12.35 shows such a fracture of a silicon chip.

Even though, no more failures occurred up to more than 38,000 cycles when the test was continued with the four remaining samples, the early failures cause by fractures in the silicon device are alarming. Presumably, the relatively large area of the silicon devices is responsible for these early failures, because this effect was not observed for power cycling tests of smaller chips (< 30 mm^2) in the same package type.

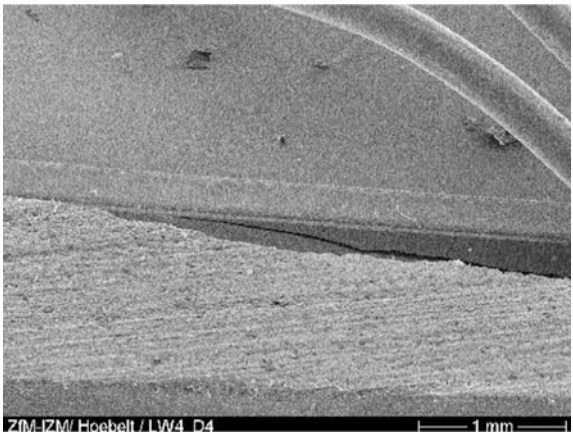


Fig. 12.35 Crack in the silicon diode chip in a transfer mold TO package on a Cu leadframe after 3800 cycles with ΔT_j = 110 K and T_m = 95 °C

12.7.7 Power Cycling of SiC Devices

When we compare the material parameters of Si and SiC, a significant difference in the mechanical characteristics emerges. The Young's modulus, which represents the stiffness of the material under mechanical stress is about three times higher in SiC (501 GPa in SiC compared to 162 GPa in Si). The temperature swings and thermal mismatch induce mechanical stress into the package which leads to fatigue. For evaluation of the expected effect in a solder layer, the plastic strain energy density ΔW according to the method of Darveaux [Dar02] is often used. An investigation of Si and SiC chips in power modules under thermo-mechanical stress was performed by FEM simulation [Pol10]. The results give evidence that crack propagation in a solder layer with SiC chips will happen about 3.5 times faster than with Si chips, and – if we use an inverse proportionality as first order approximation – the power cycling lifetime will be shortened by this factor.

For execution of power cycling tests with Schottky diodes, the junction voltage can be used as temperature sensitive parameter (TSEP) for junction temperature determination, since the current voltage characteristic Eq. (6.3) discussed in Chap. 6 has close similarities to that of a pn-junction. Thus, the same test procedure as described in Sect. 12.7.1 can be used.

Figure 12.36 shows a comparison of Si IGBTs and SiC Schottky diodes at similar power cycling conditions using the same power module package without baseplate [Hed14]. Figure 12.36a compares the course of thermal resistance for one selected IGBT and one Schottky diode. The SiC device reaches 1/3 of cycles to failure compared to the Si device. Figure 12.36b shows a metallographic preparation of a solder layer in a failed SiC device. A crack starting at the edge of the solder is propagating towards the center, this is the root cause for the result in Fig. 12.36a. Figure 12.36c displays finally a summary of all tests, which again indicates that SiC devices reach approx. 1/3 No. of cycles to failure compared to Si devices in the used standard packaging technology. This confirms the simulation results of [Pol10] and shows: To reach the same power cycling capability, SiC devices need a packaging technology with more effort. For SiC MOSFETs the definition of a suitable TSEP is difficult because some parameters show a drift which makes them unsuited as a reliable temperature indicator [Ibr16]. The on-state resistance $R_{DS,on}(T)$, which is strongly temperature dependent, is not suitable since it will increase at bond wire failures and separation of degradation effects will be complex. Further, $V_G - V_T$ which enters $R_{DS,on}$ according to Eq. (12.1) is influenced by a gate threshold voltage $V_T(T)$ drift.

The gate threshold voltage V_T is strongly temperature dependent but affected by a trapping phenomenon: After a power cycle, it is found that SiC MOSFETs need up to seconds to recover to the initial V_T value. The effect is shown in Fig. 12.37. For the MOSFET of manufacturer #1, an increase of 30 mV (measured 1 ms after power pulse) is found, for manufacturer #2 a shift of 140 mV is detected and the decay back to the initial value takes up to seconds. Used as TSEP, a measurement error of 26 K would result for this device. Additionally, a power cycling induced V_T

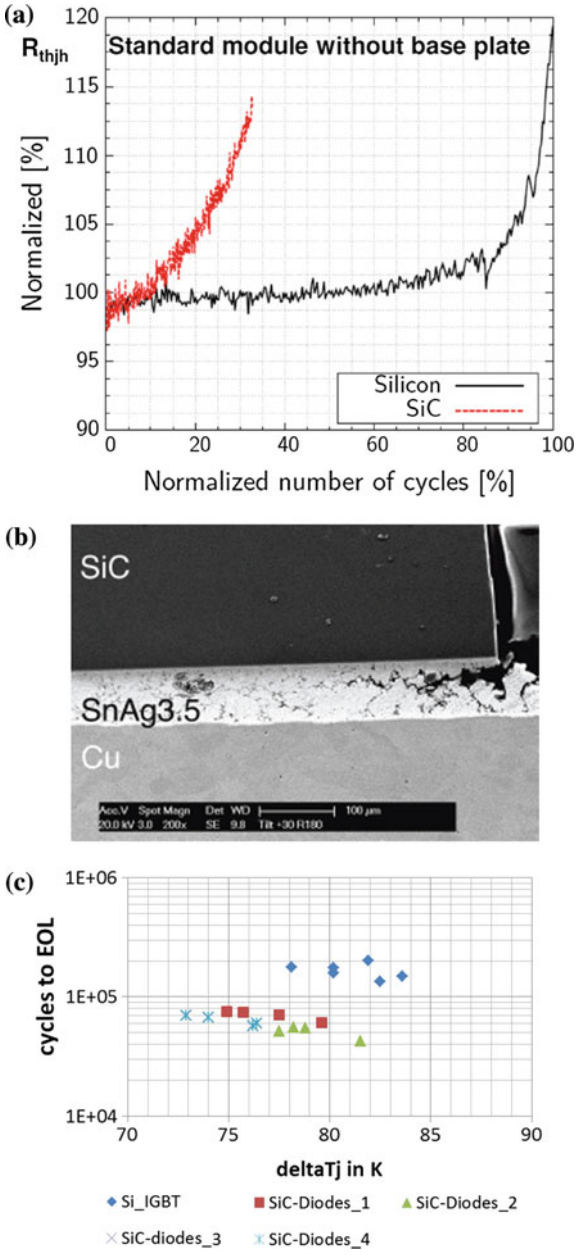


Fig. 12.36 Power cycling of a soldered 600 V SiC-Schottky diode, compared to a 1200 V IGBT with the same technology, ΔT_j = 81 K ±3 K and T_{jmax} = 145 °C **(a)** Thermal resistance R_{thjc} in dependence on number of cycles **(b)** Metallographic preparation of a solder layer in a failed SiC device. Preparation by Infineon Warstein. **(c)** Comparison of cycles to end-of-life. Figs. from [Hed14]

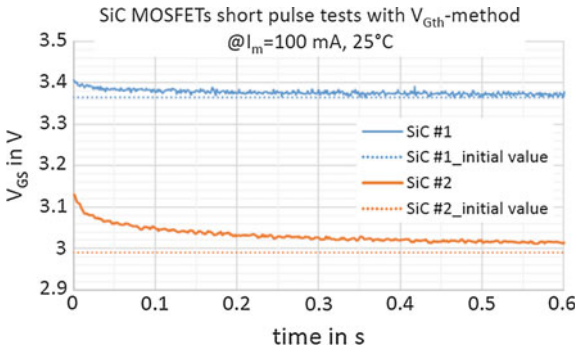


Fig. 12.37 Measurement of the threshold voltage V_T for two SiC MOSFETs after a short load pulse with V_G = 15 V

shift with SiC MOSFETs is possible. Based on these findings, V_T is regarded not suitable as TSEP.

Next, there is the voltage drop of the inverse diode $V_{SD}(T)$ at low measurement current. However, the SiC MOSFET's gate-channel is not completely off at V_G = 0 V. The voltage drop up to the built-in voltage of the pn-junction opens the channel partially and enables part of the current to pass the marginally inverted channel. The current-voltage characteristic of the inverse diode is depending on gate voltage as shown in Sect. 9.12.3, Fig. 9.32. It indicates that a gate voltage of −6 V or lower is required to obtain a stable characteristics of the diode and the junction voltage $V_j(T)$ can be measured.

Since the current through the MOS channel will depend on the threshold voltage, it must be ensured that the channel is safely turned-off. Therefore, a negative gate voltage below −6 V is recommended. Figure 12.38 shows $V_j(T)$ for a SiC MOSFET measured with V_G = −10 V.

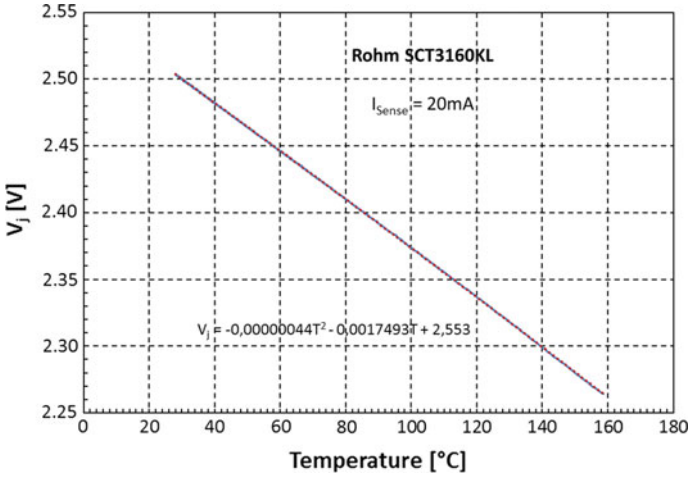


Fig. 12.38 Calibration function $V_j(T)$ for a 17 A 1200 V SiC MOSFET (Rohm) at V_G = −10 V

For power cycling of MOSFETs, the automotive standard [LV324] allows to use the body diode to generate the power losses. However, in MOSFETs of Si as well as of SiC the on-state losses of the inverse diode decrease with increasing temperature, while in forward operation the losses increase with temperature. Using the inverse diode results in a reduction of power losses when the temperature increases. This does not resemble the typical application. Therefore, it is recommended for SiC MOSFETs to use the drain-source forward current to create the power load.

The recommended test method for MOSFETs is creating power losses in forward mode and measuring $V_j(T)$ in reverse mode. Figure 12.39a shows the setup for one device under test (DUT), there may be several devices arranged in series connection. In series with the current source for I_{sense} are some diodes for protection. Figure 12.39b shows the control signals applied during a cycle. First, V_G is set on with the specified voltage V_{Guse} of the manufacturer. Next, the auxiliary switch is closed. Now the load current flows. At the end of the load pulse the auxiliary switch is turned off. After a short delay of 1–50 µs, a negative voltage is applied (here −7 V) for the temperature measurement via V_j of the inverse diode.

For measuring V_j, there is again a delay time t_d between turning-off the load current and the moment of measurement, compare Fig. 12.11 The cooling-down during t_d is higher for SiC devices compared to Si devices. For SiC, it was found in the range of 4–5 K, even up to 6 K for a t_d of 1 ms, due to the usually higher power densities [Hed16]. This is significant now, especially if packages with SiC and Si are compared. A correction is possible with the square-root-t method [Bla75]

$$T_{vj}(t) - T_{vj}(0) = \frac{2 \cdot P_v}{A \cdot \sqrt{\pi \cdot \rho \cdot \lambda \cdot c_{spec}}} \cdot \sqrt{t} \tag{12.10}$$

Equation (12.10) holds for the boundary condition of a planar heat source at the surface of a semi-infinite thick cylinder, assuming one-dimensional heat flow. Since the dominant heat source in SiC devices is in a narrow region close to the device surface, Eq. (12.10) was confirmed to be applicable for SiC devices with good accuracy [Hed16].

For an exact evaluation of power cycling tests with SiC devices, the selected delay time t_d between load current turn-off and T_{vj} measurement should be added in the documentation. It has to be mentioned when the measured values have been corrected with a Z_{th}-model or another method.

For SiC MOSFETs, a test with the method described in Fig. 12.39 with prototypes of 250 A 1200 V power modules is reported in [Hed17].

The $V_j(T)$ method with $V_G = -6$ V was used as temperature sensing method in a power cycling test in [Sct17]. For a new packaging technology with silver sintering and 125 µm aluminum bond wires topside, a very high power cycling capability can be achieved. The evolution of T_{vjhigh} in a power cycling test with constant t_{on} and t_{off} and $\Delta T_j = 110$ K is shown in Fig. 12.40. The module finally fails after more than 1.1 million cycles due to substrate failure.

The results of [Sct17] show that an excellent power cycling capability can be achieved with SiC devices, if improved packaging technologies are applied.

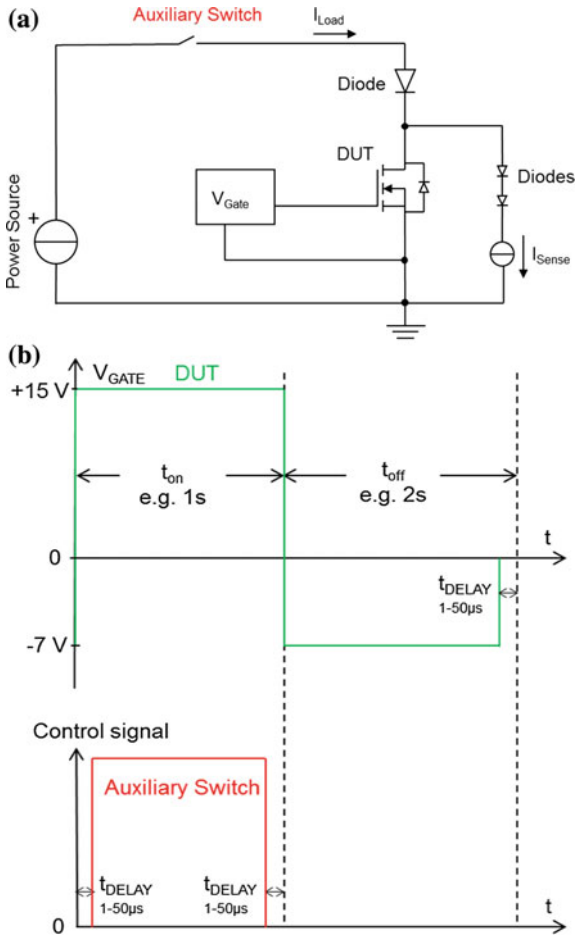


Fig. 12.39 Power cycling of SiC MOSFETs. **a** Test setup, **b** pulse pattern at the MOSFET and at the auxiliary switch

However, before design-in into a reliability-sensitive application, the power cycling capability of any new solution has to be proven.

The German automotive companies have established a mutual delivery specification for the qualification of power electronic components for automotive applications – including test methods and conditions for power cycling tests [LV324]. Even though, this delivery specification needs to be extended regarding the latest research results on Si and SiC MOSFETs discussed here, it is an important milestone on the path to a standardization of power cycling test methods and documentation. An international standard for power cycling tests, which is accepted by the major players in the field, would help to render power cycling test results more transparent and comparable.

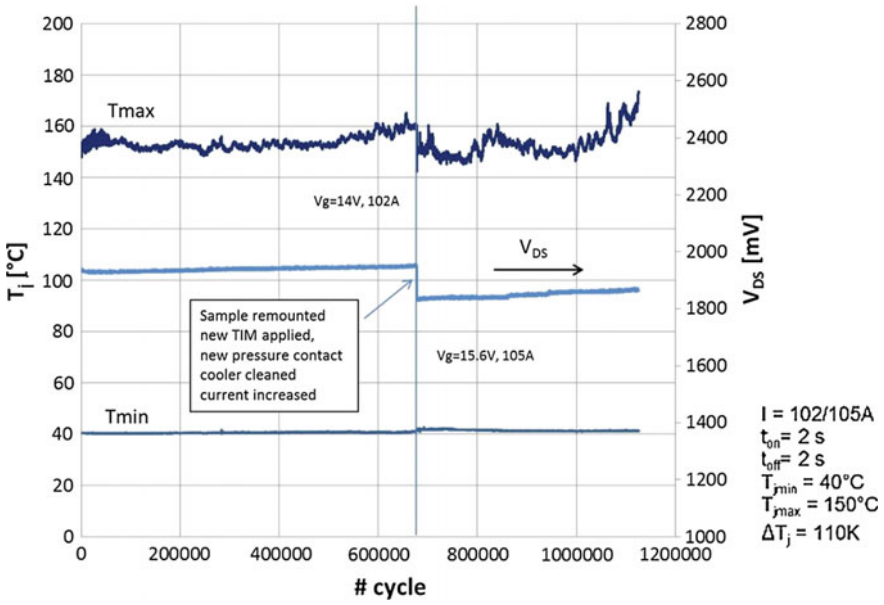


Fig. 12.40 Power cycling test with a SiC-MOSFET module: Evolution of V_{DS}, T_{vjlow} (*Tmin*) and T_{vjhigh} (*Tmax*). Figure from [Sct17], PCIM Europe 2017

12.8 Cosmic Ray Failures

12.8.1 The Salt Mine Experiment

With the introduction of high voltage semiconductors with turn-off capability in converters for electric traction in the beginning of the 1990s, failures in the application were observed which could not be explained with the available knowledge at that time. The failures occurred during the blocking mode of the devices. The application conditions were employed in the lab, and long term tests with high DC voltage in blocking direction were carried out. The tests confirmed the occurrence of spontaneous failures [Kab94]. The spontaneous character of the failures was strange, with no prior indications in device behavior, e.g. an increase of the leakage current. In some English literature, a cosmic ray failure is denoted as "Single Event Burnout" (SEB) [Alb05].

Figure 12.41 shows the results of the salt mine experiment. First, 6 failures occurred in the lab within 700 device hours. The test was interrupted and continued in a salt mine, with 140 m of solid rock above the test location. Under these conditions, no failure occurred. The test in the salt mine was interrupted again and continued in the lab, and now failures reappeared at a similar rate as in the lab before. The test setup was moved to a place in the cellar of a high-rise building with 2.5 m of concrete in total above the test location. The failures continued to occur but at a reduced rate.

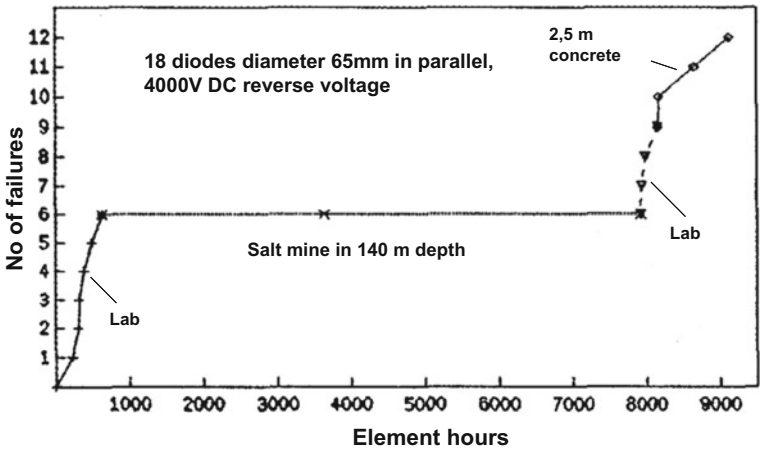


Fig. 12.41 Results of the salt mine experiment. Number of failures over the accumulated time of DC voltage stress. Figure from [Kab94] © 1994 IEEE

In addition to [Kab94], two further research groups [Mat94, Zel94] published the effect at the same conference. [Mat94] showed that the failure rate decreases by more than one decade with shielding by 2 m thick concrete. With these results, it was proven that cosmic ray was the root cause for said failures of power semiconductor devices under high voltage stress.

12.8.2 Origin of Cosmic Rays

The primary cosmic radiation consists of high energy particles: 87% protons, 12% alpha-particles and 1% heavy nuclei. Ultra-high energy photons are also part of the cosmic radiation. For particles up to 10 GeV (10^{10} eV), our sun is considered to be the main source. Up to 10^{16} eV, supernova eruptions are assumed as the main source, up to 10^{18} eV the source is expected to be located in our galaxy. Above 10^{18} eV, supernovae cannot explain the extreme energy. The cores of distant active galaxies are potential candidates for the origin of particles with extreme high energy.

A high-energy primary cosmic ray particle usually does not reach the surface of the earth directly; it rather collides with atom nuclei of atmospheric molecules. Thereby, it generates a variety of secondary high energy particles and this particle shower arrives at the earth surface as terrestrial cosmic radiation. A single primary high-energy particle can create a shower of even 10^{11} secondary particles. Most relevant for SEB at usual terrestrial altitudes are neutrons, the neutron flux density is in the range of 20 cm^{-2} h^{-1} at sea level [Nor96] and increases strongly with altitude. At 12.2 km (40,000 ft, the upper flight level of civil airplanes) there is a

neutron flux density of 7200 cm^{-2} h^{-1} for a latitude of 45° [IEC10]. The maximum neutron flux density is found at an altitude of 18 km. Further relevant parts are high-energy protons, they contribute with 20–30% at sea level and 50% at 12.2 km altitude to the total cosmic radiation. Creation and absorption compete in the showers in higher atmosphere. Also short-living high-energy particles like pions are generated, which decay to further particles. The pion flux is estimated as 1–3% of the flux of protons [IEC10].

Regarding space application, proton flux is the majority of energetic particle flux at low earth orbit [Das17]. Protons as charged particles can be shielded, however aluminum shielding of 2.5 mm thickness is no proper protection against proton flux, since a large fraction of the proton flux exhibit a high energy above 100 MeV range [Das17]. At sea level, usually only neutrons are considered for triggering cosmic ray failures in power devices. It has to be noted that the sun activity enhances the shielding effect of the Van-Allen belt. With a more intensive Van-Allen belt, less high-energy charged particles pass and collide with molecules of the atmosphere, finally less neutrons reach the earth surface. Figure 12.42 shows the correlation between sun activity and relative neutron flux on the earth´s surface.

The sun activity shows a somehow regular repetition in about 11 years, and after some time of higher activity an enforced belt becomes visible. The sun, in this respect, acts more protective than destructive. The shielding of the primary particles due to earth's magnetic field is more effective at the equator, where the flux of secondary particles at sea level amounts to $^1/_3$ compared to latitude of 45°, and the shielding effect is less at the poles, where the intensity is 3 times higher [IEC10].

For high altitude aircrafts and space applications, this is more complex. Radiation bursts of the sun were found to increase the particle flux in an altitude of 12 km up to a factor of 300, and can last several hours. Such events are estimated to happen 7 times in 67 years. [IEC10].

The high energy particles come from deep space. Figure 12.43 shows the flux density of primary cosmic radiation as a function of kinetic energy. Direct observations have been possible from satellites up to 10^{14} … 10^{15} eV. For higher energies, the data come from air-shower detectors on the ground, which observe the cascades of secondary particles in the atmosphere initiated by the high-energy

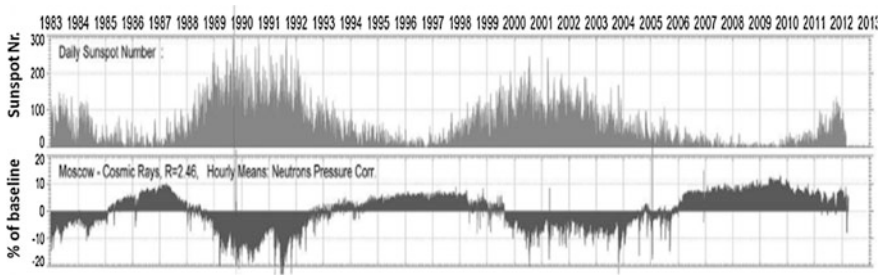


Fig. 12.42 Sun activity (daily sunspot no.) and relative neutron flux on the earth's surface. Figure from [Wil12], Figure D. Wilkinson, Wikimedia Commons license.

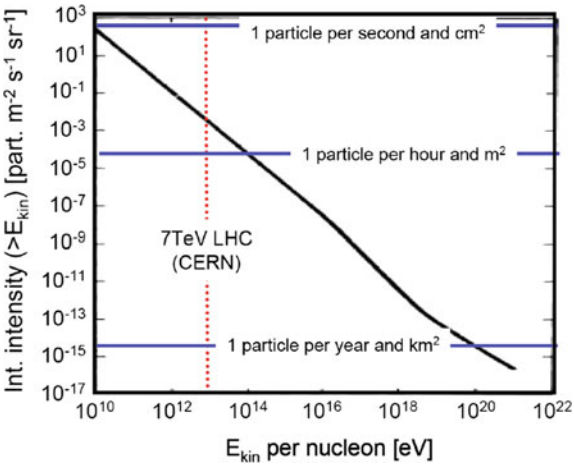


Fig. 12.43 Flux density of primary cosmic radiation as a function of kinetic energy. CERN, the largest accelerator on earth, can generate about 1×10^{13} eV. Figure from [Sti09] based on data from [Zom90]

primary particles [Gai13]. Primary particle energies above 10^{20} eV have been found. The highest energy reached by human technology at the CERN accelerator lies in the range of 10^{13} eV – far below the energy reached by cosmic particles.

For very high energy, cores of active galaxies are the only reasonable candidates as sources. Figure 12.44 shows the nearest active galaxy in our neighborhood, Centaurus A in a distance of 12 million light years.

Centraus A is a quite low-energy active galaxy compared to other active galaxies, for example Cygnus A in a distance of about 750 million light years. High energy particle impacts are monitored by the Pierre Auger observatory in Argentina which uses a large area of 3000 km^2 for detection of secondary particles and which



Fig. 12.44 Active galaxy Centaurus A. Left: optical image ESO 2.2 m telescope Silla Observatory Chile. Right: Chandra x-ray telexcope. X-ray image, showing one of two particle jets ejected from the core with relativistic velocities. Reprint of both Fig's according to Wikimedia Commons license

recalculates from these observations the path and energy of the initial particle. For some high-energy events, active galaxies are identified as origin [Blu10], but the majority of particles seem to originate from a random background.

Cosmic ray particles may be on their journey for several 10 or 100 million years before they collide with the earth atmosphere. An explanation with the simplified model of "black holes" in galactic cores has not been successful so far [Joo06], the real process creating such high energies has not been understood up to now.

12.8.3 Cosmic Ray Failure Patterns

Cosmic ray failures in power devices are single event burnout (SEB) or single event gate rupture (SEGR) in MOSFETs [IEC13] and other field controlled devices. SEGR means the gate oxide breakdown, leading first to gate leakage and finally to gate rupture. In power MOSFETs irradiated with protons, the gate ruptures are explained as caused by charge deposition from recoil ions [Tit98]. Griffoni et al. [Gri12] conducted tests with silicon IGBTs, silicon superjunction and SiC MOSFETs using neutrons, they observed SEGR only in the superjunction MOSFETs. SEGR should also be considered for power devices, however it is assumed to be a major effect for space applications only, while in all other applications SEB is the main effect. Thus, we will focus in the following on SEB effects.

Devices failed by SEB in laboratory tests exhibit a pinhole-size molten channel between the cathode and the anode side randomly distributed over the chip area. Failure pictures from device failures caused by cosmic ray in laboratory tests are shown in Fig. 12.45. On the left hand side in Fig. 12.45, a pinhole can be seen. On the right hand side, bubbles in the metallization can be seen, and below the metallization a pinhole is hidden. The failure pictures clearly show an effect occurring in a very narrow region.

Figure 12.46 shows the cosmic ray failure picture of a 3.3 kV IGBT die in a laboratory test. Again a pinhole is found, with a size in the order of the cell pitch of the IGBT die.

Figure 12.47 shows a cross section through the failure position of another IGBT die. The destroyed area is reaching through the whole drift layer of the device. The molten silicon in the *n*-drift region rapidly solidified and a crack was generated during the rapid cooling process.

Several tests of different device designs were carried out; for acceleration of the failure rate tests stations at high altitudes were arranged (Zugspitze 2964 m, Jungfraujoch 3580 m). The terrestrial cosmic particle flux increases with an altitude above sea level up to a height above 11 km [All84]. The acceleration factor with respect to sea level failure rates increases from 10 in 3000 m to approx. 45 in 5000 m [Kai05]. In accelerated experiments, tests with particle accelerators are performed. Devices at high reverse voltages are irradiated with high energy neutrons, protons and other ions accelerated with high energy. The irradiation with ions, especially the types of ions and energies which are generated at an inelastic

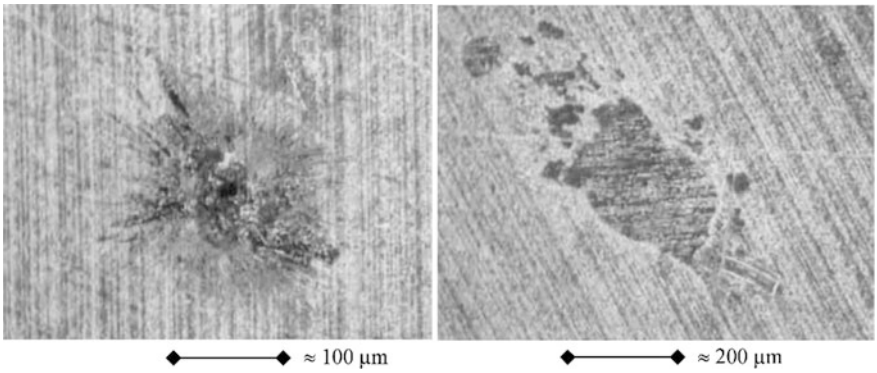


Fig. 12.45 Failure pictures after a cosmic ray destruction in a laboratory test of 4.5 kV diodes with a diameter of <50 µm. The photos are taken from the cathode side. Left hand side: small pinhole. Right hand side: Molten area in the metallization with bubbles. Pictures from Jean-Francois Serviere, Alstom

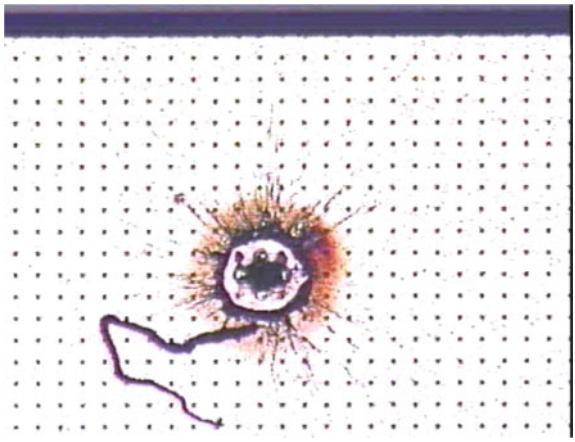


Fig. 12.46 Cosmic ray failure of a 3.3 kV IGBT in the cell area (laboratory test). Cell pitch 15 µm. Picture from G. Sölkner, Infineon

neutron-silicon-collision, is suited to study the detailed failure mechanism. A great amount of research work has been carried out in this field, see e.g. [Soe00]. Tests are executed today with "white neutron sources" having a similar energy spectrum than the cosmic neutrons or with monochromatic proton beams. It was shown that high-energy protons (>150 MeV) lead to similar results as white neutrons [IEC13].

It must be emphasized, that the failure images presented in this section are all resulting from accelerated or non-accelerated laboratory tests. In laboratory tests, only very limited energies are involved, so that the initial destruction image can be evaluated. In a real power electronic applications, the substantially higher energies in DC link capacitors involved will cause considerably more damage. It is in

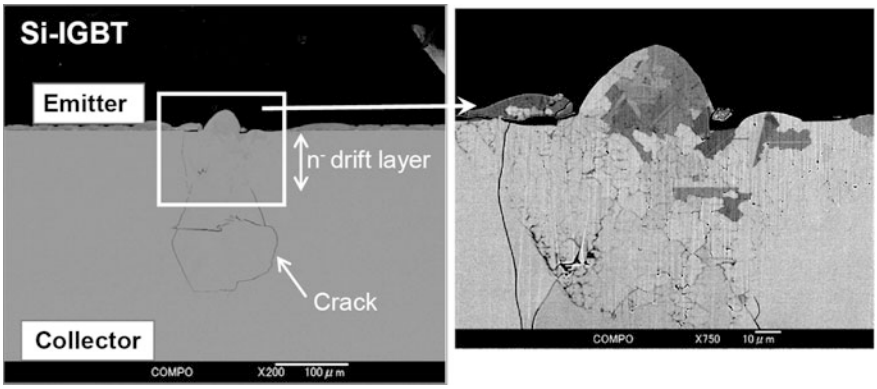


Fig. 12.47 Cross section at the failure position of a cosmic ray failure of an IGBT in a laboratory test. Figure from T.Shoji, Toyota [Sho11] © IEEJ 2011

general not possible to distinguish a cosmic ray failure from any other device failure resulting in a short circuit mode of the device.

12.8.4 Basic Failure Mechanism Model

A collision of a particle with the nucleus of a lattice atom does not destroy a power semiconductor by the SEB effect. Precondition is the presence of a high electric field and impact ionization.

Let us consider a device in the blocking state with a trapezoidal electric field as drawn in Fig. 12.48a. A neutron travelling through the space charge of a blocking semiconductor device is captured by a $^{28}_{14}$Si nucleus in a process of inelastic scattering, see Fig. 12.48b. The created $^{29}_{14}$Si nucleus is highly excited and decays fast in several light ions. The light ions have a high kinetic energy and create electron-hole pairs locally forming a dense plasma of charge carriers. All this happens on a time scale of picoseconds.

In the plasma is a high density of electrons and holes, and due to its state of almost neutrality no high fields can occur. The electric field in the plasma region is low. At the boarders between the plasma and the space charge, however, an extreme high density of charge emerges, leading to very high and steep field peaks. In silicon, it can even reach up to 1 MV/cm [Wei15]. If the electric field exceeds a certain threshold value, impact ionization creates more carriers than carriers which flow out of the plasma region by diffusion. One field peak moves fast to the anode, the other to the cathode. A so-called “streamer” is formed in analogy to a discharge in a gas. The device is flooded locally with free carriers within some hundred picoseconds, hence, a local current tube occurs. Basic failure models assumed that the very high local current density destroys the semiconductor device [Kab94, Kai04].

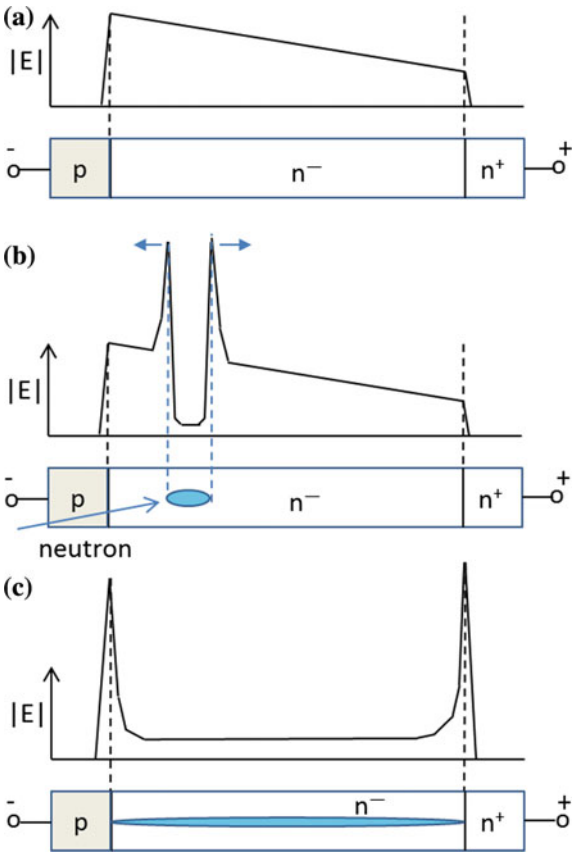


Fig. 12.48 Destruction of a power device (diode) by a neutron. **a** Device in the blocking state, **b** inelastic scattering of the incoming particle, formation of an electron-hole plasma, high field peaks at the boarders of the plasma, **c** the field peaks have run through the device, forming a streamer. At the pn^- and nn^+ junction, field maxima occur.
Figure inspired by Kaindl [Kai05] and Weiss [Wei15]

If the field peaks arrive at the pn- and nn^+ junction, then an Egawa-type field (see Chap. 13) with two peaks at the pn- and nn^+ junction is given. This phenomenon is displayed in Fig. 12.48c for the case of abrupt junctions. We will come back to this process in the discussion of extended models, Sect. 12.8.6.

12.8.5 Basic Design Rules

The basic design rule considers the field strength at the position of the impact. The lower the field strength, the lower is the probability of failures.

In Fig. 12.49 PT and NPT designs are sketched for the same rated voltage. The maximal electric field is much lower for the PT-design. The shape of the electric field is drawn for two simplified diode structures with the same thickness and for the same applied blocking voltage. The area below the line $-\mathsf{E}(x)$, which corresponds to the reverse voltage, is the same for both devices. However, the value E_0 is

much lower for the PT design. At the same voltage, impact ionization is still negligible for the device with PT design. For the occurrence of impact ionization in the PT diode, the applied voltage must be increased. Then the line of $-\mathsf{E}(x)$ in Fig. 12.49b is shifted upwards, until E_0 becomes close to a value as in Fig. 12.49a. Then impact ionization emerges in the diode with PT-dimensioning as well, but at a much higher voltage.

Experimental and simulation results on this effect are shown in Fig. 12.50, taken from [Kai04]. Diodes rated at 3.3 kV have been irradiated with ^{12}C ions, and the reverse voltage applied during irradiation was increased. Two diode designs were compared having a triangular electric field shape (NPT, see Fig. 12.49a) and another diode with a trapezoidal shape of the electric field (PT in Fig. 12.49b), which is indicated by FS (field stop). At a small voltage the charge generated by a single ^{12}C ion is small for all samples, above a defined threshold voltage a strong charge carrier multiplication arises and the created charge increases more than three decades. For the FS diode, which has a PT design, this threshold voltage is more than 700 V higher when compared to the NPT diode. A device design for increased cosmic ray stability must keep the maximum electric field E_0 at the maximum DC voltage V_{bat} required by application as low as possible. Thus, the base doping N_D

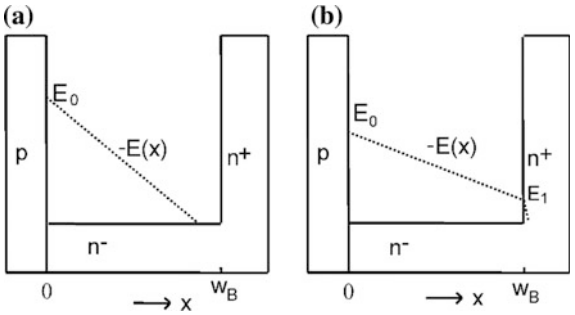


Fig. 12.49 Schematic drawing of the field shape for an NPT design (**a**) and PT-design (**b**) at the same thickness and same applied blocking voltage

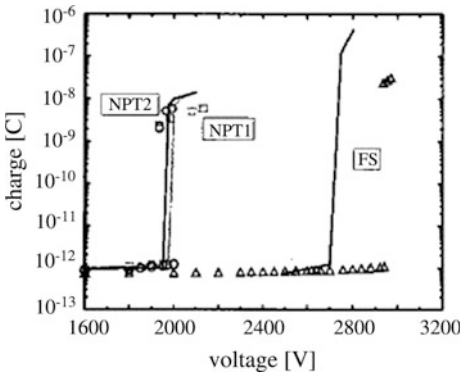


Fig. 12.50 Charge created by irradiation of a 3.3 kV diode with single ^{12}C ions of a kinetic energy of 1.7 MeV as a function of the reverse DC voltage applied during irradiation. Simulation (straight line) and experiment (symbols) for a triangular field shape NPT1 (white square), NPT2 (white circle) and a trapezoidal field shape FS (white triangle). Figure from [Kai04] © 2004 IEEE

must be lowered, so that a PT dimensioning can be achieved. Additionally, E_0 can be reduced by a wider drift region w_B. However, the latter measure is detrimental with regard to conduction and switching losses.

Furthermore, some quantitative models for this design dependency were published, the first was introduced by Zeller [Zel95]. The failure rate is given in the unit 'Failure In Time' (FIT); 1 FIT is one failure in 10^9 h. For a power electronic module in traction application for example, a failure rate below 100 FIT is required. Considering that a typical power module for such an application comprises 24 IGBT chips and 12 freewheeling diode chips, the failure rate for a single device must be more than one order of magnitude lower.

The Zeller model for the failure rate r reads as

$$r = a_1 \cdot A \cdot S^2 \cdot e^{-\frac{b_1}{S}} \quad \text{with} \quad S = 0.2786 \cdot \frac{V}{t} + 0.8972 \cdot \frac{t}{\rho} \tag{12.11}$$

With the device area A in cm^2, the specific resistivity of base doping ρ in Ω cm, the field strength factor S, the applied voltage $V = V_{DC} = V_{bat}$ in V and the thickness of the base layer t in µm. The equation has the advantage that it uses values which are known to the device designer, like thickness and resistivity. It can be shown that the results correspond to the field strength at a cosmic ray impact, and can also be written as

$$r = a \cdot e^{-\frac{b}{E_C}} \tag{12.12}$$

The model was quite in agreement with experimental results for high voltage devices >2 kV. For devices with lower voltage rating, Pfirsch und Sölkner [Pfi10] showed that for IGBTs and freewheeling diodes the failure rate is overestimated. They extended it by a correction factor $f(V_{bat})$ for V_{bat} below 2 kV

$$r = f(V_{bat}) \cdot a \cdot e^{-\frac{b}{E_C}} \tag{12.13}$$

The factor $f(V_{bat})$ is displayed as given in [Pfi10] in Fig. 12.51. For $V_{bat} > 2$ kV it approaches 1, for $V_{bat} < 500$ V the failure rate r becomes insignificant.

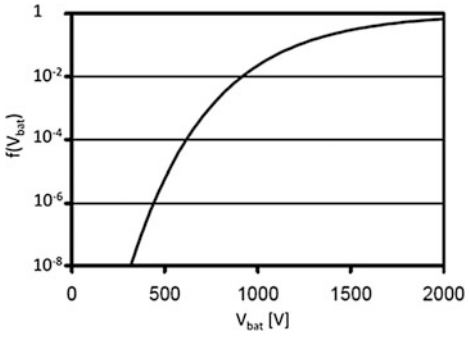


Fig. 12.51 Correction factor $f(V_{bat})$ for cosmic ray failure rate. Figure adapted from [Pfi10]

Moreover, a model by Kaminski [Kam04] was published, which is based on a fit of experimental results from IGBT power modules of one manufacturer. These equations are just the best adaption to the experiment. For T = 25 °C and sea level altitude, the results are given by the equation

$$r = C_3 \cdot a \cdot e^{\frac{C_2}{C_1 - V_{bat}}} \tag{12.14}$$

valid for $V_{bat} > C_1$.

Even though the models of Zeller, Pfirsch and Kaminski are designed for devices from different manufacturers, a quantitative comparison is given in Fig. 12.52 for a 1700 V device with an area of 0.44 cm^2.

The model of Pfirsch predicts a lower failure rate compared to that of Zeller. For a wide range, the model of Kaminski predicts an even higher failure rate than the Zeller model. The shape of the function, however, is different for Kaminski: It has a pole at the voltage C_1 in Eq. (12.14). Below C_1 the model is not valid and the failure rate is zero.

In addition, the model of Kaminski contains the dependency on the temperature and the altitude above sea level. The full equation is

$$r = C_3 \cdot a \cdot e^{\frac{C_2}{C_1 - V_{bat}}} \cdot e^{\frac{25 - T_{vj}}{47.6}} \cdot e^{\frac{1 - \left(1 - \frac{h}{44{,}300}\right)^{5.26}}{0.143}} \tag{12.15}$$

with the temperature T_{vj} in °C and the elevation above sea level h in m. Terms 2 and 3 of this equation are very useful to transfer results from one condition to other conditions and they are, if no further details are known, also used for devices of other manufacturers. The failure rate is highest at low temperatures and it decreases with temperature due to the decreasing avalanche ionization rates at increased temperature, see Chap. 2. The dependency on altitude is due to the increasing neutron flux density with height and is in reasonable agreement with the data in [IEC10] for a height up to 12,000 m. For altitudes above 18 km, the neutron flux is

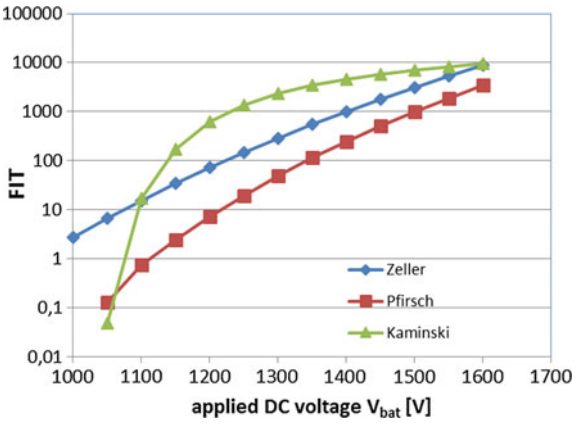


Fig. 12.52 Prediction of different models for a 1700 V device, area 0.44 cm^2

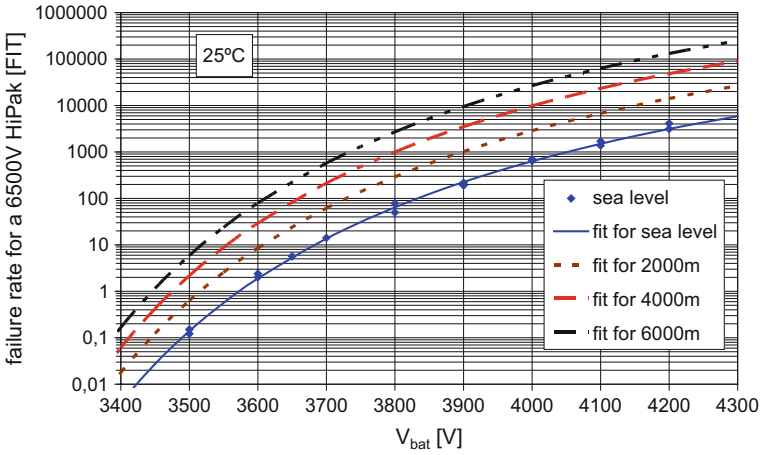


Fig. 12.53 Cosmic ray failure rate at $T = 25$ °C for the 6.5 kV IGBT module 5SNA0600G650100 from ABB. Figure from [Kam04]

decreasing again. Figure 12.53 shows an example for the dependency on height above sea level for a 6.5 kV IGBT module from [Kam04].

Following these rules and data, the probability of cosmic ray induced failures can be estimated for specific combination of application and device design and measures for a reduction of failure rates can deduced if necessary. Also a noticeable effect on the cosmic radiation induced failure rate from voltage peaks at switching is found [Hae12].

However, coming back to Fig. 12.52, the state of knowledge is unsatisfactory. Referring to the same area, the difference in failure rates resulting from different models amounts to almost two decades. Hence, the uncertainty is very large. Experimental data are necessary to reduce this uncertainty.

Furthermore, some of the models are for modules containing IGBTs and freewheeling diodes, and these models do not distinguish between IGBT and diode. The device with higher failure rate will dominate the total failure rate, in one case it can be the IGBT, in another case it may be the freewheeling diode.

For high voltage devices, the dimensioning rules for cosmic ray stability are in contradiction with other rules for optimizing the device characteristics; e.g. for diodes it contradicts to the requirement for soft recovery behavior, which is hard to achieve if a strong PT dimensioning is used. A trade-off between different requirements must be made. To meet said demands, most of today's high-voltage devices use designs with a layer thickness w_B much higher than necessary for the required blocking capability. However, this leads to higher losses in the forward conduction mode and/or increased turn-off losses.

In a MOSFET, the failure mechanism is explained to be finally the activation of the parasitic bipolar transistor and second breakdown of the bipolar transistor [Was86]. In diodes, no parasitic component of any transistor type is present. Even a strong local avalanche breakdown should be stable, see Sect. 13.4. Field

redistribution effects in analogy to dynamic avalanche of the third degree are considered in deep submicron CMOS devices for improved ruggedness against radiation-caused single event pulses [DaG07].

In general no experimental data are publicly available for most power device manufacturers. Large customers received some data under the condition of non-disclosure agreements which required to keep the data confidential. Thus, no open scientific comparison was possible. However, new groups have started to work on cosmic ray stability. Si devices with a much higher onset voltage than predicted by the models discussed before were found. Meanwhile, some publications present not only experimental data but also extended explanations giving more insight.

12.8.6 Extended Model Considering the nn$^+$ Junction

Schulze and Lutz [Scu06] claimed the failure mechanism in diodes to be the occurrence of an Egawa-type electric field with 2 field peaks, similar to the third-degree dynamic avalanche. Egawa-type fields exhibit two field peaks, one at the pn-, one at the nn$^+$ junction. In Fig. 12.48c, the electric field in a streamer with high current density is shown. In this case, both pn- and nn$^+$ junction are abrupt. For this Egawa-type field, impact ionization occurs at both junctions. The figure has clear similarities to a dynamic avalanche of the third degree which will be discussed in detail in Sect. 13.4.2, compare Figs. 13.16 and 13.17 at the position of the filament. Figure 12.54a adapted from [Scu06] is very similar to Fig. 12.48c. Figure 12.54 is the key figure for understanding the extended model.

An electric field in the streamer as in Fig. 12.54b will be obtained, if the junction between low doped n$^-$ layer and n$^+$ layer shows a very slow increase of the doping, resp. if its gradient dN_D/dx is low. Since in a streamer an electron density of about 10^{17} cm^{-3} or more is expected, the dN_D/dx must be low especially in this range. Figure 12.55 shows a doping profile of the cathode layer which leads to an electric field in a streamer as in Fig. 12.54b.

The doping profile in Fig. 12.55 is taken from Fig. 7.3, it was used as collector layer of a bipolar transistor. The reason for this type of profile was to increase the stability of the bipolar transistor against second breakdown. The same type of profile hinders the formation of Egawa-type fields at dynamic avalanche, as will be discussed in Sect. 13.4. It also provides an increased cosmic ray stability.

According to Weiss [Wei15], the time in which the streamer formation happens is very short. The field peaks move with a velocity that is five times the saturation velocity. It takes only 200 ps, a time too short for a significant temperature increase that would lead to destruction. After arrival at the pn- and nn$^+$ junction, the streamer shortens both sides of the diode and is, at the first moment, combined with an Egawa-type avalanche at both sides. For an abrupt nn$^+$ junction, the cathode-side injection delivers the main part of the generated current. In the streamer, carriers diffuse laterally lowering the carrier density. It takes up to 20 ns until the channel,

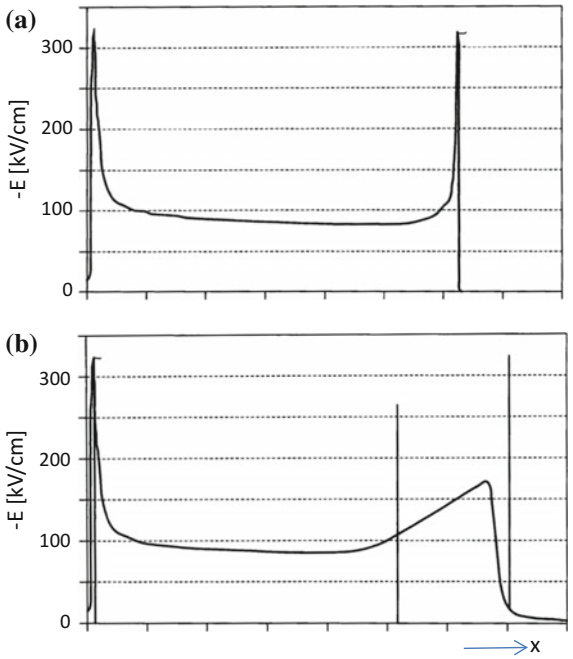


Fig. 12.54 Electric field at high current density in a streamer **a** abrupt pn^- and nn^+ junction **b** abrupt pn junction, nn^+ junction with low gradient dN/dx. Fig. from [Scu06]

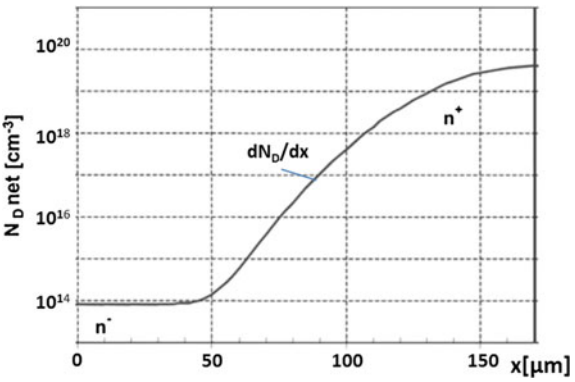


Fig. 12.55 Doping profile of the n^+ layer which leads to a field as in Fig. 12.54b in case of a streamer

which shortens anode and cathode, vanishes. Within this time, strong heating-up takes place, destroying the device. For the investigated diodes with an abrupt nn^+ junction, the highest temperature occurs at the nn^+ junction. However, for other device structures the temperature maximum can appear at the pn junction or inside the streamer close to the pn junction [Wei15].

In conclusion, the basic model considering the field strength at the neutron impact is not complete, it must be extended. Occurrence of impact ionization at the pn junction and formation of a streamer are necessary, but not sufficient conditions

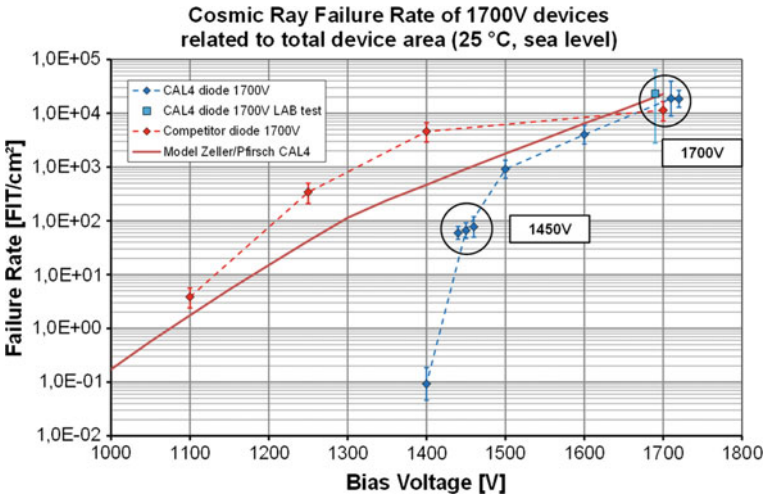


Fig. 12.56 Cosmic ray failure rate at $T = 25$ °C for a 1700 V CAL diode compared to model (Zeller with correction by Pfirsch) and to a competitor diode. Figure from [Scn15b]

for cosmic ray failures in diodes. The streamer is destructive if additional impact ionization at the nn^+ junction occurs, this is a sufficient condition that leads to failure. If a second avalanche in diodes or another amplifying mechanism in other devices can be avoided, a higher stability against cosmic ray failures is given.

Meanwhile, the extended model has been proven. For the 1700 V CAL diode [Lut94] it was shown that the threshold for cosmic ray destruction lies in the range of 1400 V [Scn15b]. Between 1000 and 1400 V, the failure rate is negligible or far below the prediction of the Pfirsch/Zeller model, and the detected threshold voltage is far above the threshold C1 of 983 V in Eq. (12.14) given for a 1700 V device in [Kam04]. The results for 1700 V are summarized in Fig. 12.56.

For the 1200 V CAL diode, a threshold in the range of 1200 V was found, and for the 600 V CAL diode no failures occurred even at 750 V.

Similar results have been published before by Shoji et al. [Sho13]. Failure rates of IGBTs and diodes are shown in Fig. 12.57a. The results indicate a threshold voltage for every device. Figure 12.57b shows the cosmic ray failure threshold voltage depending on the device base layer thickness for IGBTs with three different designs and one diode.

It is remarkable that a diode with 70 µm base width – which is typically used for a 600 V … 700 V diode – is better than an IGBT with approx. 160 µm wide n^- layer, which is typical for a 1700 V device.

Figure 12.58 shows the electric field for the diode from Fig. 12.57. For the onset of avalanche or moderate current density the field marked as line (1) is obtained. For an increased current density, the field has the shape of line (2). In a streamer, the device simulation shows line (3) for the electric field. The streamer front consists of electrons when it arrives at the cathode side. The higher the density of electrons, the

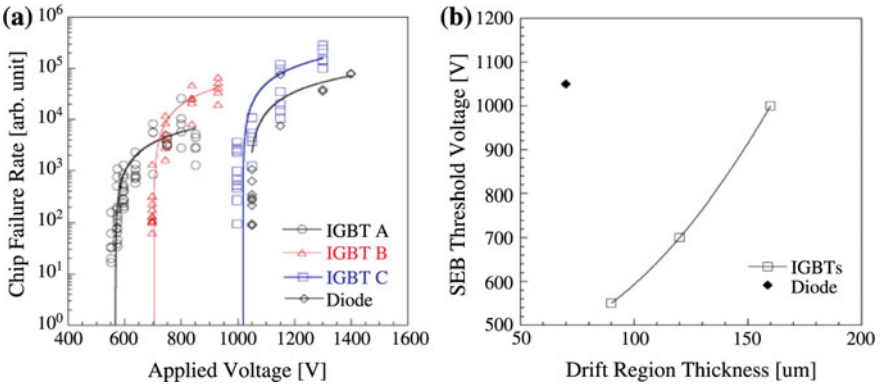


Fig. 12.57 Failure rate of IGBTs and diodes. **(a)** Failures, indicating a threshold voltage, **(b)** cosmic ray failure (SEB) threshold voltage depending on device base layer thickness, for IGBTs and one diode. Figure from [Sho13] © 2013 The Japan Society of Applied Physics

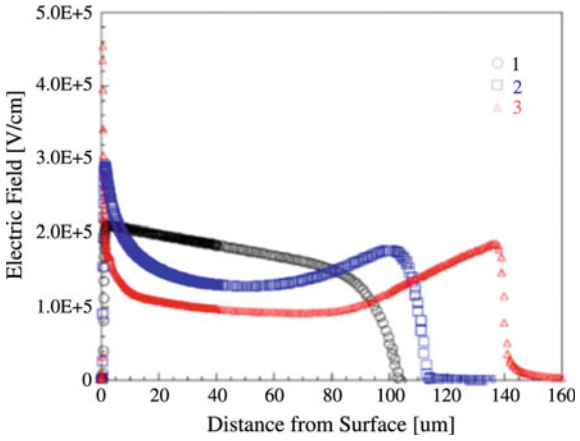


Fig. 12.58 Electric field in a diode, simulated for low current density (1), high current density (2), very high current density in a streamer (3). Figure from [Sho13] © 2013 The Japan Society of Applied Physics

more the doping of the n^+-layer is compensated, and the deeper the electric field penetrates into the n^+ layer. Figure 12.58 is very similar to Fig. 12.54b.

Both designs with very high ruggedness against cosmic ray, the CAL-diode [Scn15b] as well as the diode from [Sho13], contain similar n^-n^+ junctions with low doping gradients dN_D/dx. At high density of arriving electrons, they are partially compensated by the increasing density of positively charged donors, and the field can expand further into this compensation layer.

In summary, it can be stated that the failure mechanism in diodes with abrupt nn^+ junction is due to the occurrence of an Egawa-type electric field with two field peaks, similar to the dynamic avalanche of the third degree, as claimed in [Scu06]. This is experimentally confirmed by Scheuermann 2015 [Scn15b] and Shoi 2013 [Sho13] and supported with device simulation in [Wei15]. Designs with slowly

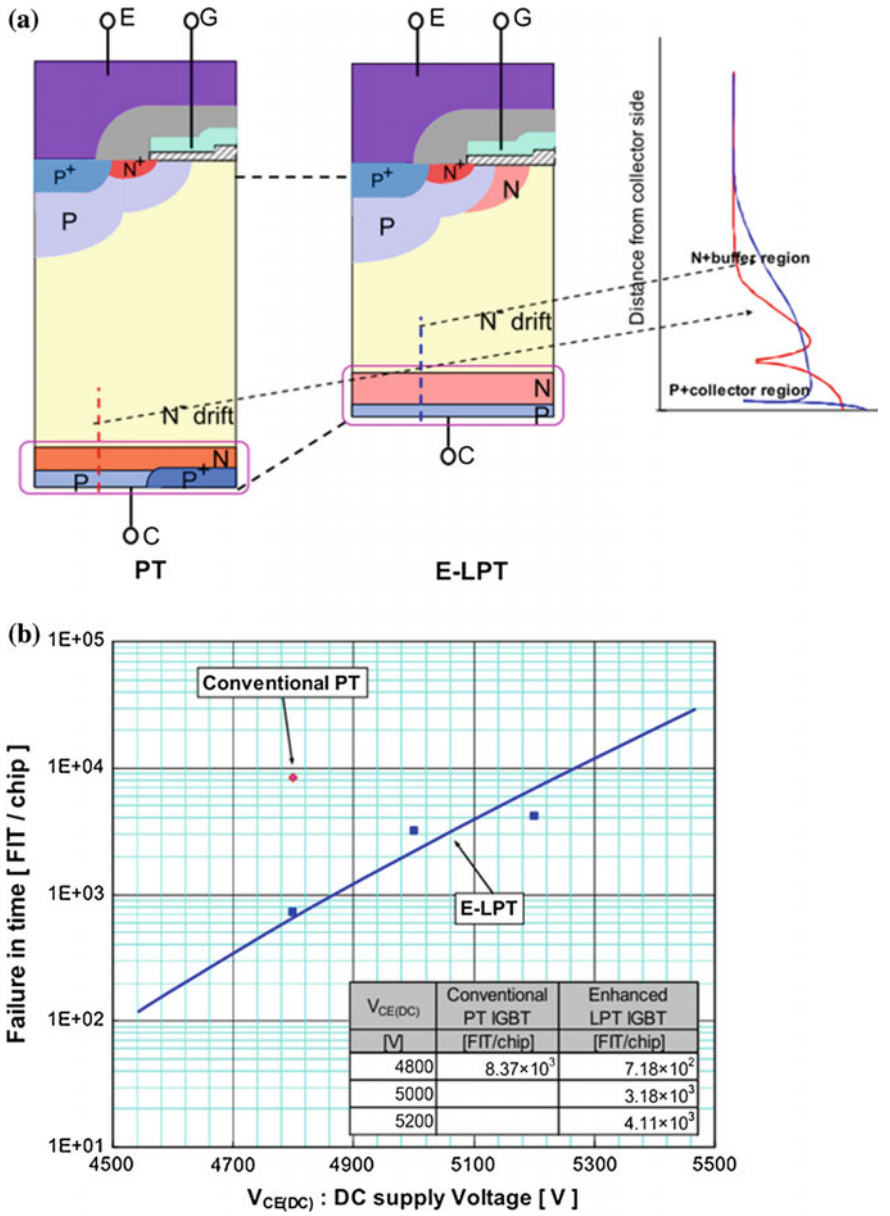


Fig. 12.59 Effect of the buffer doping on cosmic ray stability of a 6.5 kV IGBT. **a** Structure of the former (PT) and advanced generation (E-LPT), **b** failure rate for the former advanced generation. Figure from [Uem11] PCIM Europe 2011

increasing doping in the buffer can be very rugged. The threshold voltage can even be above the rated voltage.

The failure rate in modern IGBTs was found to be higher than that of cosmic ray stable diodes, yet much better than the competitor diode in Fig. 12.56 [Scn15b]. Figure 12.57 from [Sho13] also confirms a lower threshold voltage for IGBTs than for a well-designed diode. It is not possible to transfer the deep n^+ profile from Fig. 12.55 in the same form to an IGBT, since such a high buffer doping would lead to a very low current gain of the pnp-transistor. However, the buffer is of influence, results are shown in Fig. 12.59 [Uem11]. The recent design denoted as E-LPT has a more flat buffer layer with a lower gradient of doping dN_D/dx. Despite the new design having a lower base width, its failure rate is about one order of magnitude lower than for the conventional design denoted as PT.

The formation of an Egawa-field with a second field peak at the nn^+ junction to the buffer is supposed to be involved in the root cause for cosmic ray failures in IGBTs [Sho13]. The generated carriers activate the pnp-transistor. The onset of impact ionization at n^-n^+ interface can cause the parasitic transistor of IGBT to switch on; this phenomenon is known as destructive latching in IGBTs.

12.8.7 Further Design Aspects in Extended Models

Additionally to the impact of the nn^+ junction on the cosmic ray failure rate, the pn junction is of influence as well. Figure 12.60 compares two p layers [Wei15], where one layer is denoted as standard and the second is of lower depth and lower integral doping density G_n, compare Eq. (5.107). The shallow anode shows a reduced threshold for cosmic ray failures by approx. 100 V.

It has to be considered that in a streamer a carrier density in the range of 10^{17} cm^{-3} is to be expected. At the pn junction, holes of this density arrive and are capable to compensate the negative charged acceptors of the p layer. The electric field penetrates the p layer and may locally reach the semiconductor surface. Then a large number of carriers is injected by the metal layer. The effect is shown by device simulation in [Wei15].

It is reported there that the effect occurs suddenly for too shallow doping profiles, and above a certain limit other effects dominate the failure rate. There are clear similarities to the second-degree dynamic avalanche, where a too low doped anode layer can be compensated by the hole density in a current filament, compare Sect. 13.4.2. However, in a cosmic ray induced streamer even higher current densities can be expected than in case of dynamic avalanche.

With an additional very shallow and highly doped p^+ layer directly located at the metal-semiconductor interface (contact implantation), the electric field will no longer reach the semiconductor surface, and it is reported that diodes with shallow emitter no longer show a decreased threshold voltage for cosmic ray failures [Wei15]. A very similar result is published in [Mit15], where such an additional shallow p^+ layer was investigated and the higher doping demonstrated improved stability.

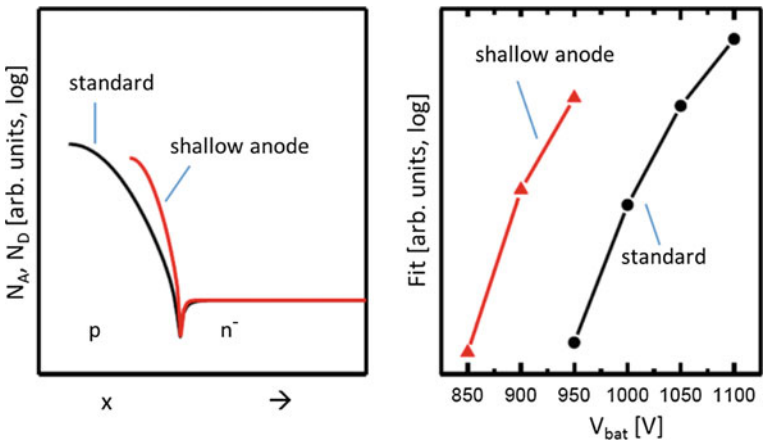


Fig. 12.60 Influence of anode design in diodes. Left: Comparison of standard and very shallow doped anode. Right: Failure rate measured with a proton beam. Figure adapted from [Wei15]

In a MOSFET, the failure mechanism is explained to be the activation of the parasitic bipolar transistor and second breakdown of the bipolar transistor [Was86]. For superjunction MOSFETs, it is reported that the distance between regions compensated with p columns and the n^+ substrate is of influence, compare Fig. 9.10. The higher this distance, the more cosmic ray stability was achieved [Wei15]. However, also the cell structure was found to be of influence. Regarding cosmic ray stability, superjunction devices are challenging.

A detailed final model of the failure mechanism related to 'Single Event Burnout' in IGBTs and MOSFETS is still a subject of research.

12.8.8 Cosmic Ray Stability of SiC Devices

SiC is sometimes considered as "radiation hard", however, there was never any experimental evidence for a statement regarding higher stability against cosmic ray failure or 'Single Event Burnout' (SEB). In fact, for the same voltage in SiC, the width of the space charge is 10 times lower, see Fig. 6.8. The volume where high field strength is given is also 10 times smaller for the same device area. On the other hand, the electric field strength at the same condition is 10 times higher. Which of these factors provide the higher influence?

First published experimental results were contradictory. For 600 V SiC Schottky diodes, the failure rate was found to be higher than for Si pin diodes. For 1200 V SiC Schottky diodes and JFETs, the failure rate was significantly below a Si pin diode. Therefore, early results were difficult to evaluate. Meanwhile, we know that there are pin diodes with failure rates varying over several orders of magnitude, see last paragraph. This might explain the conflicting results. For SiC, only since recently we

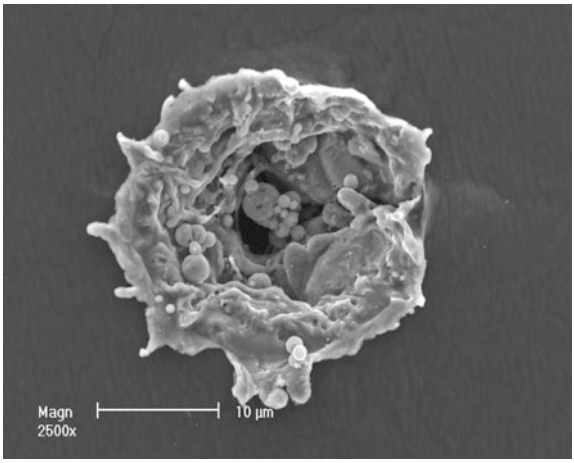


Fig. 12.61 Cosmic ray failure pattern of a SiC MPS diode. Figure from [Sho14]

are having results that allow a first evaluation. The defects in SiC resulting from cosmic ray failures look quite similar to that of Si, see Fig. 12.61 [Sho14].

Failure rates of different Si devices and one SiC MOSFET are compared in Fig. 12.62. The failure rate is normalized to the on-resistance, however, it is significant that there is a different threshold voltage for the different devices, especially some of the superjunction MOSFETS are sensitive. While for the 1200 V Si IGBT the threshold appears between 750 and 800 V, no failure is found for the SiC MOSFET up to 950 V.

So far, SiC seems to have an advantage. However, in an experimental investigation using the neutron beam in Uppsala with a spectrum comparable to the cosmic ray spectrum at sea level altitude, Consentino et al. [Con15] found the threshold voltage for a 1200 V SiC MOSFET in the range of 1020 V. In [Scn15b]

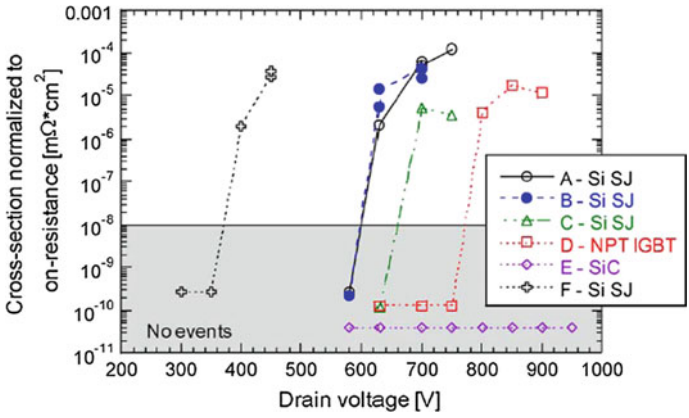


Fig. 12.62 Cosmic ray failure rates of four Si superjunction structures, Si IGBT and SiC MOSFET. The superjunction devices are rated 600 V (F), 900 V (C), 1000 V (B), 1100 V (A), Si IGBT and SiC MOSFET are rated 1200 V. Figure from [Gri12] © 2012 IEEE

the threshold voltage was found at 850 V for an Si IGBT of a more recent design (fieldstop) than the NPT IGBT in Fig. 12.60. So for a first quantitative comparison, the threshold voltage for cosmic ray failures can be assumed as 70% of V_{rated} for the Si IGBT and as 85% of V_{rated} for the SiC MOSFET (Fig. 12.63).

In simulations of cosmic ray failures in SiC MPS diodes by Shoji, similar effects like in Si diodes were found [Sho14]. A streamer shortens anode and cathode and the impact ionization generates higher charge at the cathode side. So we can conclude: The SEB current produced by impact ionization at the n^-n^+ interface is common to both Si and SiC power devices. SEB in power diodes thus corresponds to thermal destruction caused by an Egawa-type field (dynamic avalanche of the third degree, Sect. 13.4.2) In [Sho14], the same effect is denominated as "local second breakdown".

For diodes, a comparison between Si pin-diodes and SiC diodes is given in [Feg16a]. The results are shown in Fig. 12.64. The Si-Diode-C has an abrupt nn^+ junction where the electric field in the streamer will be comparable to Fig. 12.54a, the Si-Diode-A (CAL) has a soft nn^+ junction with low gradient dN/dx and the electric field in the streamer is expected to be similar to Fig. 12.54b. The results for the two Si-diodes are very different, the SiC diode is more robust than the Si-Diode-C but less robust than the Si-Diode-A.

A detailed comparison for SiC MOSFETs and Si IGBTs is presented in [Feg16b]. The devices are compared not only for the rated voltage, but also for the measured breakdown voltage. The rated voltage compared to the measured breakdown voltage V_{BD} for the investigated 1200 V Si IGBTs was found to be 88–89%, for the 1200 V SiC MOSFET it is 73%. For 1700 V, the Si IGBT used 79% of the breakdown voltage as rated voltage, the SiC MOSFET used 64%. This shows that the high critical field strength of SiC is only partially used in the investigated SiC designs. The results analyzed in dependence of V_{DC}/V_{BD}, meaning normalized to applied DC voltage V_{DC} compared to the measured breakdown voltage V_{BD}, are as follows: For 1200 V devices there is a small advantage of the SiC-MOSFET (Fig. 12.65a), for 1700 V the failure rate becomes significant at similar V_{DC}/V_{BD} (Fig. 12.65b).

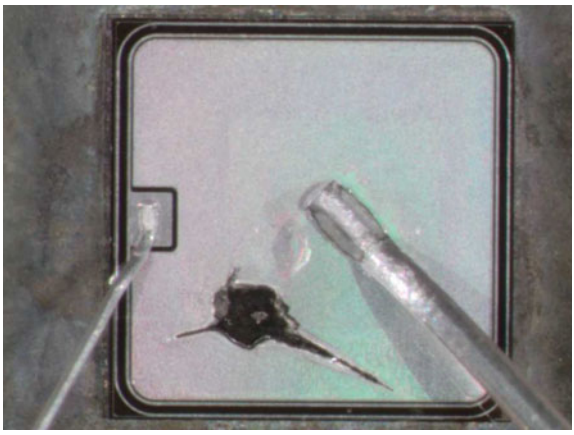


Fig. 12.63 Cosmic ray failure of a SiC MOSFET. Figure from [Con15] PCIM Europe 2015

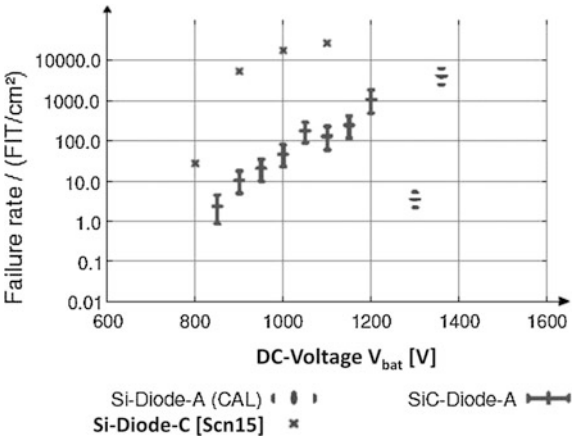


Fig. 12.64 Cosmic ray failure rate of two different Si diodes and one SiC diode. All diodes are rated 1200 V. Figure adapted from [Feg16a] PCIM Europe 2016

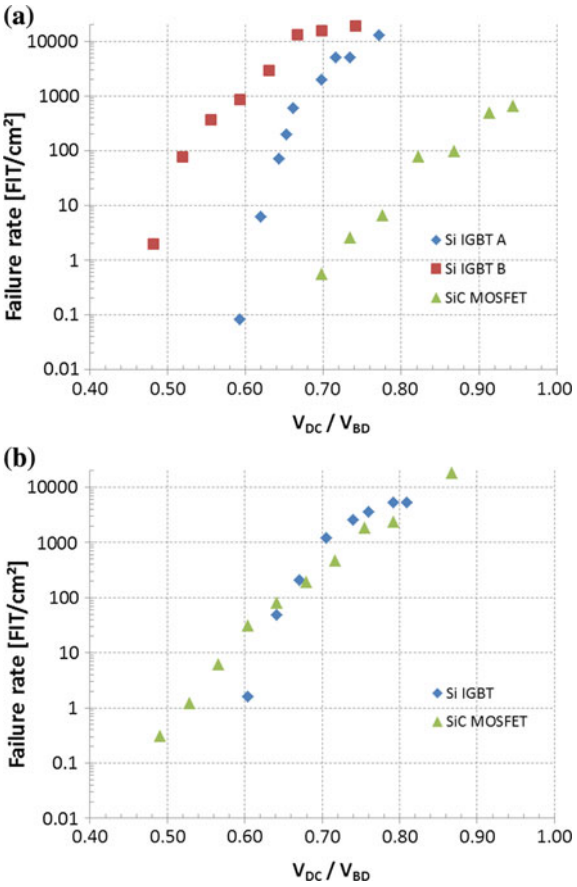


Fig. 12.65 Comparison of cosmic ray failure rates for Si IGBTs and SiC MOSFETs normalized to applied DC voltage V_{DC} compared to the measured breakdown voltage V_{BD}. **a** 1200 V rated devices, **b** 1700 V rated devices. Figures according to data from [Feg16b]

The physics of cosmic ray failures seems to be very similar for Si and SiC. However, with increased crystal quality in the future, for SiC the full potential of the material will also be exploited. More data for SiC and for Si are of high interest. For the same rated current there is a lower device area for SiC. Therefore, it still may remain a small advantage for SiC.

12.9 Statistical Evaluation of Reliability Test Results

The evaluation and interpretation of reliability test results can be considerably enhanced by statistical methods. Statistic software collections (e.g., Minitab® Statistical Software [MIN18]) provide powerful tools for the analysis of reliability test data or for the investigation of field failures. While statistical methods can be very helpful if applied correctly, they can also be misuse or misinterpreted by unexperienced engineers. It would exceed the frame of this book to provide fundamental knowledge on statistics, thus we will give some examples of statistical interpretations and we will also discuss some cases of incorrect application or misinterpretation of statistics, which are sometimes encountered in reliability investigations.

A good example for statistical analysis is the evaluation of end-of-life power cycling test results of power modules. If we only have a single result, we cannot say anything about the expected result of a second test performed under the same conditions. Often such a single test result is associated with a 50% failure probability. However, since we have no information on the statistical distribution of failures, this interpretation is not supported by statistics.

If several power cycling tests are conducted at the same test condition, much more information on the statistical character of failures can be extracted by statistical tools. The example in Table 12.6 shows the results of 6 power cycling tests which were conducted at identical test conditions. Power pulses with t_{on} = 1 s and t_{off} = 1 s were imposed by a DC load current of 244 A on the IGBTs of a 1200 V SKiM63 module bonded with an aspect ratio of ar = 0.31. The tests were conducted until the total failure of all wire bonds occurred and the number of cycles to reach this failure is indicated by N_{ex}.

Using a statistic software like Minitab allows to analyze the data obtained for N_{ex}. Minitab [MIN18] provides a distribution identification tool, which compares the probability that a given sample belongs to a specific statistical distribution by an Anderson-Darling-Test. For the values N_{ex}, this test confirms that distribution is most probably a Weibull distribution. The Weibull distribution is described by the probability density function

Table 12.6 Power cycling results for 6 IGBTs of a 1200 V SKiM63 module with temperature swing parameters, experimental results N_{ex} and corrected values N_{corr}

	T_{jmin} (°C)	T_{jmax} (°C)	ΔT_j (K)	T_{jm} (°C)	N_{ex}	N^1_{mod}	$f_{corr} = N_{ex}/N_{mod}$	$N_{corr} = N_{ref} \cdot f_{corr}$
1T	80	148	68	114	521,940	499,549	1.0448	459,829
1B	80	150	70	115	451,810	440,103	1.0266	451,810
2T	83	152	69	117.5	464,440	461,578	1.0062	442,832
2B	80	149	69	114.5	537,050	468,641	1.1460	504,346
3T	82	153	71	117.5	454,520	409,564	1.1098	488,411
3B	82	150	68	116	594,860	494,492	1.2030	529,432
Ref	80	150	70	115		440,103		

[1]SKiM63 lifetime model estimation with $ar = 0.31$ and $t_{on} = 1$ s

$$f(x, \alpha, \beta) = \frac{\alpha}{\beta^{\alpha}} \cdot x^{\alpha-1} \cdot \exp\left(-\left(\frac{x}{\beta}\right)^{\alpha}\right) \tag{12.16}$$

with the shape parameter α and the scale parameter β. The parameter x represents in this context the number of cycles to failure. The accumulated number of failures at a given number of cycles is obtained by the integration over the probability density function up to x which gives the cumulative distribution function of the Weibull distribution

$$F(x, \alpha, \beta) = 1 - \exp\left(-\left(\frac{x}{\beta}\right)^{\alpha}\right) \tag{12.17}$$

From the cumulative distribution function, the parameters of the Weibull distribution can be extracted as shown in Fig. 12.66. By plotting $\ln[-\ln(1 - F)]$ over $\ln(x/\beta)$ the Weibull distribution is represented by a straight line with the slope given by the shape parameter α. The two additional curves on both sides of estimated Weibull distribution indicate the confidence intervals for a 95% confidence level, i.e. the straight line for the real distribution can be located anywhere in the area between the curves with a confidence of 95%.

However, if we look at the temperature limits T_{jmin} and T_{jmax} of the swing in Table 12.6, we can see that the power cycling conditions are not exactly identical for all 6 tests. This observation, which is typical for power cycling tests, is caused by the active role of the device under test (DUT) in power cycling testing: The temperature swing is not only determined by external parameters like current and load pulse duration, but also by internal characteristics, i.e. the forward voltage drop and the thermal resistance of the DUT. These device parameters exhibit variations due to material tolerances and fluctuations in the production process, which results in small differences in the temperature swing. The parameters of the temperature swing must be determined in the stable phase of the test when all initial transient processes have saturated. These transient processes are the heating of the heat sinks and module parts, the redistribution of thermal interface materials between module and heat sink

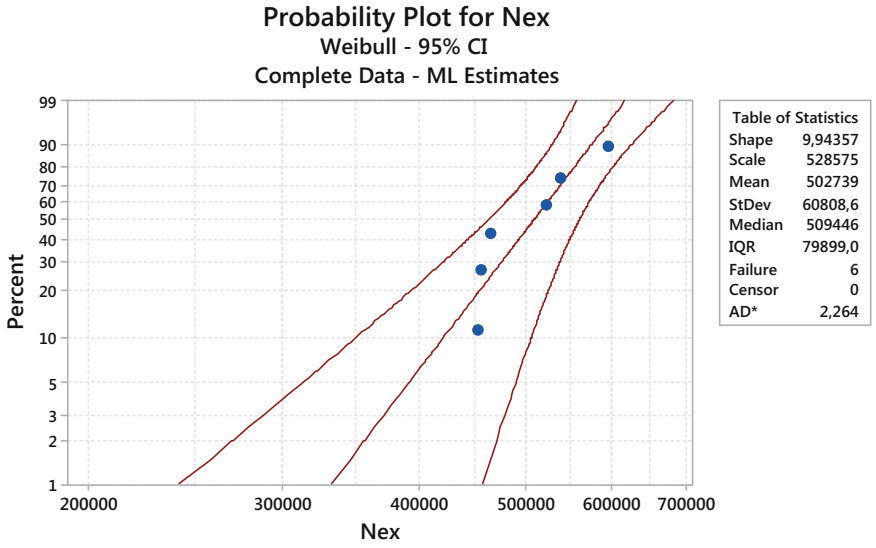


Fig. 12.66 Weibull distribution of N_{ex} in Table 12.6 with parameters and 95% confidence intervals extracted by maximum likelihood estimation with Minitab®

and the stabilization of the cooling liquid temperature controller. On the other hand, the initial temperature swing parameters must be determined before any degradation effects occur, so that no general rule can be applied. For the tests listed in Table 12.6, the initial values are determined by averaging T_{jmin} and T_{jmax} over several hundred cycle values after a few thousand cycles and rounding to integer values.

We can assume that the small variations of the temperature swing ΔT_j and medium temperature T_{jm} in Table 12.6 will have an impact on the test results. If a lifetime model is available that allows to calculate this impact on the number of cycles to failure, the test results can be corrected for this parameter variation. For the given example in Table 12.6, a lifetime model is available: The SKiM63 model introduced in Sect. 12.7.4. Applying the model parameters given in Table 12.4 to the temperature swing data, we can calculate the number of cycles to failure N_{mod} resulting from the lifetime model. The correction factor $f_{corr} = N_{ex}/N_{mod}$ describes the experimental results relative to the model prediction at the experimental temperature swing. If we would have tested at the reference temperature swing, we should have obtained for the cycles to failure $N_{corr} = N_{ref} \cdot f_{corr}$ if the lifetime model is correct. Thus, we can eliminate the variation in the temperature swing and can statistically analyze the corrected data as shown in Fig. 12.67.

To evaluate the correction process, we can compare the probability density functions of the two Weibull distributions in Fig. 12.68. The Weibull distribution for the uncorrected values exhibit a wider probability density function with a shape factor of 9.9 compared to 16.4 for the corrected values. This must be expected for a valid lifetime model, since the corrected values are more exact and the uncorrected values feature a broadening of the distribution by parameter variation.

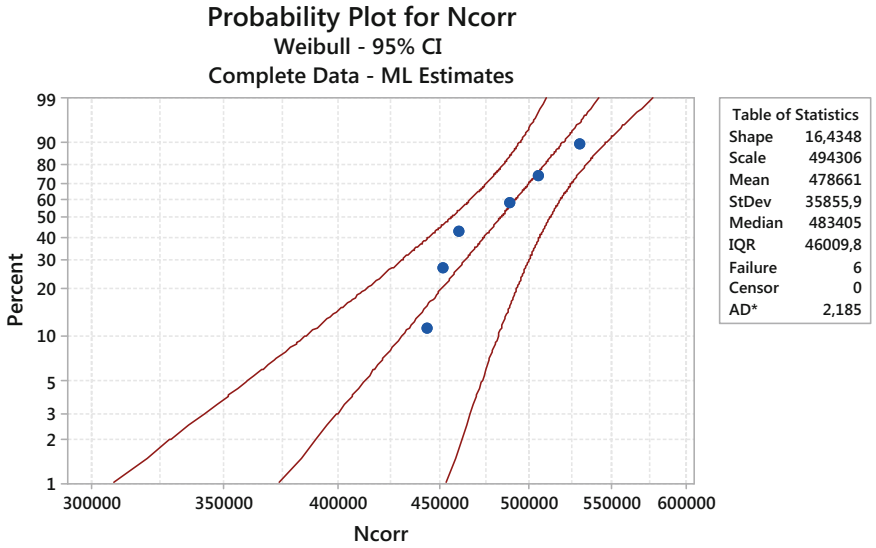


Fig. 12.67 Weibull distribution of corrected values N_{corr} in Table 12.6 extracted by maximum likelihood estimation with MiniTab®

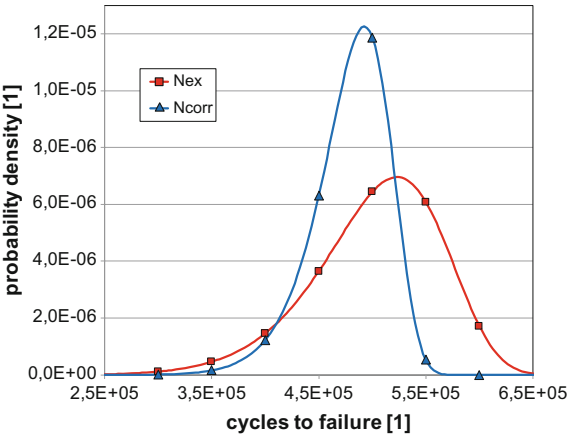


Fig. 12.68 Comparison of the probability density functions for the experimental result N_{ex} and the corrected results N_{corr} with the SKiM63 lifetime model

This correction process can be applied to perform a statistical evaluation of a complete lifetime model. Therefore, we could map all power cycling results with a wide variation of parameters to a single reference condition as we have done in Table 12.6. However, since the selected reference point is arbitrary, we can simply select the correction factor f_{corr} for the evaluation. This was done for the complete data set of more than 100 power cycling test results that was the basis for the parametrization of the SKiM63 lifetime model [Scn13]. Since the model coefficients (see Table 12.4) were determined by a least square fit, half of the data values

should be >1. Since the failure probability in Fig. 12.69 is shown as a function of the relative number of cycles to failure minus the threshold value (0.507) of a 3-parameter Weibull distribution, the 50% failure probability is reached at a value of ∼0.5 as expected (long arrow in Fig. 12.69). When the lifetime model predictions are multiplied with a margin factor of 0.8 (short arrow in Fig. 12.69), only 15% of parameter combinations would exhibit a failure before the estimated lifetime, while 85% would reveal a longer lifetime.

It should be emphasized, that the evaluation of a complete lifetime model has a higher significance than the evaluation of a single combination of test parameters in Figs. 12.66 and 12.67. The result for a single test condition provides no information on the statistical distribution at a different set of test parameters. The evaluation of a lifetime model gives a statistical information for the whole parameter space covered by the data set that was applied to determine the lifetime model coefficients.

The statistical analysis of test results delivers a correlation between the cycles to failure and the expected accumulated failure probability. However, an important question remains: Which failure probability should be defined for an estimated lifetime. Often, lifetime estimations for an accumulated failure probability of 1% or even lower are requested for a component by system designers. This can be understood as a quest for the lowest possible failure probability in application, however, this definition is problematic from a statistical point of view: The estimation of lifetime from a sample of tests run to end-of-life exhibits the highest accuracy for a 63% failure probability for a Weibull distribution, as can be seen in the probability plots shown before. At 63% probability, the difference between the

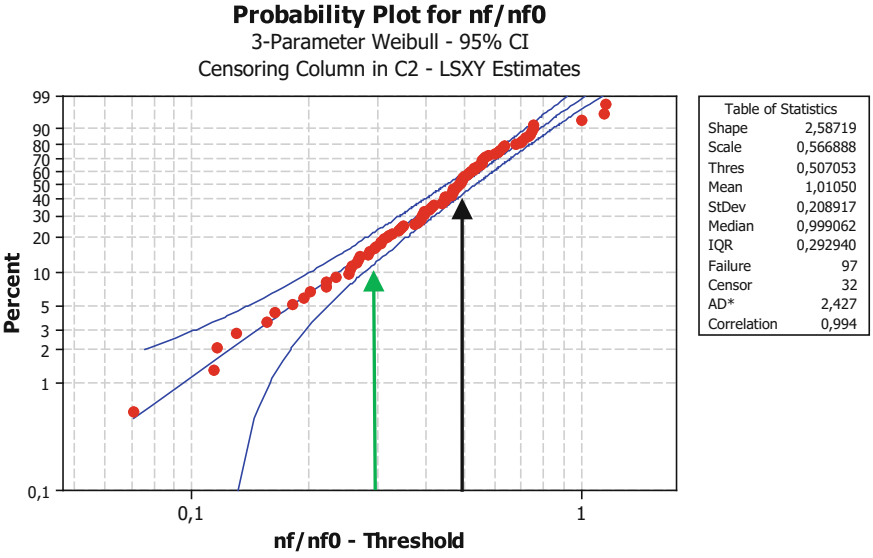


Fig. 12.69 Statistical evaluation of the SKiM63 lifetime model with a 3-parameter Weibull distribution. A margin factor of 0.8 (short arrow) indicates a failure probability of 15% [Scn13]

confidence intervals has its minimum. For higher or lower failure probabilities, the difference between the confidence intervals increases. Thus, the lower the defined failure probability, the higher the uncertainty of the medium distribution parameters indicated by the straight line in the probability plots. This results from the fact, that in a statistical sample a large fraction of elements will be in proximity of the maximum of the probability density function and only few elements will be found at the extremes. Therefore, failure probability of 10–15% seems to be a good compromise between low failure probability and uncertainty of prediction.

The statistical analysis is not restricted to power cycling tests; it can be applied to all failures observed in reliability testing. Figure 12.70 shows the probability plot of cosmic ray failures of 1700 V CAL diodes in an accelerated test with a mono-energetic 200 MeV proton beam of a flux density of 1.8×10^6 ps^{-1} cm^{-2} at a bias voltage of 1450 V. This test exhibits an acceleration factor of 6.48×10^8 related to the natural cosmic ray flux density at sea level (10 ph^{-1} cm^{-2}).

At the first glance, there seems to be no difference to the probability plots shown before. However, we can see in the data given in Fig. 12.70 that the shape factor is very close to 1.

To interpret the results, we have to look at the statistical definition of the failure rate. The reliability function or accumulated survival rate for a Weibull distribution is

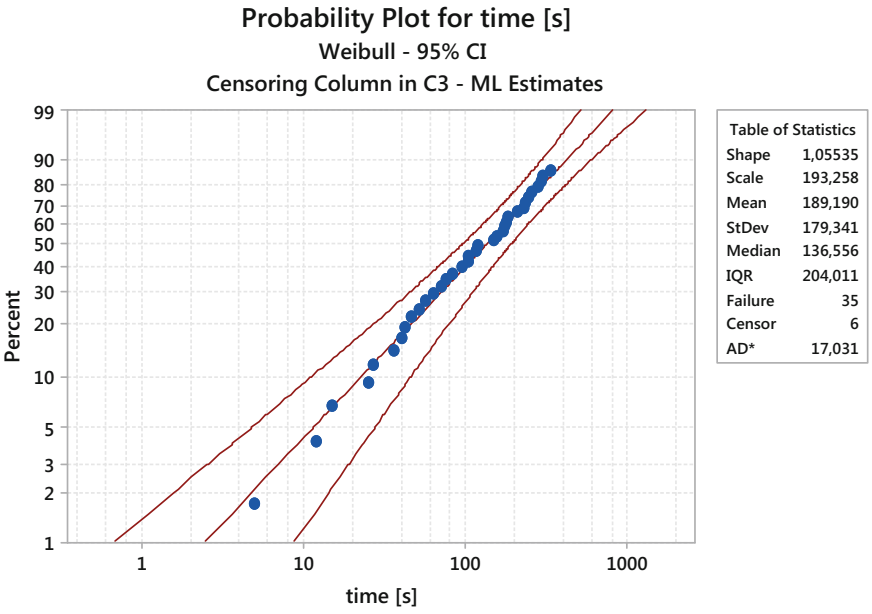


Fig. 12.70 Statistical analysis of cosmic ray failures of 1700 V CAL 4 diodes (area 44 mm^2) at 1450 V bias voltage in an accelerated test with a 200 MeV proton beam with a flux density of 1.8×10^6 $ps^{-1}cm^{-2}$ (middle value in Fig. 12.56) by MiniTab®

$$R(x,\alpha,\beta) = 1 - F(x,\alpha,\beta) = \exp\left(-\left(\frac{x}{\beta}\right)^{\alpha}\right) \tag{12.18}$$

The failure rate at any value x is given by the probability density function divided by the probability of survival at x, so that we obtain for a Weibull distribution

$$\lambda(x,\alpha,\beta) = \frac{f(x,\alpha,\beta)}{R(x,\alpha,\beta)} = \frac{\alpha}{\beta^{\alpha}} \cdot x^{\alpha-1} \tag{12.19}$$

From Eq. 12.19 we can see, that for a Weibull distribution with a shape factor $\alpha = 1$ the failure rate is a constant $\lambda = 1/\beta$ independent from the parameter x. Thus, we can determine from the scale factor $\beta = 193$ s the failure rate at sea level $\lambda = 29$ FIT for this 44 mm^2 diode. This is equivalent to a normalized failure rate of 65 FIT/cm^2 as shown in Fig. 12.56 for the middle value at 1450 V. The statistical analysis also allows to calculate defined confidence intervals for the measured data as described in standard textbooks on statistics, e.g. [Rau04].

Statistical methods also facilitate the calculation of the maximum resolution achievable with an accelerated test and the selection of sample size and test duration. An example is shown in Fig. 12.71 for accelerated cosmic ray tests. If at least 5 failures are required for statistical evaluation, the maximum acceleration factor given by the proton irradiation beam limits the detectable failure rate to $\sim$0.01 FIT for practical test conditions of 100 samples and 30 min beam time.

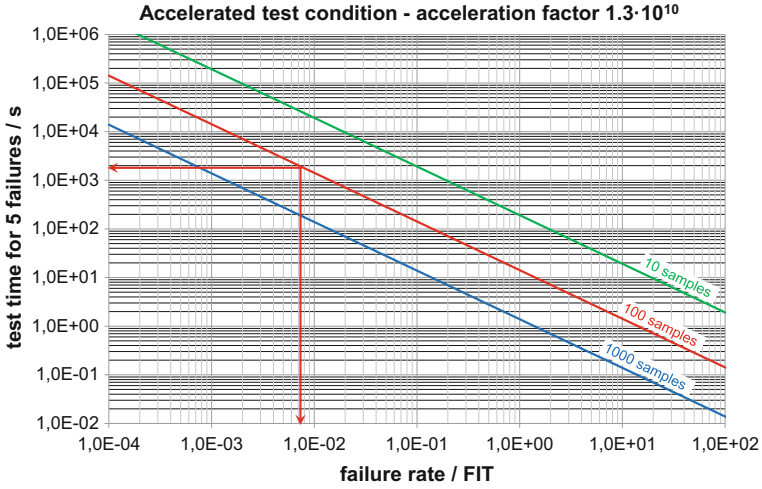


Fig. 12.71 Average test duration to obtain 5 failures at a specific failure rate for different sample sizes with an acceleration factor of 1.3×10^{10}. Arrows mark the practical detection limit for 100 samples and 30 min test time. Figure from [Scn16]

The analysis of the cosmic ray failures has shown, that they follow a Weibull distribution with a shape factor of 1, in contrast to the failures in power cycling tests which are characterized by $\lambda > 1$. The reason for this difference is, that they belong to different failure categories or failure types.

The stress during power cycling causes degradation and fatigue and is associated with accumulating damage in the system. Since it takes time to accumulate damage, a low failure rate at the beginning, which increases over time, can be expected. Therefore, power cycling failures belong to the failure type of 'end-of-life' failures in Fig. 12.72.

Cosmic ray failures are not increasing over time; the failure rates remain constant throughout the system life. The reason is, that an event originating from outside of the system is the root cause of failure. It does not matter, if a system has operated for 1 h or for 1 million hours; the probability of an impact of a cosmic ray particle remains constant. This failure type is referred to as random failure.

The third failure type, the 'early-life' failures, is characterized by a failure rate that decreases over time. This failure type is related to weak systems, which are selected from the population by sudden failures. When weak elements are removed, the remaining population becomes stronger and the failure rate decreases.

Since the bathtub curve in Fig. 12.72 is constructed by Weibull distributions, the reliability function or survival probability can be calculated from the distribution parameters and Eq. (12.18). The result is shown in Fig. 12.73. As discussed before, a survival probability of 99% is often requested for the limitation of useful life by degradation. This condition is reached in Fig. 12.73 after approx. 17.5 years of continuous operation. However, there could be considerably more failures after 17.5 years resulting from early-life and random failures. This must be kept in mind when targeting for so-called health monitoring concepts. The idea is to monitor degradation effects in power electronic components and issue a warning before the

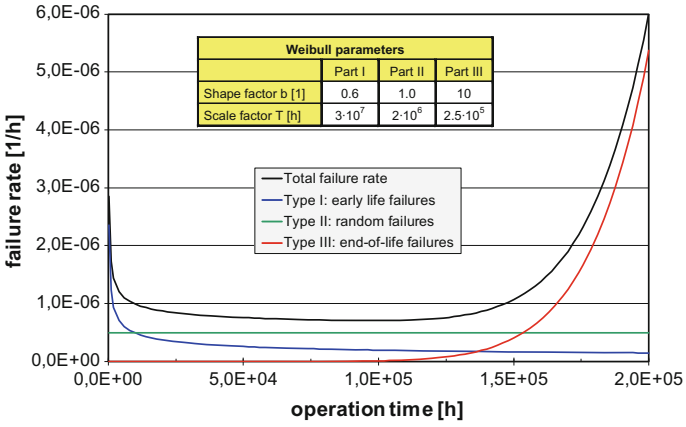


Fig. 12.72 Bathtub curve for a hypothetical system as a sum of failure rates for early-life-failures, random failures and end-of-life failures [Scn15a]

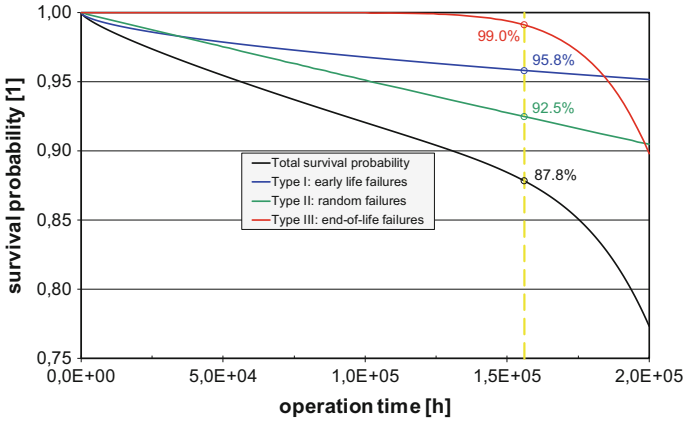


Fig. 12.73 Survival probability for the bathtub curve in Fig. 12.72 and contributions of the different failure types for a hypothetical system [Scn15a]

component fails. Since only end-of-life failures are associated with degradation, these concepts must be blind with respect to early-life and random failures. Therefore, health monitoring concepts will only make sense, when early-life and random failures are negligible.

In reliability assessments, MTTF (Mean Time To Failure) is often considered as an appropriate measure for the reliability of components or systems. However, MTTF is often misinterpreted. Let us assume a constant failure rate of 500 FIT or 0.5×10^{-6} h^{-1}. For a constant failure rate, the shape factor of the related Weibull distribution must be 1 and the MTTF is simply MTTF = β=1/λ = 2×10^6 h. Statements are sometimes heard like: "An MTTF of 2 million hours is a good value, because our inverter is designed for 2×10^5 h lifetime. Since the MTTF is 10 times higher than the lifetime, this inverter will not fail". This interpretation assumes, that the MTTF-value can make predictions about failures of an individual inverter, which is not the case. MTTF is a characteristic of a statistical entity. An MTTF = 2×10^6 h predicts that for an entity comprising 1000 systems, one failure is expected every 2000 h of operation of all system.

MTTF is the expectancy value for a failure:

$$MTTF = \int_0^\infty R(x)dx \tag{12.20}$$

For a Weibull distribution, this expectancy value can be calculated with the gamma function $\Gamma(x)$:

$$MTTF = \beta \cdot \Gamma\left(1 + \frac{1}{\alpha}\right) \tag{12.21}$$

We can now calculate the MTTF values for the Weibull distributions in Fig. 12.72 with Eq. (12.21) and the MTTF value of the resulting bathtub curve using Eq. (12.20). We can also calculate the reliability function R(MTTF) using Eq. (12.18) for the 3 Weibull distributions. For the bathtub curve, the reliability function is determined by the product of the 3 contributing functions at the MTTF value of the bathtub curve. The results are collected in Table 12.7.

The results show, that at a time MTTF = 2×10^6 h, only $\sim$37% of a population of inverters is expected to survive assuming a constant failure rate of 500 FIT. The survival probability at MTTF is with $\sim$28% even worse for the early-life failures. This example demonstrates, that MTTF alone is not a good measure for the system reliability. Without additional information on the distribution of failures, the MTTF value gives no information about the survival probability of a statistical entity.

Finally, we will discuss the failure type of early-life failures. This failure category comprises failures with a decreasing failure rate. A way to reduce this failure rate is to establish appropriate tests that can screen out weak components and thus reduce the initial failure rate. However, it must be validated that the test conditions, which are often beyond the specification limits of the components, exhibit a decreasing failure rate as a function of test time.

An example of such a screening test is published in [Sie17]. SiC trench MOSFETs were subjected to a gate stress test at a temperature of 150 °C. The gate voltage applied was beyond the specification limit of 15 V. The observed failures as a function of test time are shown in Fig. 12.74. The results in this image show that the failure rate at each test level decreases over time.

Table 12.7 MTTF values and survival probabilities at MTTF for the Weibull distributions and the resulting bathtub curve in Fig. 12.72

Failure type	Shape α	Scale β [h]	MTTF [h]	R(MTTF) (%)
Early-life failures	0.6	3×10^7	4.5×10^7	27.9
Random failures	1	2×10^6	2×10^6	36.8
End-of-life failures	10	2.5×10^5	2.4×10^5	54.5
Bathtub curve	–	–	2.2×10^5	67.1

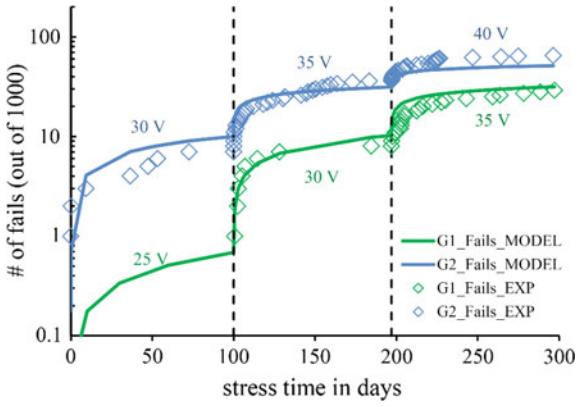


Fig. 12.74 Results of a long-time gate stress test at 150 °C on 1000 SiC trench MOSFETs with a rated maximum gate voltage of 15 V. Figure from [Sie17]

A statistical evaluation of the failures observed in group G1 at 30 V gate voltage (lower curve in Fig. 12.74) is shown in Fig. 12.75. In contrast to the previously discussed probability plots, the majority of the elements have not failed. Only 8 failed devices out of 1000 MOSFETs failed in 100 days test time at 30 V gate voltage. The MOSFETs that survived the test are taken into account as censored values. Elements without failure are marked as right-censored in the analysis. The shape factor of the Weibull distribution is 0.37, which verifies that this test produces early-life failures. The poor accuracy of the estimated shape factor is attributed to the small fraction of failed elements; it could be considerably improved by an increased test duration with more failures or by an increased stress, i.e. a higher gate voltage in this example

When the censored values are not taken into account, a totally different result is obtained, as Fig. 12.76 shows. This probability plot would be correct, if 7 MOSFETs were tested and all devices failed. However, in the test published in [Sie17] only a small fraction of elements had failed while more than 99% of devices survived the test level at 30 V gate voltage.

Statistical tools can also be applied to analyze field failures. However, if a short circuit of a device occurs in a wire bonded power module, the high energy involved in power electronic systems in general causes massive destruction and the identification of the root cause seldom possible. Therefore, only the accumulated bathtub curve in Fig. 12.72 can be determined. Furthermore, field failures are only found in a small fraction of total systems in operation. This is same situation as discussed in Figs. 12.75 and 12.76. The challenge in analyzing field failures is to collect the information on the operation time of all functional systems in the same application. If this information is

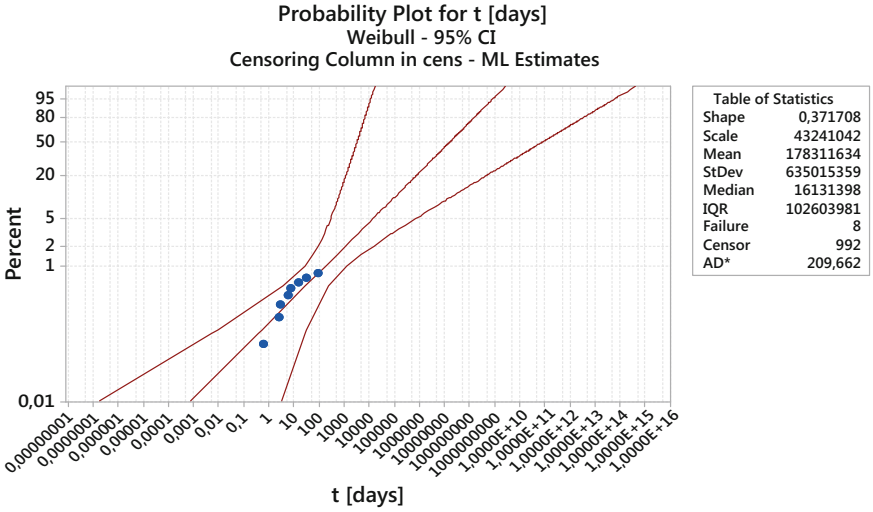


Fig. 12.75 Probability plot for the failures at 30 V gate voltage of SiC MOSFET of group G1 (lower curve) in Fig. 12.74 by MiniTab®

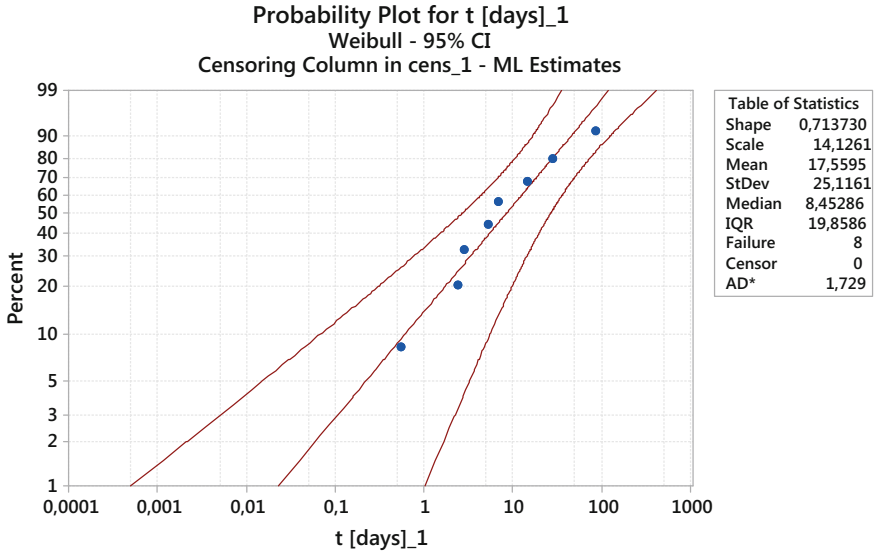


Fig. 12.76 Probability plot for the failures at 30 V gate voltage of SiC MOSFET of group G1 (lower curve) in Fig. 12.74 without censored values by MiniTab®

available or if it can be approximated by valid assumptions, statistical analysis can deliver estimations on the number of expected failures in the future.

12.10 Further Reliability Tests

A test sequence similar to the example given in Table 12.1 is mandatory for all series products of a power module manufacturer. However, additional test sequences can be negotiated for specific applications.

For applications in extreme environmental conditions, special tests under corrosive atmospheres are recommended. Corrosive gases can interfere seriously with the reliable function of power modules. The silicone soft mold represents almost no protection against corrosive gases. SO_2 interacts with all metal surfaces except noble metals, H_2S is highly corrosive for silver and silver alloys and unprotected Cu surfaces and Cl_2 together with high humidity will produce HCl, which corrodes non-noble metals, especially Al. The contribution of NO_x is not fully understood today, but its corrosion effect in connection with humidity is comparable to the impact of SO_2 and H_2S. The reliability of non-hermetically sealed modules must be verified in accelerated corrosive atmosphere tests in single or mixed corrosive gas environments for these applications.

Similar corrosion effects are connected with the impact of salt spray. This specific stress condition is found particularly in seaside or off-shore applications, which are typical environments for wind generator systems. The NaCl dissociates in

aqueous solution and produces HCl, which over time can penetrate the soft mold cover layer and dissolve the Al wire bonds. If non-hermetically sealed modules are used in these environments, an additional protection must be implemented in the system to prevent corrosive degradation of power modules and driver PCBs.

Additional tests can be conducted for special application conditions to verify the suitability of power electronics modules in extreme environments.

The standard test procedures discussed in the previous sections always use new power modules fresh from production for every reliability test. Some experts in the field of reliability testing propose the combination of tests to increase the reliability of power electronic components. Even though the consecutive or simultaneous application of different stress conditions could accelerate the time to failure of a power module, the number of possible sequences and combinations is quite high and there is no basis of experience for such test conditions.

However, the idea was adapted in the concept of HALT/HASS testing. The 'Highly Accelerated Lifetime Test' (HALT) is actually not a test, but much more a test philosophy, that accompanies the development process. During the HALT procedure, a single device parameter is selected and it is stepwise increased even far beyond the specification limit until the sample finally fails. Typical examples are the stepwise increase of the insulation test voltage until failure or the stepwise increase of the temperature swing during passive temperature cycling test. The goal of this procedure is to increase the stress to the destruction limit. From the analysis of the failure, small modifications can sometimes improve the robustness of a design considerably.

In a second step, a 'Highly Accelerated Stress Screening' (HASS) can be applied. Using the failure limits resulting form the HALT procedure, which typically are far beyond the specification limit, a HASS stress level can be defined, which is selected higher than the specification limit but a sufficient safety margin away from the destruction limit. For a fixed period of time 100% of a production is then subjected to an accelerated stress screening test, which allows to identify weak parts in a series production and improve the production process to enhance the product reliability. In this test sequence, combined stress levels are common. A typical example is the operation of a power module with high load currents under varying ambient temperature swings, sometimes even combined with the stress of mechanical vibration.

The general philosophy is developed further in the proposed concept of robustness validation [SAE08]. This concept combines experimentally determined failure limits and simulation results and compares these limits with the reliability requirements of applications, defined in mission profiles and specifications to increase the reliability of automotive electrical or electronic modules.

A general problem is the limited knowledge of the stress levels and the distribution of stress requirements in real applications. The implementation of the digital driver technology in power electronic systems gives the opportunity to monitor and record characteristic parameters during field operation and therefore has the potential to deliver more realistic data on the stress requirements of realistic field applications.

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